Solutions Manual

Fundamentals of Acoustics

FOURTH EDITION

Lawrence E. Kinsler
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Alan B. Coppens
James V. Sanders

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TO ACCOMPANY

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Introduction

This Solutions Manual includes Solutions to Selected Problems, Answers to Even-Numbered Problems and some additional homework problems for the textbook *Fundamentals of Acoustics*, Fourth Edition.

ADDITIONAL HOMEWORK PROBLEMS THAT MAY BE OF INTEREST

The following are either new problems that were inadvertantly omitted from the manuscript for the fourth edition of this text or certain revisions that may be of interest.

1.2.2. An ideal *torsional* pendulum consists of a body of moment of inertia I attached to the end of a weightless wire of length L. A static torque applied to the end of the wire is found to be proportional the angular displacement of the end of the wire with proportionality constant η. (*a*) Show that the equation of angular motion for the mass is that of a simple harmonic oscillator. (*b*) Find the frequency of torsional oscillations of this pendulum.

1.2.3. An ideal *gravitational* pendulum consists of a small mass m hung from a weightless string of length L. The acceleration of gravity is g. (*a*) For small displacements from the vertical, show that the equation of motion for the mass is that of a simple harmonic oscillator. (*b*) Find the frequency of oscillation of this pendulum. (*c*) Show that for large displacements from the vertical, the motion is no longer simple harmonic.

1.15.10C. A sawtooth wave can be represented as a repeating waveform with each cycle having the same shape as

$$f(t) = a(1 - 2t/T) \qquad 0 \le t < T$$

(*a*) Show that the Fourier series coefficients are $A_n = 0$ and $B_n = a/(n\pi)$ for $n = 1, 2, 3, \ldots$. (*b*) Make plots of the Fourier series retaining the same number of harmonics as in each of the plots for the square wave in Fig. 1.14.1. Comment on the relative importance of overshoot for the two waveforms.

7.6.10. An acoustic fish finder has a piston source with a radius of 5.0 cm. It radiates 50 W of acoustic power at 100 kHz.. (*a*) What is its beamwidth between the down 6 dB points? (b) What is the axial pressure level in dB *re* 1 μPa at a distance of 10 m from the face of the transducer?

7.8.12C. Plot the directional factor for the "shotgun" microphone of Problem 7.8.11 for various value of kd and N and compare with the patterns for a circular piston at the same values $ka = kd$.

7.10.12. Assume a spherical source with uniform density $\mathbf{q}(\vec{r}_0) = Q/(4\pi a^3/3)$ for $r_0 < a$ and 0 for $r_0 \ge a$. Calculate the multipole expansion for the acoustic field at large distances.

8.3.4. (*a*) Show that the intensity of a traveling wave $\mathbf{p} = P\exp(-\alpha x)\exp[j(\omega t - kx)]$ satisfies the equation $(1/I)(dI/dx) = -2\alpha$. (*b*) Show that for a standing wave $\mathbf{p} = P\sin(kx)\exp(-\beta t)\exp(j\omega t)$ the intensity satisfies $(1/I)(dI/dt) = -2\beta$. (*c*) If both waves satisfy the same lossy Helmholtz equation, show that $\beta \approx \alpha c$.

16.3.4. A spherical source of radius a radiates high intensity sound of angular frequency ω. The surface of the source has a peak Mach number $M = U/c$. (*a*) For sufficiently large ka, using appropriate approximations and ignoring losses, follow Earnshaw's reasoning to find the discontinuity distance $\ell \equiv r - a$ at which a shock will form. (*b*) Show that your result reduces to (16.3.6) in the limit $ka \gg 1/(\beta M)$. (*c*) Using the definition $\beta M/(\alpha/k)$ for the Goldberg number, express Γ in terms of α, ℓ, and a. (*d*) Estimate the smallest source for which these results are valid at 1 kHz in air.

SOLUTIONS TO SELECTED PROBLEMS

Because of software limitations, in a few equations characters that should appear in script font will be written as (*Script*"X"). For example, (*ScriptP*) = $\mathscr{P}$ and (*ScriptT*) = $\mathscr{T}$.

1.3.2. $\omega_0 = 5$ rad/s $\quad x(0) = 0.03$ m $\quad \dot{x}(0) = 0$ $\quad$ Solving $\ddot{x} + \omega_0^2 x = 0$ with $x = A\cos\omega_0 t + B\sin\omega_0 t$ we get $x = 0.03\cos\omega_0 t$

(*a*) $\ddot{x} = -0.03\,\omega_0^2\cos\omega_0 t$ and $\ddot{x}(0) = -0.03\cdot 5^2 = \underline{-0.75\text{ m/s}^2}$

(*b*) Amplitude $= A = \underline{0.03\text{ m}}$

(*c*) $\dot{x} = -0.03\,\omega_0\sin\omega_0 t$ and $\dot{x}_{\max} = 0.03\,\omega_0 = \underline{0.15\text{ m/s}}$

1.5.3. (*a*) $\mathbf{AB} = AB\exp[j(2\omega t + \theta + \phi)]$ and $\mathrm{Re}\{\mathbf{AB}\} = \underline{AB\cos(2\omega t + \theta + \phi)}$

(*b*) $\mathbf{A}/\mathbf{B} = (A/B)\exp[j(\theta - \phi)]$ and $\mathrm{Re}\{\mathbf{A}/\mathbf{B}\} = \underline{(A/B)\cos(\theta - \phi)}$

(*c*) $\mathrm{Re}\{\mathbf{A}\}\mathrm{Re}\{\mathbf{B}\} = \underline{AB\cos(\omega t + \theta)\cos(\omega t + \phi)}$

(*d*) From (*a*), phase $= \underline{2\omega t + \theta + \phi}$

(*e*) From (*b*), phase $= \underline{\theta - \phi}$

1.6.1. $m = 0.5$ kg $\quad m' = 0.2$ kg $\quad \tau = 1$ s $\quad g = 9.8$ m/s^2 $\quad \Delta x = 0.04$ m

$$\Delta F = s'\Delta x = m'g$$

$$s = 0.2\cdot 9.8/0.04 = 49\text{ N/m}$$

$$\omega_0 = \sqrt{s/m} = \sqrt{49/0.5} = 9.90\text{ rad/s}$$

$$\beta = 1/\tau = 1\text{ s}^{-1}$$

$$R_m = 2m\beta = \underline{1.0\text{ kg/s}\quad\text{or}\quad 1.0\text{ N}\cdot\text{s/m}}$$

$$\omega_d = \sqrt{\omega_0^2 - \beta^2} = \sqrt{98 - 1} = \underline{9.85\text{ rad/s}}$$ or, alternatively,

$$\omega_d = \omega_0\sqrt{1 - (\beta/\omega)^2} \approx \omega_0\left[1 - \tfrac{1}{2}(\beta/\omega)^2\right] \approx 9.90[1 - 1/(2\cdot 9.8)]$$

$$\approx 9.90(1 - 0.05) \approx \underline{9.85\text{ rad/s}}$$

$$x = Ae^{-\beta t}\cos(\omega_d t + \phi)$$

$$\dot{x} = -A\omega_d e^{-\beta t}\sin(\omega_d t + \phi) - A\beta e^{-\beta t}\cos(\omega_d t + \phi)$$

$$x(0) = A\cos\phi = 0.04$$

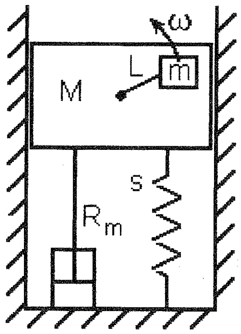
$$\dot{x}(0) = -A\omega_d\sin\phi - \beta A\cos\phi = 0$$

$$\therefore\ \tan\phi = -\beta/\omega_d = -0.1015$$

$$\phi = \underline{-5.8^\circ}$$

$$A = 0.04/\cos\phi = \underline{0.0402\text{ m}}$$

1.7.3.



x = upward displacement of M
$y = L\cos\omega t + x$ = upward displacement of m

Vertical force on m = $+m\ddot{y}$
Vertical force on M = $-m\ddot{y} - R_m\dot{x} - sx = +M\ddot{x}$

Generalize x and y to $\mathbf{y} = Le^{j\omega t} + \mathbf{x}$

and substitute into the equation for the motion of M

$$\omega^2 Lm\,e^{j\omega t} = m\frac{d^2\mathbf{x}}{dt^2} + M\frac{d^2\mathbf{x}}{dt^2} + R_m\frac{d\mathbf{x}}{dt} + s\mathbf{x}$$

$$\omega^2 Lm\,e^{j\omega t} = \left\{R_m + j\left[\omega(M+m) - \frac{s}{\omega}\right]\right\}\frac{d\mathbf{x}}{dt} = \mathbf{Z}_m\mathbf{u}$$

and thus $\quad \underline{|\mathbf{u}| = \omega^2\dfrac{Lm}{Z_m}}$

1.8.1. $f = F\sin\omega_0 t \quad t \geq 0 \quad$ and $\quad \beta \ll \omega_0$

$\mathbf{f} = -jF e^{j\omega_0 t}$

In this approximation $\quad \omega_d = \omega_0\sqrt{1-(\beta/\omega_0)^2} \approx \omega_0 - \frac{1}{2}\beta^2/\omega_0$

The steady state (homogeneous) solution is $\quad \mathbf{u}_h = \mathbf{f}/\mathbf{Z}_m = j\,\omega_0\mathbf{x}_h \quad$ whence

$$\mathbf{x}_h = \frac{1}{j\omega_0}\frac{-jF e^{j\omega_0 t}}{\mathbf{Z}_m} = -\frac{F}{\omega_0\mathbf{Z}_m}e^{j\omega_0 t} \quad \text{But} \quad \mathbf{Z}_m = R_m \quad \text{for} \quad \omega = \omega_0 \quad \text{so}$$

$$x_h = -\frac{F}{\omega_0 R_m}\cos\omega_0 t$$

Substituting into (1.8.1) yields the complete solution

$$x = A\,e^{-\beta t}\cos(\omega_d t + \phi) - \frac{F}{\omega_0 R_m}\cos\omega_0 t \quad \text{with constants to be determined}$$

Applying the initial conditions $\quad x(0) = 0 \quad$ and $\quad \left.\dfrac{dx}{dt}\right|_{t=0} = 0 \quad$ gives

$$A\cos\phi - \frac{F}{\omega_0 R_m} = 0$$

$$-\beta A\cos\phi - \omega_d A\sin\phi = 0$$

From the second equation $\dfrac{\sin\phi}{\cos\phi} = -\dfrac{\beta}{\omega_d}$ but $\dfrac{\beta}{\omega_d} \ll 1$ so that $\phi \approx -\dfrac{\beta}{\omega_d}$

and then the first gives $A \approx \dfrac{F}{\omega_0 R_m \cos(\beta/\omega_d)} \approx \dfrac{F}{\omega_0 R_m}$

Substitution into the complete solution gives

$$x \approx \frac{F}{\omega_0 R_m}\left[e^{-\beta t}\cos(\omega_d t - \beta/\omega_d) - \cos\omega_0 t\right]$$

$$\approx \frac{F}{\omega_0 R_m}\left[e^{-\beta t}\left(\cos\omega_d t + \frac{\beta}{\omega_d}\sin\omega_d t\right) - \cos\omega_0 t\right]$$

Now for $\omega_0 t < 1$ with $\beta/\omega_0 \ll 1$ this becomes

$$\underline{x \approx \frac{F}{\omega_0 R_m}(e^{-\beta t} - 1)\cos\omega_0 t}$$

and for $\omega_0 t \geq 1$ the $\exp(-\beta t)$ makes the first term in the complete solution negligibly small, so the above approximation is valid for all t.

1.8.2. Let $\mathbf{u} = A e^{j(\omega_0 t + \phi)} - \dfrac{F}{\omega m - s/\omega} e^{j\omega t}$

Then $\mathbf{x} = \dfrac{A}{j\omega_0} e^{j(\omega_0 t + \phi)} - \dfrac{1}{j\omega}\dfrac{F}{\omega m - s/\omega} e^{j\omega t}$

(a) Apply the initial conditions on the real parts

$$u(0) = 0 = \mathrm{Re}\left\{A e^{j\phi} - \frac{F}{\omega m - s/\omega}\right\}$$

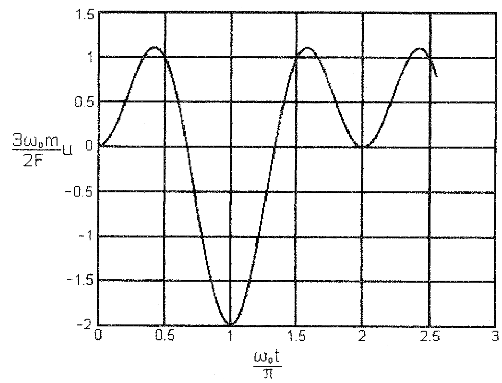
$$x(0) = 0 = \mathrm{Re}\left\{\frac{A}{j\omega_0} e^{j\phi} - \frac{1}{j\omega}\frac{F}{\omega m - s/\omega}\right\}$$

These give $A\cos\phi = \dfrac{F}{\omega m - s/\omega}$ and $A\sin\phi = 0$ so that

$\phi = 0$ and $A = \dfrac{F}{\omega m - s/\omega}$

Thus, $\underline{u = \dfrac{F}{\omega m - s/\omega}(\cos\omega_0 t - \cos\omega t)}$

(b)



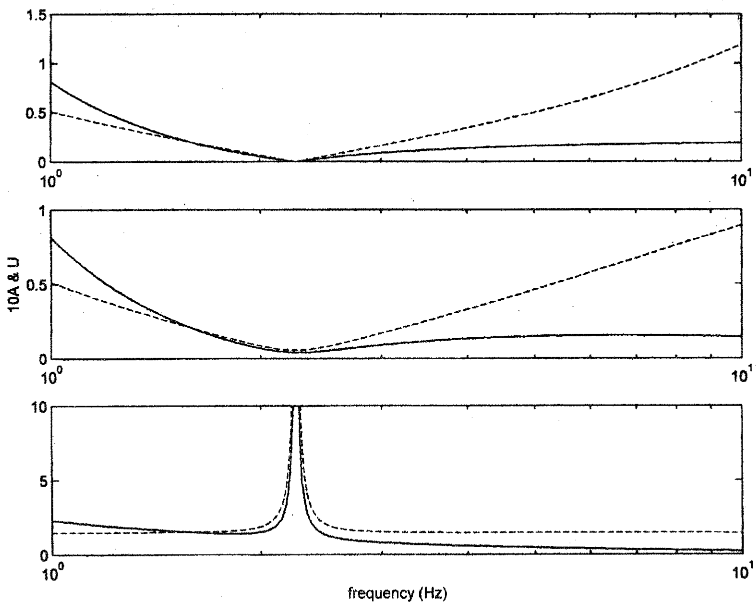
(c) If $\omega \ll \omega_0$ then $\omega m \ll s/\omega$ and for the steady state, $t \gg 1/\beta$ and $u = \dfrac{\omega F}{s}\cos\omega t$

1.10.1. $Z_m^2 = R_m^2 + (\omega m - s/\omega)^2 = (\omega_0 m/Q)^2 + (\omega m - s/\omega)^2$

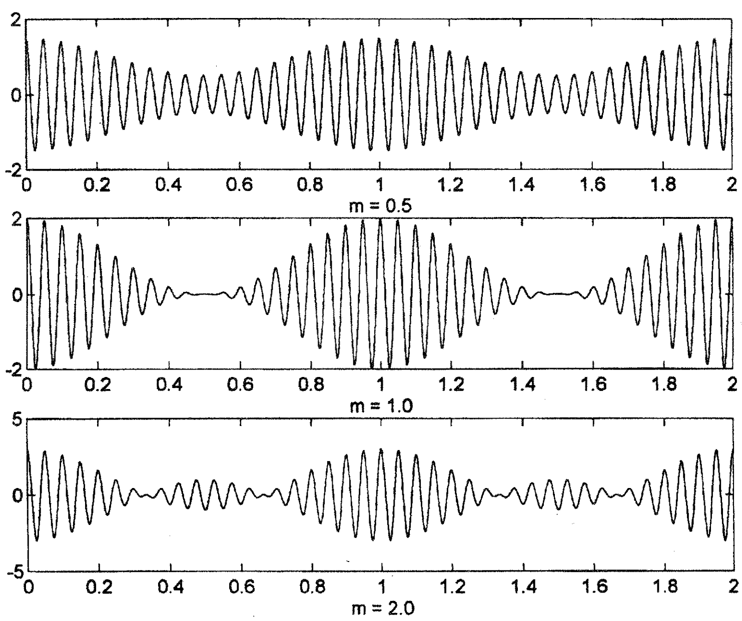
$$= (\omega_0 m)^2\left[\frac{1}{Q^2} + \left(\frac{\omega}{\omega_0} - \frac{s}{m\omega\omega_0}\right)^2\right] = (\omega_0 m)^2\left[\frac{1}{Q^2} + \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)^2\right]$$

1.10.4. At the half-power points $X_m = \pm R_m$ so that $\omega m - s/\omega = \pm R_m$ Solving this pair of equations for ω and using $\beta = R_m/2m$ gives the quadratic $\omega^2 \mp 2\beta\omega - \omega_0^2 = 0$ which has solutions $\omega = \pm\sqrt{\omega_0^2 + \beta^2} \pm \beta$ Since the angular frequency must be positive, $\omega_\pm = \sqrt{\omega_0^2 + \beta^2} \pm \beta$ For $\beta \ll \omega_0$ this simplifies to $\omega_\pm \approx \omega_0 \pm \beta$

1.12.4C.



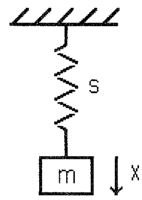
1.13.6C.



1.15.1.

$$\mathbf{U}(t) = \int_{-\infty}^{\infty} \frac{\delta(w-\omega)}{\mathbf{Z}(w)} F e^{jwt}\, dw = \underline{\frac{F}{\mathbf{Z}} e^{j\omega t}}$$

1.15.7.



$\mathrm{f} = mg \cdot 1(t)$
$\mathbf{Z}(w) = j(wm - w/s)$
$\omega_0 = (s/m)^{1/2}$
With the help of Table 1.15.1, we have

$$\mathbf{G}(w) = \frac{mg}{2\pi}\frac{1}{jw}$$

Then, again usingTable 1.15.1

$$\mathbf{U}(t) = \int_{-\infty}^{\infty} \frac{mg}{2\pi}\frac{1}{jw}\frac{e^{jwt}}{j(wm - s/w)}\,dt = \frac{-mg}{2\pi m}\int_{-\infty}^{\infty}\frac{e^{jwt}}{w^2 - \omega_0^2}\,dt$$

$$= g\int_{-\infty}^{\infty}\frac{1}{2\pi}\frac{1}{\omega_0^2 - w^2}e^{jwt}\,dt = \frac{g}{\omega_0}\sin\omega_0 t \cdot 1(t)$$

and integration over time yields the displacement

$$X(t) = \int_0^t U(t)\,dt = \frac{g}{\omega_0}\frac{-1}{\omega_0}\cos\omega_0 t\Big|_0^t = \underline{\frac{mg}{s}\left(1 - \cos\omega_0 t\right)}$$

2.3.1. (a) $\rho_L(x)dx\dfrac{\partial^2 y}{\partial t^2} = T\dfrac{\partial^2 y}{\partial x^2}dx$ gives $\underline{\dfrac{\partial^2 y}{\partial t^2} = c^2(x)\dfrac{\partial^2 y}{\partial x^2}}$ with $c^2(x) = \dfrac{T}{\rho_L(x)}$

(b) $\rho_L dx\dfrac{\partial^2 y}{\partial t^2} = \dfrac{\partial}{\partial x}\left(\rho_L x g\dfrac{\partial y}{\partial x}\right)dx$ gives $\underline{\dfrac{\partial^2 y}{\partial t^2} = g\dfrac{\partial}{\partial x}\left(x\dfrac{\partial y}{\partial x}\right)}$

2.8.1 $y = 4\cos(3t - 2x)$ $\quad\rho_L = 0.1$ g/cm

(a) $A = \underline{4\text{ cm}}$ $\quad c = \omega/k = 3/2 = \underline{1.5\text{ cm}}$ $\quad f = \omega/2\pi = 3/2\pi = \underline{0.48\text{ Hz}}$

$\lambda = 2\pi/k = \pi = \underline{3.1\text{ cm}}$ $\quad k = \underline{2\text{ cm}^{-1}}$

(b) $u = \partial y/\partial t = -12\sin(3t - 2x)$ so $u(0,0) = \underline{0\text{ m/s}}$

2.9.3 Let the midpoint of the string be at $x = 0$

(a) The impedances are in series and ∴ add, so twice the impedance of each,

$$\underline{\mathbf{Z}_{m0} = -j2\rho_L c\cot(kL/2)}$$

(b) $$A = \frac{U}{\omega} = \frac{F}{\omega Z_{m0}} = \frac{F}{2kc^2\rho_L}\tan\frac{kL}{2} = \underline{\frac{F}{2kT}\tan\frac{kL}{2}}$$

(c) Let $y = A'\sin kx$ for $0 \le x \le L/2$ Then we must have

$$A'\sin\frac{kL}{2} = \frac{F}{2kT}\tan\frac{kL}{2} \quad \text{so that} \quad A' = \frac{F}{2kT}\frac{1}{\cos(kL/2)}$$

Evaluating at $x = L/4$ we obtain $y(L/4) = \underline{\dfrac{F}{2kT} \dfrac{\sin(kL/4)}{\cos(kL/2)}}$

2.9.5. (*a*) At $x = 0$ we have $\mathbf{u} = U_0 e^{j\omega t}$

Let $\mathbf{y} = A\sin[k(x-L)]e^{j\omega t}$ and solve for A

$$\mathbf{u}(0,t) = j\omega A\sin[k(0-L)]e^{j\omega t} = U_0 e^{j\omega t} \quad \text{so that} \quad A = \frac{jU_0}{\omega \sin kL}$$

$A \to \infty$ when $kL = n\pi$ which gives $f = \underline{\dfrac{1}{2}\dfrac{nc}{L}}$

(*b*) At $x = 0$ we have $Y_0 e^{j\omega t} = -A\sin kL\, e^{j\omega t}$ so that $A = -\dfrac{Y_0}{\sin kL}$

$A \to \infty$ when $kL = n\pi$ which gives $f = \underline{\dfrac{1}{2}\dfrac{nc}{L}}$

(*c*) $\mathbf{Z} = -j\rho_L c\cot kL$ and $Z = 0$ when $kL = \dfrac{\pi}{2}, 3\dfrac{\pi}{2}, 5\dfrac{\pi}{2}, \ldots$

$$f = \underline{\frac{1}{2}\left(n - \frac{1}{2}\right)\frac{c}{L}}$$

(*d*) Comparison of the above results shows that the answer is <u>no</u>.

2.11.2. $m_L = 0.2$ kg, $m_s = 0.05$ kg, $L = 1.0$ m, so that $\rho_L = 0.05$ kg/m

(*a*) $c^2 = T/\rho_L = 0.2\cdot 9.8/0.05 = 39.2$ so $c = \underline{6.26 \text{ m/s}}$

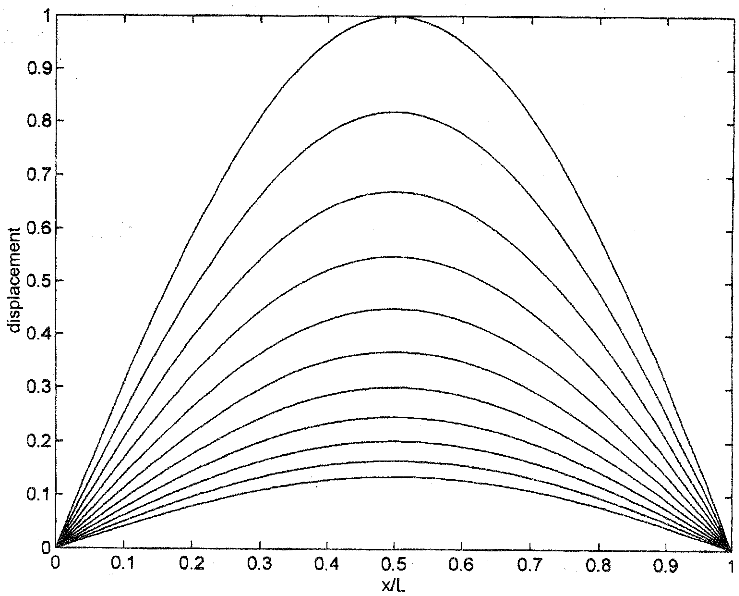
(*b*) $\cot kL = \dfrac{m_L}{m_s}kL = \dfrac{0.2}{0.05\cdot 1}kL = 4kL$

By trial and error, or tables, $kL = 0.48$ so $f = 0.48c/2\pi L = \underline{0.48 \text{ Hz}}$

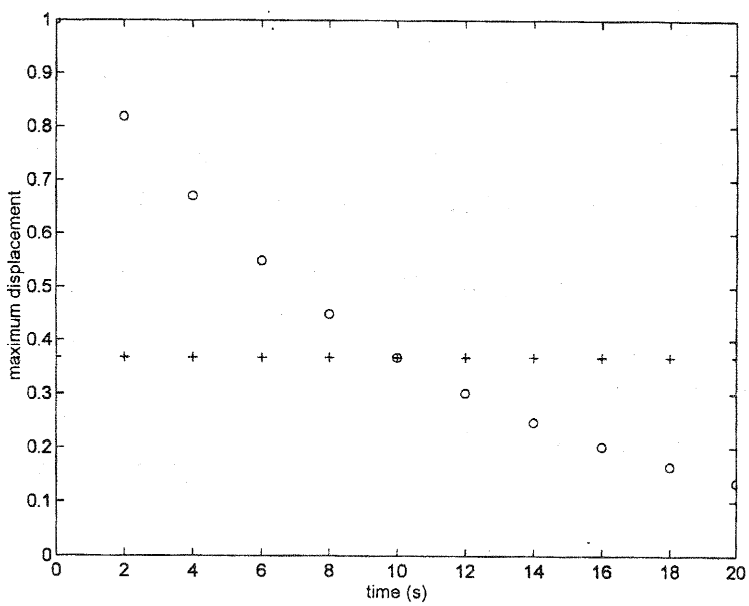
The first overtone has $kL = 3.219$ so $f = \underline{3.21 \text{ Hz}}$

(*c*) $\mathbf{y} = A\sin kL\, e^{j\omega t}$ from which we obtain $\dfrac{A}{|\mathbf{y}(L)|} = \dfrac{1}{|\sin 3.219|} = \underline{12.9}$

2.11.7C (*a*)



(*b*)



2.12.2. $y = 2\sin(x/5)\cos 3t$ with all units in cgs

(*a*) $c = \omega/k = 3/(1/5) = \underline{15\ \text{cm/s}}$

$f = 3/(2\pi) = \underline{0.48\ \text{Hz}}$

$k = 1/5 = \underline{0.2\ \text{cm}^{-1}}$

(*b*) $x = L/2 = 31.4/2 \approx 5\pi$

$$y = 2\sin\pi\cos\omega t = 0$$

amplitude = $\underline{0\ \text{cm}}$ speed = $\underline{0\ \text{cm/s}}$

$x = L/4 = 31.4/4 \approx 5\pi/2$

$$y = 2\cos 3t = 0$$

amplitude = $\underline{2\ \text{cm}}$ speed = $\underline{6\ \text{cm/s}}$

(*c*)

$$\frac{dE}{dx} = \frac{1}{2}\rho_L\left[c^2\left(\frac{\partial y}{\partial x}\right)^2 + \left(\frac{\partial y}{\partial t}\right)^2\right]$$

$$= \frac{1}{2}\rho_L\left(\frac{225\cdot 4}{25}\cos^2\frac{x}{5}\cos^2 3t + 36\sin^2\frac{x}{5}\sin^2 3t\right)$$

$$= 1.8\left(\cos^2\frac{x}{5}\cos^2 3t + \sin^2\frac{x}{5}\sin^2 3t\right)$$

at $x = L/2 = 5\pi$ $\qquad \dfrac{dE}{dx} = \underline{-1.8\cos^2 3t}$

at $x = L/4 = 5\pi/2$ $\qquad \dfrac{dE}{dx} = \underline{1.8\sin^2 3t}$

(*d*) Since the total energy remains constant, we can choose any time for the integration over x. For convenience, choose $3t = \pi/2$.

$$E = 1.8\int_0^L \sin^2\frac{x}{5}\,dx = 1.8\int_0^{10\pi}\sin^2\frac{x}{5}\,dx$$

$$= 1.8\cdot 5\left[\frac{1}{2}\frac{x}{5} - \frac{1}{4}\sin\frac{2x}{5}\right]_0^{10\pi} = 9\pi$$

$$= 28.3\ \text{ergs} = \underline{2.83\times 10^{-6}\ \text{J}}$$

3.5.1. $S = 1\times 10^{-4}\ \text{m}^2$ free at $x = 0$

$L = 0.25\ \text{m}$ loaded with 0.15 kg at L

$m = 0.15\ \text{kg}$

$m_b = 7700\times 10^{-4}\cdot 0.25 = 0.1925\ \text{kg}$

(*a*)

$$\tan kL = -\frac{m}{m_b}kL = -0.78kL \quad \text{so that} \quad kL = 2.12 = \frac{2\pi f}{c}L$$

$$f = 2.12\frac{5050}{2\pi\cdot 0.25} = \underline{6.8\ \text{kHz}}$$

(*b*) $\xi = 2A\cos kx\, e^{j\omega t}$ and clamp the bar where $\cos kx = 0$

$$kx = \frac{\pi}{2} \qquad x = \frac{\pi\cdot 0.25}{2\cdot 2.12} = \underline{0.185\ \text{m}}$$

(c) $\dfrac{\cos 0}{\cos kL} = \dfrac{1}{\cos 2.12} = \underline{1.91}$

(d) $\tan kL = -0.78kL \qquad kL = 4.965 \qquad f = 6800\dfrac{4.965}{2.12} = \underline{15.9\ \text{kHz}}$

3.6.4. Assume the bar is free at $x = 0$ and fixed at $x = L$ so that

$\xi = Ae^{j\omega t}\cos kx$ Then at $x = L$

$$\frac{\mathbf{f}}{\mathbf{u}} = -j\frac{s}{\omega} = -\frac{SY}{j\omega\xi}\frac{\partial\xi}{\partial x}\bigg|_L \quad \text{so that} \quad SY\frac{\partial\xi}{\partial x}\bigg|_L = -s\xi(L)$$

Thus, $-SYAe^{j\omega t}k\sin kL = -sAe^{j\omega t}\cos kL$ and so $\underline{\tan kL = \dfrac{s}{\omega}\dfrac{L}{mc}}$

3.8.2. $\kappa^2 = \dfrac{1}{S}\displaystyle\int r^2 dS$

(a) $\kappa^2 = \dfrac{2}{wt}\displaystyle\int_0^{t/2} r^2 w\,dr = \dfrac{t^2}{12}$ so $\underline{\kappa = \dfrac{t}{\sqrt{12}}}$

(b) Exchange the roles of t and w so $\underline{\kappa = \dfrac{w}{\sqrt{12}}}$

(c) The area of integration is the right triangle with hypotenuse $\sqrt{w^2+t^2}$ and r is measured perpendicularly from the hypotenuse. If h is the height of the triangle (perpendicular to the hypotenuse), then $h = wt/\sqrt{w^2+t^2}$. The length l of the incremental area $dS = l\,dr$ of height dr and distance r from the hypotenuse is $l = (1 - r/h)\sqrt{w^2+t^2}$ and the integral becomes

$$\kappa^2 = \frac{1}{wt}2\int_0^h r^2\left(1-\frac{r}{h}\right)\sqrt{w^2+t^2}\,dr = \frac{\sqrt{w^2+t^2}}{wt}\frac{1}{6}h^3$$

Substitution for h results in $\underline{\kappa = \dfrac{1}{\sqrt{6}}\dfrac{wt}{\sqrt{w^2+t^2}}}$

3.9.2. $\omega c\kappa = [\text{rad/s}][\text{m/s}][\text{m}] = [\text{m}]^2/[\text{s}]^2$ so $\underline{\sqrt{\omega c\kappa}\ \ \text{has the dimensions of speed}}$

$v = \sqrt{\omega c\kappa}$ and if $v = c$ then $\omega = \dfrac{c}{\kappa}$

A circular rod of radius a has $\kappa = a/2$ so that $f = c/\pi a$ and for this aluminum rod $f = \dfrac{5.15\times10^3}{\pi\cdot 0.005} = \underline{328\ \text{kHz}}$

3.11.2. Clamped, clamped bar. $L = 1$ m. κ and c constants

With $g = \omega/v$ and $v = \sqrt{\omega\kappa c}$
The trial solution is $Y(x) = A\cosh gx + B\sinh gx + C\cos gx + D\sin gx$

with boundary contitions, $Y(0) = Y(L) = 0$ and $\left.\frac{dY}{dx}\right|_0 = \left.\frac{dY}{dx}\right|_L = 0$

Application at $x = 0$ gives $A + C = 0$ and $B + D = 0$ so that

$$Y(x) = A(\cosh gx - \cos gx) + B(\sinh gx - \sin gx)$$

Application at $x = L$ then gives

$$A(\cosh gL - \cos gL) = -B(\sinh gL - \sin gL)$$

$$A(\sinh gL + \sin gL) = -B(\cosh gL - \cos gL)$$

Divide one by the other, cross multiply, and simplify, $\cosh gL \cos gL = 1$

Trial and error solutions should be $\approx \frac{(2n+1)}{2}\pi$ for $n = 1, 2, \ldots$

where the $n = 0$ solution is the trivial one of no motion.

$$(gL)_n = 4.7300,\ 7.8532,\ \ldots = 1.506\pi,\ 2.500\pi,\ \ldots$$

so that $(gL)_n \approx \frac{(2n+1)}{2}\pi$ for $n = 1, 2, \ldots$. The frequencies are given by

$$f_n = \frac{1}{2\pi}\frac{\kappa c}{L}(gL)_n^2$$

$$\frac{f_n}{f_1} = \frac{(gL)_n^2}{(gL)_1^2} \approx \left(\frac{2}{3}\right)^2\left(\frac{2n+1}{2}\right)^2 = \left(\frac{2n+1}{3}\right)^2$$

$$\underline{f_n/f_1 = 1,\ 2.76,\ 5.40,\ \ldots}$$

The phase speeds are given by $v = \sqrt{\omega\kappa c}$ so that

$$\underline{v_n/v_1 = \sqrt{f_n/f_1} = 1,\ 1.66,\ 2.32,\ \ldots.}$$

Wwavelengths $\lambda_n = v_n/f_n = 2\pi v_n/\omega_n = 2\pi L/(gL)_n = 4L/(2n+1)$

$$\underline{\lambda_n = 132.8,\ 80.0,\ 57.1,\ \ldots \text{ cm}}$$

Nodal positions

- For the fundamental mode, there is one loop on the bar between the endpoints, so the nodes are at <u>0 and 100 cm</u>.
- For the first overtone, there are two loops so by symmetry the nodal points are at <u>0, 50, and 100 cm</u>.
- For the second overtone, the *true nodes* are found where

$$\frac{d^2Y_3}{dx^2} = 0 = g^2[A(\cosh gx + \cos gx) + B(\sinh gx + \sin gx)]$$

With the first boundary condition at $x = L$ this gives

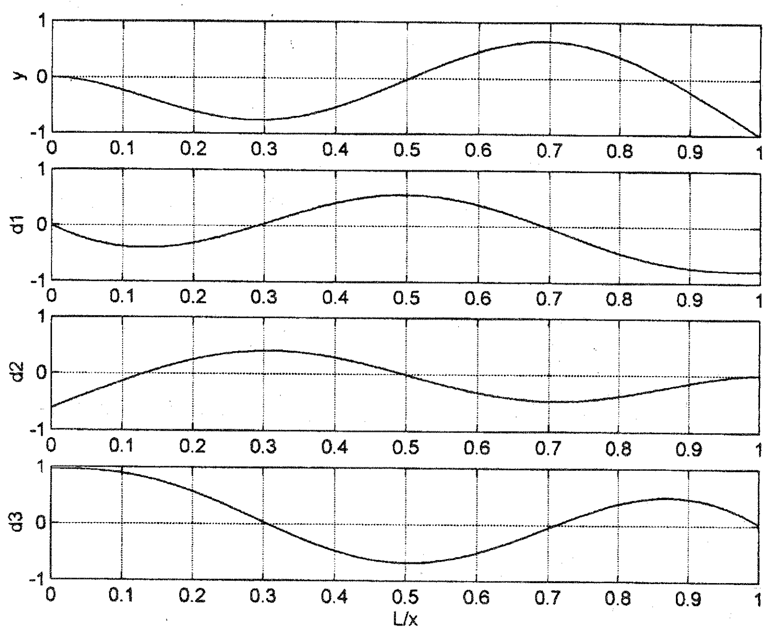
$$(\sinh gL - \sin gL)(\cosh gx + \cos gx) = (\cosh gL - \cos gL)(\sinh gx + \sin gx)$$

and using $gL = 7\pi/2$ results in

$$\frac{\sinh gx + \sin gx}{\cosh gx + \cos gx} \approx 1$$

Soilution by trail and error yields $gx = 3.91, 7.08$ These nodes are found at $(gx/gL) \cdot L = 35.6$ and 64.4 cm. Thus, nodes are at 0, 35.6, 64.4, and 100 cm.

3.11.3C.



4.3.2. Let $L_x = a$ and $L_y = 2a$ Then $f_{nm} = \frac{c}{2}\sqrt{\left(\frac{n}{a}\right)^2 + \left(\frac{m}{2a}\right)^2} = \frac{c}{2a}\sqrt{n^2 + \frac{m^2}{4}}$

$$f_{11} = \frac{c}{2a}\sqrt{1.25}$$

$f_{11}/f_{11} = \sqrt{1.25/1.25} = 1.000$ $\qquad$ $f_{21}/f_{11} = \sqrt{4.25/1.25} = 1.844$

$f_{12}/f_{11} = \sqrt{2/1.25} = 1.265$ $\qquad$ $f_{14}/f_{11} = \sqrt{5/1.25} = 2.000$

$f_{13}/f_{11} = \sqrt{3.25/1.25} = 1.612$ $\qquad$ $f_{22}/f11 = \sqrt{5/1.25} = 2.000$

4.4.2. (*a*) $\mathbf{y} = AJ_m(kr)\cos m\theta\, e^{j\omega t}$ with boundary condition $\left.\frac{\partial y}{\partial r}\right|_a = 0$

This gives $\left.\frac{d}{dr}J_m(k_{mn}r)\right|_a = 0$ so that $k_{mn} = j'_{mn}/a$

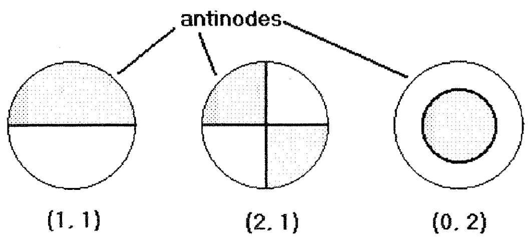
with $j'_{11} = 1.84$ $\quad$ $j'_{21} = 3.05$ $\quad$ $j'_{02} = 3.83$

If we define $c^2 = \mathscr{T}/\rho_S$ then the normal modes are

$$\mathbf{y} = A_{mn} J_m\left(j'_{mn} r/a\right) \cos m\theta \, e^{j\omega_{mn} t}$$

$$\omega_{mn} = j'_{mn} c/a$$

(b)



(c) $f_{11} = k_{11} c/2\pi = 0.293(\mathscr{T}/\rho_S)^{1/2}$

$f_{21} = 0.486(\mathscr{T}/\rho_S)^{1/2}$ $\quad f_{02} = 0.610(\mathscr{T}/\rho_S)^{1/2}$

4.5.1. $\mathscr{T} = 4 \times 10^3$ N/m^2 $\quad a = 0.01$ m $\quad \rho_S = 0.2$ kg/m^2 $\quad A_{01} = 0.01$ cm

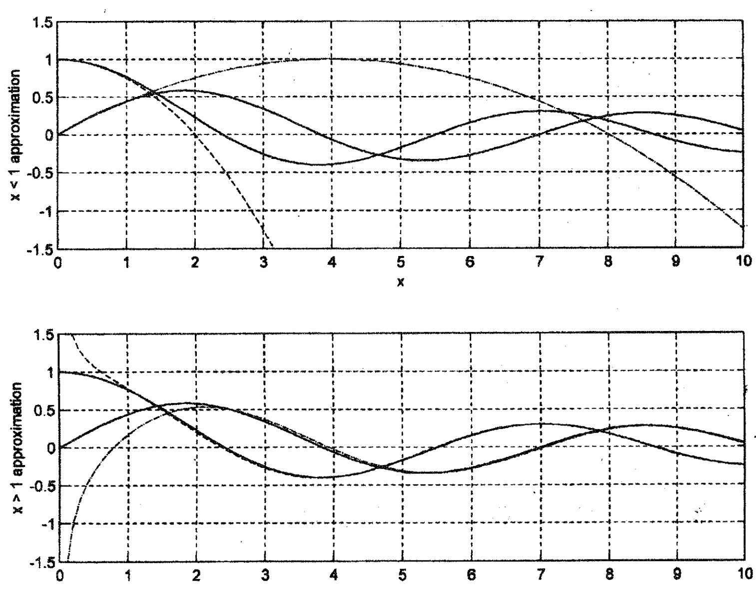
(a) $c = (\mathscr{T}/\rho_S)^{1/2} = \sqrt{4000/0.2} = 141.4$ m/s

$f_1 = 141.4 \cdot 2.405/(2\pi \cdot 0.01) = \underline{5.415 \text{ kHz}}$

(b) $\langle \Psi_1 \rangle_S = 0.432 A_{01}$

$V = \pi a^2 \langle \Psi_1 \rangle_S = \pi \cdot 1.0^2 \cdot 0.01 \cdot 0.432 = \underline{0.0136 \text{ cm}^3}$

4.5.5C.



4.7.1. $a = 0.25$ m $\quad \rho_S = 1$ kg/cm^2 $\quad \mathscr{T} = 1 \times 10^4$ N/m

(*a*) $c = (\mathscr{T}/\rho_S)^{1/2} = \sqrt{10^4 \cdot 1} = 100$ m/s

$f_1 = 2.405/(2\pi \cdot 0.25) = \underline{153 \text{ kHz}}$

(*b*) $V_0 = \frac{2}{3}\pi a^3 = 0.0328$ m^3 $\quad \gamma = 1.4$

$\mathscr{P}_0 = 1 \times 10^5$ N/m^2 $= 1 \times 10^5$ Pa

$B = \pi a^4 \gamma \mathscr{P}_0 / \mathscr{T} V_0 = \pi(0.25)^4 1.4 \cdot 1 \times 10^5/(1 \times 10^4 \cdot 0.0328) = 5.28$

Trial and error solution of (4.7.6) gives $ka \approx 3.05$

$$f_1 = \frac{kc}{2\pi} = \frac{3.05}{0.25}\frac{100}{2\pi} = \underline{194 \text{ Hz}}$$

4.7.3. (*a*) $J_m(k_m a) = 0$

$\underline{j_{m1} = k_m a = 3.83, 5.14, 6.38, 7.59, 8.77}$ for $m = 1, 2, 3, 4, 5$

(*b*) If these are to be brought into concordance with the 2, 3, 4, 5, 6 harmonics, with the 6th harmonic frequency the same as for j_{51}, then the shifts are

$$s = \frac{m+1}{6}\frac{j_{51}}{j_{m1}} - 1 = \underline{-0.283, -0.148, -0.083, -0.037, 0}$$

Not uniform, but regular. Well approximated by $s = 0.21 \ln(n/6)$

4.9.2. $a = 0.015$ m $\quad$ tensile stress $= 2 \times 10^8$ N/m^2 $\quad d = 2 \times 10^{-5}$ m

(*a*) $\mathscr{T} = (2 \times 10^{-5})(2 \times 10^8) = \underline{4 \times 10^3 \text{ N/m}}$

(*b*)

$$f = \frac{2.405}{2\pi \cdot 0.015}\sqrt{\frac{4 \times 10^3}{2700 \cdot 2 \times 10^{-5}}} = \underline{6.945 \text{ kHz}}$$

(*c*) At 500 Hz $\quad ka = 2.405 \cdot 500/6945 = 0.173 \ll 1$

so that $J_0(ka) \approx 1 - \frac{1}{4}(ka)^2$ and $J_0(0.173) = 0.9925$

$$y = \frac{P}{k^2(ScriptT)}\left[\frac{1 - J_0(ka)}{J_0(ka)}\right] \approx \frac{P}{k^2(ScriptT)}\frac{1}{4}\frac{(ka)^2}{J_0(ka)}$$

$$\approx \frac{Pa^2}{4 \cdot (ScriptT) \cdot J_0(ka)} = \frac{2 \cdot 0.015^2}{4 \cdot 4 \times 10^3 \cdot 0.9925} = \underline{2.84 \times 10^{-8} \text{ m}}$$

(*d*)
$$\frac{Pa^2}{8\,(ScriptT)}\left[1 + \frac{1}{6}(ka)^2\right] = \frac{2 \cdot 0.015^2}{8 \cdot 4000}\left(1 + \frac{0.173^2}{6}\right) = \underline{1.41 \times 10^{-8} \text{ m}}$$

4.11.1. $a = 2.0$ cm $\quad d = 0.02$ cm $\quad \rho = 7700$ kg/m^3

$Y = 19.5 \times 10^{10}$ N/m^2 $\quad \sigma = 0.28$

(*a*)

$$f_1 = 0.47\frac{d}{a^2}\sqrt{\frac{Y}{\rho(1-\sigma^2)}} = 0.47\frac{2 \times 10^{-4}}{0.02^2}\sqrt{\frac{19.5 \times 10^{10}}{7700 \cdot (1 - 0.28^2)}} = \underline{1230 \text{ kHz}}$$

(*b*) If $d = 0.04$ cm then $\underline{f_1 = 2460 \text{ Hz}}$

(*c*) If $a = 4$ cm then $\underline{f_1 = 307 \text{ Hz}}$

4.11.2. $a = 2.0$ cm $\quad \rho_S = 7700 \cdot 2 \times 10^{-4} = 1.54 \text{ kg/m}^2$

$$f_1 = \frac{2.405}{2\pi a}\sqrt{\frac{(ScriptT)}{\rho_S}} \quad \text{so}$$

$$(ScriptT) = \left(\frac{2\pi \cdot 0.02 \cdot 1230}{2.405}\right)^2 1.54 = \underline{6.36 \times 10^3 \text{ N/m}}$$

5.2.1. (*a*) $\mathscr{P}/\mathscr{P}_0 = (\rho/\rho_0)^\gamma = (1+s)^\gamma \approx 1 + \gamma s \quad \therefore\ p = \mathscr{P} - \mathscr{P}_0 = \mathscr{P}_0 \gamma s$

Then from (5.2.6) $\quad \|\ \mathscr{B} \approx \gamma \mathscr{P}_0$

(*b*) $\mathscr{B} = \gamma \mathscr{P}_0 = \gamma \rho_0 r T_K$ so $\mathscr{B} \ \underline{\propto T_K}$

5.4.2. Expanding (5.3.3) gives $\dfrac{\partial \rho}{\partial t} + \rho \nabla \cdot \vec{u} + \vec{u} \cdot \nabla \rho = 0$

(*a*) For constant density, the first and last terms on the LHS of the above equation vanish (or use the relation given in the problem statement) so that the middle term must be zero, hence $\underline{\nabla \cdot \vec{u} = 0}$

(*b*) For constant density $s = 0$ and (5.4.9) becomes

$$\underline{-\nabla(ScriptP) = \rho \frac{\partial \vec{u}}{\partial t} + (\vec{u} \cdot \nabla)\vec{u}}$$

(*c*) From (5.2.6) p can change but $s = 0$ Therefore $\underline{\mathscr{B} \to \infty}$

For motion in one dimension, (*a*) gives $\vec{u} = \text{constant}$ and $\underline{c \to \infty}$

5.5.2. (*a*) For the stated assumptions

(5.2.3), (5.2.6) and (5.5.5) give $\quad p = \rho_0 c^2 s$

Use this to eliminate s from (5.3.4) to get $\quad \dfrac{1}{c^2} \nabla \dfrac{\partial p}{\partial t} + \rho_0 \nabla \cdot \vec{u} = 0$

Take the gradient of both sides $\quad \nabla \dfrac{\partial p}{\partial t} = -\rho_0 c^2 \nabla(\nabla \cdot \vec{u})$

But taking $\partial/\partial t$ of (5.4.10) gives $\quad \nabla \dfrac{\partial p}{\partial t} = -\rho_0 \dfrac{\partial^2 \vec{u}}{\partial t^2}$

and then elimination of the pressure term between the above two results in

$$\underline{\nabla(\nabla \cdot \vec{u}) = \frac{1}{c^2}\frac{\partial^2 \vec{u}}{\partial t^2}}$$

(*b*) From Appendix A7 $\quad \nabla^2 \vec{u} = \nabla(\nabla \cdot \vec{u}) - \nabla \times \nabla \times \vec{u}$ and if $\nabla \times \vec{u} = 0$ then

$$\underline{\frac{1}{c^2}\frac{\partial^2 \vec{u}}{\partial t^2} = \nabla^2 \vec{u}}$$

(c) Spherical coordinates with spherical symmetry $\vec{u} = u(r,t)\hat{r}$ and $p = p(r,t)$
With the help of Appendix A7

$$\nabla \cdot \vec{u} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 u) = \frac{1}{r^2}\left[2ru + r^2\frac{\partial u}{\partial r}\right] = \frac{2}{r}u + \frac{\partial u}{\partial r}$$

$$\nabla(\nabla \cdot \vec{u}) = \hat{r}\frac{\partial}{\partial r}\left[\frac{2}{r}u + \frac{\partial u}{\partial r}\right] = \hat{r}\left[-\frac{2}{r^2}u + \frac{2}{r}\frac{\partial u}{\partial r} + \frac{\partial^2 u}{\partial r^2}\right]$$

$$\underline{\frac{\partial^2 u}{\partial r^2} + \frac{2}{r}\frac{\partial u}{\partial r} - \frac{2}{r^2}u = \frac{1}{c^2}\frac{\partial^2 u}{\partial t^2}}$$

$$\nabla^2 p = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial p}{\partial r}\right) = \frac{1}{r^2}\left(2r\frac{\partial p}{\partial r} + r^2\frac{\partial^2 p}{\partial r^2}\right) = \frac{2}{r}\frac{\partial p}{\partial r} + \frac{\partial^2 p}{\partial r^2}$$

$$\underline{\frac{\partial^2 p}{\partial r^2} + \frac{2}{r}\frac{\partial p}{\partial r} = \frac{1}{c^2}\frac{\partial^2 p}{\partial t^2}}$$

5.6.2. $T = 30°\text{C} \qquad t = 0.3 \qquad \mathscr{P}_G = 0$

(a) $c = 1402.7 + 488\cdot(0.3) - 482\cdot(0.3)^2 + 135\cdot(0.3)^3 = \underline{1509.4\ \text{m/s}}$

(b) $$\frac{dc}{dT} = \frac{dc}{dt}\frac{dt}{dT} = (488 - 2\cdot 482t + 3\cdot 135t^2)\frac{1}{100}$$

$$\left.\frac{dc}{dT}\right|_{T=30°} = 4.88 - 9.64\cdot(0.3) + 4.05\cdot(0.3)^2 = \underline{2.35\ \text{m/(s}\cdot\text{C°)}}$$

5.7.3. $\mathbf{p} = Pe^{j(\omega t - kx)}$ Using $\mathbf{p}/\mathbf{u} = \rho_0 c$

$\mathbf{u} = \dfrac{P}{\rho_0 c}e^{j(\omega - kx)}$ so $U_0 = \dfrac{P}{\rho_0 c}$

(a) $$\underline{\frac{U_0}{c} = \frac{P}{\rho_0 c^2}}$$

(b) $$|\mathbf{s}| = \frac{|\mathbf{p}|}{\rho_0 c^2} = \frac{P}{\rho_0 c^2}$$

$$\underline{\frac{U_0}{c} = |\mathbf{s}|}$$

5.9.2. (a) For the acoustic process $\mathscr{P}/\mathscr{P}_0 = 1 + p/\mathscr{P}_0 = (\rho/\rho_0)^\gamma$
But in general $\mathscr{P} = \rho r T_K$ so $\rho/\rho_0 = (\mathscr{P}/\mathscr{P}_0)(T_0/T)$ [the subscript K has been dropped for convenience]. Combining to eliminate the density ratio,

defining the temperature change $\Delta T = T - T_0$ and expanding for $\Delta T \ll T_0$ gives

$$1 + \frac{p}{(ScriptP)_0} = \left(\frac{T}{T_0}\right)^{\gamma/(\gamma-1)} \approx 1 + \frac{\gamma}{\gamma-1}\frac{\Delta T}{T_0}$$

$$\underline{\Delta T = \frac{\gamma-1}{\gamma} p \frac{T_0}{(ScriptP)_0}}$$

(*b*)

$$I = 10\ \text{W/m}^2 = \frac{P^2}{2\rho_0 c} \quad \text{so} \quad P = \sqrt{2\cdot 415\cdot 10} = 91.1\ \text{Pa}$$

$$\Delta T = \frac{1.4-1}{1.4} 91.1 \frac{293.15}{1.01325\times 10^5} = 0.075\ \text{K} = \underline{0.075\ \text{C}^\circ}$$

5.11.3. $\mathbf{p} = \frac{A}{r}\cos kr\, e^{j\omega t}$ and so $\mathbf{\Phi} = -\frac{A}{j\omega\rho_0 r}\cos kr\, e^{j\omega t}$

(*a*)

$$\mathbf{u} = \nabla\mathbf{\Phi} = \frac{\partial \mathbf{\Phi}}{\partial r} = \underline{-j\frac{A}{\rho_0 c r}\left[\sin kr + \frac{\cos kr}{kr}\right] e^{j\omega t}}$$

(*b*)

$$\mathbf{z} = \frac{\mathbf{p}}{\mathbf{u}} = \underline{j\rho_0 c \frac{\cos kr}{\left[\sin kr + \frac{\cos kr}{kr}\right]}}$$

(*c*)

$$I(t) = pu = \frac{A}{r}\cos kr \cos\omega t \frac{A}{r}\frac{1}{\rho_0 c}\left[\sin kr + \frac{\cos kr}{kr}\right]\sin\omega t$$

$$= \underline{\left(\frac{A}{r}\right)^2 \frac{1}{\rho_0 c}\cos kr\left[\sin kr + \frac{\cos kr}{kr}\right]\cos\omega t \sin\omega t}$$

(*d*) Because $\frac{1}{T}\int_0^T \cos\omega t \sin\omega t\, dt = 0$ we have $\langle I(t)\rangle_T = \underline{I = 0}$

5.12.3. $P = 2$ Pa $\quad f = 100$ Hz $\quad$ in air

(*a*) $I = \frac{P^2}{2\rho_0 c} = \frac{2^2}{2\cdot 415} = \underline{4.8\times 10^{-3}\ \text{W/m}^2}$

$IL = 10\log(4.8\times 10^{-3}/10^{-12}) = \underline{96.8\ \text{dB}\ re\ 10^{-12}\ \text{W/m}^2}$

(*b*) $U/\omega = P/(\rho_0 c\omega) = 2/(415\cdot 2\pi\cdot 100) = \underline{7.7\times 10^{-6}\ \text{m}}$

(*c*) $U = P/(\rho_0 c) = 2/415 = \underline{4.82\times 10^{-3}\ \text{m/s}}$

(*d*) $P_e = P/\sqrt{2} = \underline{1.41\ \text{Pa}} = 14.1\ \mu\text{bar}$

(*e*) $SPL = 20\log(14.1/0.0002) = \underline{97\ \text{dB}\ re\ 20\ \mu\text{bar}}$

5.12.7. (*a*) From (5.2.1) $\rho_0 \propto 1/T_K$ for constant $\mathscr{P}$ and from (5.6.5)

$c \propto (T_K)^{1/2}$ so $\underline{\rho_0 c \propto 1/\sqrt{T_K}}$

(*b*) $\rho_0 c(0°\text{C}) = 415/\sqrt{293/273} = \underline{430 \text{ Pa·s/m}}$

$\rho_0 c(80°\text{C}) = 415/\sqrt{293/353} = \underline{378 \text{ Pa·s/m}}$

(*c*)
$$\frac{I(80°\text{C}) - I(0°\text{C})}{I(0°\text{C})} = \frac{P^2}{\rho_0 c(80°\text{C})}\frac{\rho_0 c(0°\text{C})}{P^2} - 1 = \frac{430}{378} - 1 = 0.138 = \underline{13.8\%}$$

(*d*) $\Delta IL = 10 \log 1.138 = \underline{0.56 \text{ dB}}$ $\qquad \Delta SPL = \underline{0 \text{ dB}}$

5.12.11. (*a*) $\mathcal{ML} = -80$ dB *re* 1 V/μbar $= 20 \log[\mathcal{M}/(1 \text{ V}/\mu\text{bar})]$

$\mathcal{M} = 10^{-4}$ V/μbar $= 10^{-9}$ V/μPa

$\mathcal{ML} = 20 \log[10^{-9}/(1 \text{ V}/\mu\text{Pa})] = \underline{-180 \text{ dB } re \text{ 1 V}/\mu\text{Pa}}$

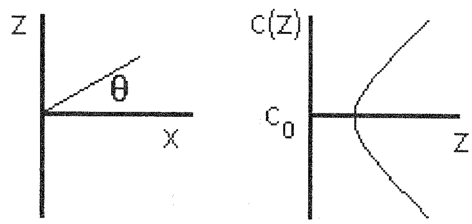
(*b*) $SPL = 80$ dB *re* 1 μbar $= 20 \log(P/1\ \mu\text{bar})$

$P = 10^4\ \mu$bar

$\mathcal{ML} = -80 = 20 \log[(V/10^4)/(1 \text{ V}/\mu\text{bar})] = 20 \log(V/10^4)$

$V = \underline{1 \text{ V}}$

5.14.2.



$$c(z) = \frac{c_0}{\sqrt{1 - (\epsilon z)^2}}$$

$$\frac{\cos\theta_0}{c_0} = \frac{\cos\theta}{c(z)}$$

(*a*)
$$\frac{dz}{dx} = \tan\theta = \frac{\sqrt{1 - \cos^2\theta}}{\cos\theta} = \sqrt{\frac{1 - (\epsilon z)^2}{\cos^2\theta_0} - 1} = \frac{1}{\cos\theta_0}\sqrt{\sin^2\theta_0 - (\epsilon z)^2}$$

$$x = \frac{\cos\theta_0}{\epsilon}\int_0^{\epsilon z} \frac{du}{\sqrt{\sin^2\theta_0 - u^2}} \quad \text{where} \quad u = \epsilon z$$

$$x = \frac{\cos\theta_0}{\epsilon}\sin^{-1}\left(\frac{\epsilon z}{\sin\theta_0}\right) \quad \text{and invert}$$

$$\underline{z = \frac{\sin\theta_0}{\epsilon}\sin\left(\frac{\epsilon x}{\cos\theta_0}\right)}$$

(*b*) Rays cross the channel axis $(z = 0)$ for values of x_n such that

$$\frac{\epsilon x_n}{\cos\theta_0} = 2\pi n \qquad \text{where} \qquad n = 0,1,2,3,\ldots.$$

The change in range is $\Delta x_n = x_{n+1} - x_n$ and the time of flight Δt_n for a ray between adjacent axis crossings is found from (5.14.19) to be

$$\Delta t_n = \int_{x_n}^{x_{n+1}} \frac{dx}{c\cos\theta} = \int_{x_n}^{x_{n+1}} \frac{c_0}{c^2\cos\theta_0}\,dx$$

$$= \frac{1}{c^2\cos\theta_0}\int_0^{\Delta x_n}\left[1 - \sin^2\theta_0 \sin^2\left(\frac{\epsilon x}{\cos\theta_0}\right)\right]dx$$

$$= \frac{\Delta x_n}{c_0\cos\theta_0}\left(1 - \frac{1}{2}\sin^2\theta_0\right)$$

The average speed of sound between axis crossings is thus

$$\langle c(\theta_0)\rangle = \frac{\Delta x_n}{\Delta t_n} = \frac{c_0\cos\theta_0}{\underline{1 - \frac{1}{2}\sin^2\theta_0}} \approx \underline{c_0\left(1 - \tfrac{1}{8}\theta_0^4 + \cdots\right)} \quad \text{for small } \theta_0$$

A greater value of ε causes the speed of sound to increase more rapidly as a function of height above and below the channel axis so that for a ray with initial elevation or depression angle θ_0, both the maximum excursion height and distance between exis crossings decrease; it is less an explanation than an observation that for this specific quasi-parabolic profile the decrease in total travel distance along the ray is exactly compensated for by the decrease in travel time along the ray, regardless of how "steep" the profile is.

(*c*)

$$c(z) = \frac{c_0}{\sqrt{1-(\epsilon z)^2}} \approx \frac{c_0}{1 - \frac{1}{2}(\epsilon z)^2} \approx \underline{c_0\left[1 + \tfrac{1}{2}(\epsilon z)^2\right]}$$

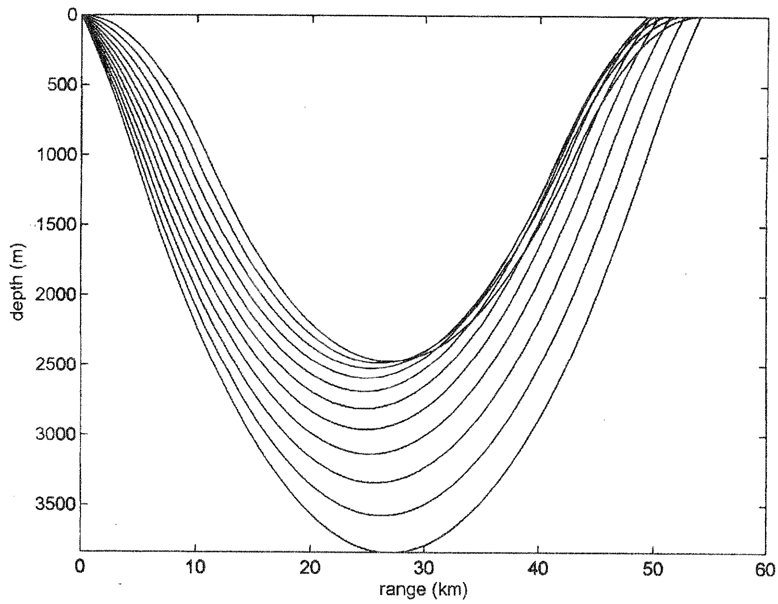
At $\theta_0 = \pi/8$ we have $(\varepsilon z)_{\max} = 0.382$ and c(exact)/c(approx) $= 1.0085$ so that the values agree within about 1%.

(*d*), (*e*)

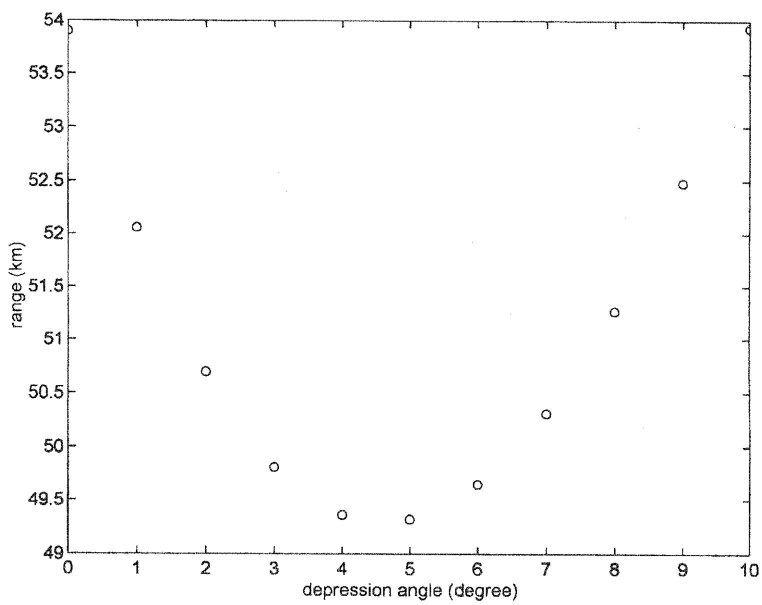
θ_0 (°)	$\langle c(\theta_0)\rangle$ (m/s)	z_{max} (m)	Δx (km)
0	1475.0 m/s	0	--
1	1475.0	116	22.4
2	1475.0	233	22.4
5	1475.0	581	22.4
10	1474.8	1158	22.1

(*f*) The higher local sound speeds encountered by the steeper rays are almost exactly balanced by the longer ray paths so that the times of flight remain essentially the same.

5.14.11C. (*a*)



(*b*)



6.2.2. A reduction in level of 20 dB corresponds to a factor of 10.

$$\frac{1 - r_1/r_2}{1 + r_1/r_2} = \pm\frac{1}{10} \qquad \text{with} \qquad r_1 = 1.54\times10^6 \text{ Pa·s/m} \quad \text{(seawater)}$$

Solving for r_2 gives two solutions,

$$r_2 = \underline{1.88\times10^6 \text{ Pa·s/m}} \quad \text{and} \quad \underline{1.26\times10^6 \text{ Pa·s/m}}$$

6.3.2. $r_1 = 1.48 \times 10^6$ Pa·s/m (water) $r_3 = 4.7 \times 10^7$ Pa·s/m (steel)
$\rho_2 = 1500$ kg/m^3 (plastic)

(a) From the discussion following (6.3.16) $r_2 = \sqrt{r_1 r_3}$ gives $T_I = 1$ if

$k_2 L_2 = \dfrac{2n-1}{2}\pi$ For $n = 1$ this gives $k_2 L_2 = \pi/2$ and $L_2 = \lambda_2/4$

Thus,

$$r_2 = \sqrt{1.48 \cdot 47} \times 10^6 = 8.35 \times 10^6 \text{ Pa·s/m}$$

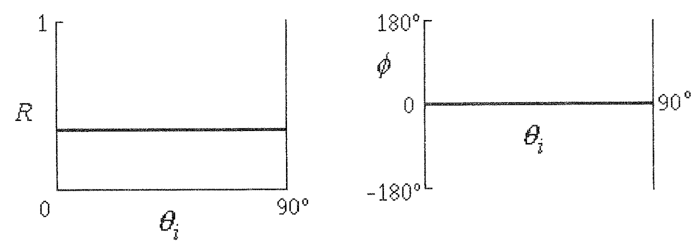
$$c_2 = r_2/\rho_2 = 8.35 \times 10^6/1500 = \underline{5550 \text{ m/s}}$$

$$L_2 = \frac{\lambda_2}{4} = \frac{1}{4}\frac{5550}{20000} = \underline{0.0694 \text{ m}}$$

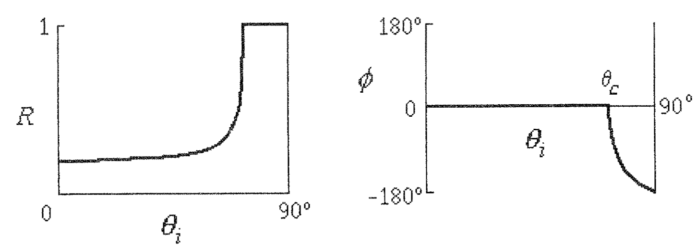
(b) $$R_I = \left(\frac{r_2 - r_1}{r_2 + r_1}\right)^2 = \left(\frac{8.35 - 1.48}{8.35 + 1.48}\right)^2 = \underline{0.49}$$

6.4.1. Write $\mathbf{R} = Re^{j\phi}$

(a)



(b)



6.4.3. From the appendix, values at 0°C $r_1 = 429$ Pa·s/m $c_1 = 331.5$ m/s

$r_2 = 114$ Pa·s/m $c_2 = 1269.5$ m/s

Assume that the membrane is acoustically transparent (*a*) $$\frac{\sin\theta_i}{\sin\theta_r} = \frac{c_1}{c_2} = \frac{331.5}{1269.5} = 0.261 = \frac{\sin\theta_i}{\sin(\theta_i + 40^\circ)}$$

Trial and error gives $\theta_i = \underline{11.9^\circ}$

(*b*) $T_{II} = 1 - R_{II} = 1 - |\mathbf{R}|^2$

$$\mathbf{R} = \frac{r_2/r_1 - \cos\theta_t/\cos\theta_i}{r_2/r_1 + \cos\theta_t/\cos\theta_i} = \frac{0.266 - 0.631}{0.266 + 0.631} = -0.407$$

$$T_{II} = 1 - 0.407^2 = \underline{0.83}$$

6.6.1. $\mathbf{z}_n = 900 - j1200$ Pa·s/m

$r_1 = 415$ Pa·s/m

Use (6.6.5) to calculate R_{II}

(*a*)

$$R_{II} = |\mathbf{R}|^2 = \frac{(r_n\cos\theta_i - r_1)^2 + (x_n\cos\theta_i)^2}{(r_n\cos\theta_i + r_1)^2 + (x_n\cos\theta_i)^2}$$

Plug in numbers, divide by 415, and define $y = \cos\theta_i$

$$R_{II} = \frac{(2.17y - 1)^2 + (2.9y)^2}{(2.17y + 1)^2 + (2.9y)^2} = \frac{y^2 - 0.33y + 0.076}{y^2 + 0.33y + 0.076}$$

Setting $dR_{II}/dy = 0$ and solving (or trial and error) yields $y = 0.29$ and thus $\theta_i = \underline{73^\circ}$

(*b*)

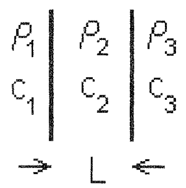
$$R_{II} = \frac{(2.17\cos 80^\circ - 1)^2 + (2.9\cos 80^\circ)^2}{(2.17\cos 80^\circ + 1)^2 + (2.9\cos 80^\circ)^2} = \frac{(0.376 - 1)^2 + 0.503^2}{(0.376 + 1)^2 + 0.503^2}$$

$$= \frac{0.642}{2.148} = \underline{0.30}$$

(*c*)

$$R_{II} = \frac{1.17^2 + 2.9^2}{3.17^2 + 2.9^2} = \frac{9.75}{18.4} = \underline{0.53}$$

6.7.1. (*a*)



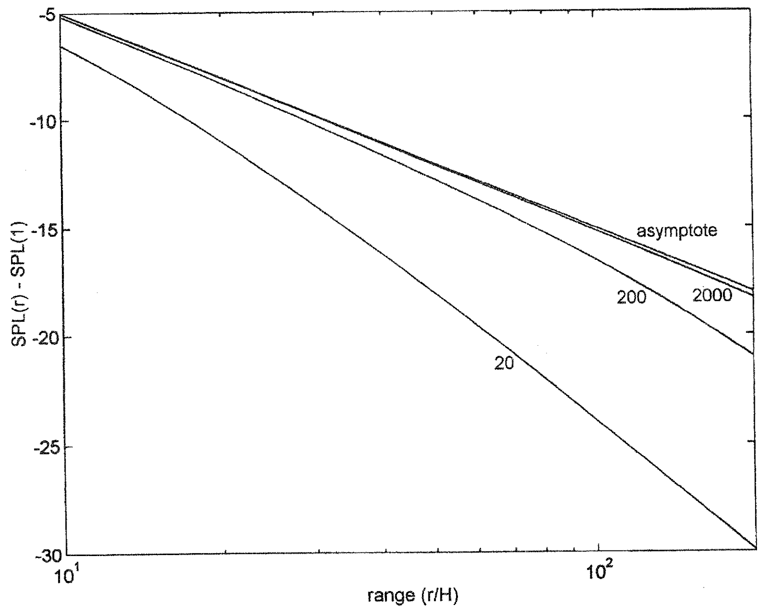
$$\underline{\frac{\omega\rho_S}{r_1} = \frac{k_2 c_2 \rho_2 L}{r_1} = \frac{r_2}{r_1}k_2 L}$$

(*b*) At normal incidence $T_I = T_{II}$ Comparing (6.3.10) and (6.7.4) at normal incidence ($\theta = 0$) shows they are the same if

$$\underline{k_2 L \approx n\pi \quad \text{for} \quad n = 1, 2, 3, \ldots}$$

(*c*) Approximation for (6.3.12) is $k_2 L \ll 1$ Consistent with (*b*) for $n = 0$.

6.8.7C.



7.1.1. $a = 0.1$ m

$I = 50\times10^{-3}$ W/m^2 at $r = 0.1$ m

$f = 100$ Hz so $\omega = 200\pi$ rad/s

$\rho_0 c = 415$ Pa·s/m

$c = 343$ m/s

(*a*) $\Pi = 4\pi r^2 I = 4\pi\cdot1^2\cdot50\times10^{-3} = \underline{0.628 \text{ W}}$

(*b*) $k = \omega/c = 200\pi/343 = 1.83$ m^{-1} and from $\cot\theta = ka$ we get $\theta = 1.390$ rad

$$I = \frac{\Pi}{4\pi r^2} = \frac{0.628}{4\pi\cdot0.1^2} = \underline{5 \text{ W/m}^2}$$

$I = P^2/2\rho_0 c$ so that $P = \sqrt{2\cdot415\cdot5} = \underline{64.4 \text{ Pa}}$

$$U = \frac{P}{z} = \frac{P}{\rho_0 c\cos\theta} = \frac{64.4}{415\cos(1.390)} = \underline{0.863 \text{ m/s}}$$

$\Xi = U/\omega = \underline{1.37\times10^{-3} \text{ m}}$ $\qquad \Xi/a = \underline{0.0137}$

$|s| = P/\rho_0 c^2 = \underline{4.52\times10^{-4}}$ $\qquad U/c = \underline{2.52\times10^{-3}}$

(*c*) $I(0.5) = I(0.1)/5^2 = \underline{0.216 \text{ W/m}^2}$ $\qquad kr = 0.915$ so that $\theta = 0.830$ rad

$P(0.5) = P(0.1)/5 = \underline{13.4 \text{ Pa}}$ $\qquad U = \dfrac{13.4}{415\cos(0.830)} = \underline{0.0479 \text{ m/s}}$

$\Xi = 0.0479/200\pi = \underline{7.62\times10^{-5} \text{ m}}$ $\qquad \Xi/r = \underline{1.52\times10^{-4}}$

$|s| = 13.4/(415\cdot343) = \underline{0.941\times10^{-4}}$ $\qquad U/c = 0.0479/343 = \underline{1.40\times10^{-4}}$

7.1.4. $f = 400$ Hz $\quad \Pi = 10$ mW $\quad r = 0.5$ m
$c = 343$ m/s $\quad \rho_0 c = 415$ Pa·s/m

(a) $I = \Pi/4\pi r^2 = 0.01/(4\pi \cdot 0.5^2) = \underline{3.18 \text{ mW/m}^2}$

(b) $P = \sqrt{2\rho_0 c I} = \sqrt{2 \cdot 415 \cdot 0.00318} = \underline{1.63 \text{ Pa}}$

(c) $kr = \dfrac{2\pi f}{c} r = \dfrac{2\pi \cdot 400}{343} 0.5 = 3.66$ so $\theta = \tan^{-1}\left(\dfrac{1}{3.66}\right) = 15.3°$

$$U = \frac{P}{\rho_0 c \cos\theta} = \frac{1.63}{415 \cdot 0.964} = \underline{0.0041 \text{ m/s}}$$

(d) $|\xi| = U/\omega = 0.0041/(2\pi \cdot 400) = \underline{1.63 \times 10^{-6} \text{ m}}$

(e) $|s| = P/(\rho_0 c^2) = 1.63/(415 \cdot 343) = \underline{1.15 \times 10^{-5}}$

7.4.1. (a) Let $k = \dfrac{\omega}{c}$ Then $ka \sin\theta_1 = 3.83$ so that $\underline{\theta_1 = \sin^{-1}\left(\dfrac{3.83}{ka}\right)}$

(b) From (7.4.8) $\frac{1}{2} kr\left[\sqrt{1 + (a/r)^2} - 1\right] = \pi$ Rearrange and square both sides

to get $1 + \left(\dfrac{a}{r}\right)^2 = 1 + 2\left(\dfrac{2\pi}{kr}\right) + \left(\dfrac{2\pi}{kr}\right)^2$ Now substitute $y = \dfrac{r}{a}$ and get

$\dfrac{1}{y^2} = 2\left(\dfrac{2\pi}{ka}\right)\dfrac{1}{y} + \left(\dfrac{2\pi}{ka}\right)^2 \dfrac{1}{y^2}$ Multiply by y^2 and then solve for y

$$\underline{\frac{r}{a} = \frac{1}{2}\left(\frac{ka}{2\pi} - \frac{2\pi}{ka}\right)}$$

(c) From the above for $\theta_1 \ll 1$ we must have $ka \gg 1$
but this makes $r/a \gg 1$ Thus, impossible.

7.4.3. $f = 10$ kHz $\quad a = 0.5$ m $\quad \Pi = 5000$ W $\quad c = 1500$ m/s

$ka = \dfrac{2\pi f}{c} a = 2\pi \dfrac{10000}{1500} 0.5 = 21$

10 dB down means $[2J_1(x)/x]^2 = 0.1$ From Appendix A6

$x = ka \sin\theta \approx 2.7$ so $\sin\theta = 2.7/21 = 0.129$ and $\underline{2\theta = 14.8°}$

7.4.4. From A4 on Bessel functions, the simple trigonometric approximations show that the zeros of $J_1(x)$ occur when $x \approx (m + 1/4)\pi$ This yields

$$\underline{ka \sin\theta'_m \approx \left(m + \frac{1}{4}\right)\pi}$$

For $m = 1$ the error is $\dfrac{\sin\theta'_1}{\sin\theta_1} = \dfrac{5\pi/4}{3.83} = 1.025$ or $\underline{2.5\% \text{ high}}$

7.6.6. $a = 0.15$ m $\quad f = 330$ Hz $\quad \Pi = 0.5$ W $\quad m = 0.015$ kg
$s = 2000$ N/m $\quad R_m \approx 0 \quad c = 343$ m/s $\quad \rho_0 c = 415$ Pa·s/m

(*a*)
$$ka = \frac{2\pi f}{c}a = \frac{2\pi \cdot 330}{343}0.15 = 0.907$$
$$R_r = \rho_0 c \pi a^2 R_1(2ka) = 415\pi \cdot 0.15^2 R_1(1.81) = 29.33 \cdot 0.357 = 10.5 \text{ N·s/m}$$
$$\Pi = \tfrac{1}{2} R_r U_0^2 \qquad U_0 = \sqrt{2 \cdot 0.5/10.5} = \underline{0.31 \text{ m/s}}$$

(*b*) $\omega = 2\pi f = 2073$ rad/s
$$X_m = (\omega m - s/\omega) = (2073 \cdot 0.015 - 2000/2073) = 30.13 \text{ N·s/m}$$
$$X_r = \rho_0 c \pi a^2 X_1(2ka) = 29.33 \cdot X_1(1.81) = 29.33 \cdot 0.615 = 18.04 \text{ N·s/m}$$
$$F/U_0 = |\mathbf{Z}_r + \mathbf{F}_m| = |10.5 + j(30.13 + 18.04)| = \sqrt{10.5^2 + 48.2^2} = 49.3 \text{ N·s/m}$$
$$F = 49.3 \cdot 0.31 = \underline{15.3\ N}$$

7.6.7. Water $\quad a = 0.1$ m $\quad f_1 = 15$ kHz $\quad f_2 = 3.5$ kHz
$$k_1 a = \frac{2\pi \cdot 15 \times 10^3}{1500}\frac{1}{10} = 6.28$$
$$k_2 a = (3.5/15)\,6.28 = 1.47$$
$$D = \frac{(ka)^2}{1 - 2J_1(2ka)/(2ka)}$$
$$D_1 = \frac{3.94}{1 + 0.027} = 38.4 \quad \text{and} \quad D_2 = \frac{2.16}{1 - 0.26} = 2.9$$
$$\Pi = \frac{4\pi}{D}\frac{P_e^2(1)}{\rho_0 c} \qquad \therefore \ \Pi_1 D_1 = \Pi_2 D_2$$
$$\Pi_2/\Pi_1 = 38.4/2.9 = \underline{13.2}$$

7.8.3. Solving for (*b*) first and then obtaining (*a*) as a special case is convenient.

(*b*) From (7.8.16),

major lobes occur when $\frac{1}{2}kd\left(\sin\theta - \sin\theta_0\right) = 0, \pm\pi, \pm 2\pi, \ldots$

nodal surfaces occur when $\frac{N}{2}kd\left(\sin\theta - \sin\theta_0\right) = \pm m\pi \qquad 1 \le m \le N-1$

Thus, $\left|\frac{1}{2}kd\left(\sin\theta - \sin\theta_0\right)\right| \le \frac{N-1}{N}\pi \qquad \text{for} \qquad -\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$

There are two cases, one each for θ_0 positive or negative--

For $0 \le \theta_0 \le \pi/2$ maximum of $|\ldots|$ must occur for $\theta = \pi/2$

$$\left|\frac{1}{2}kd\left(-1 - \sin\theta_0\right)\right| = \frac{N-1}{N}\pi \qquad \text{where} \qquad 0 \le \sin\theta_0 \le 1$$

$$\frac{1}{2}\frac{2\pi}{\lambda}d\left(1 + \sin\theta_0\right) = \frac{N-1}{N}\pi \qquad \text{or} \qquad \frac{\lambda}{d} = \frac{N}{N-1}\left(1 + \sin\theta_0\right)$$

For $-\pi/2 \le \theta_0 \le 0$ maximum must occur for $\theta = -\pi/2$

$$\left|\frac{1}{2}kd\left(1 - \sin\theta_0\right)\right| = \frac{N-1}{N}\pi \qquad \text{but since} \qquad -1 \le \sin\theta_0 \le 0$$

$$\frac{1}{2}\frac{2\pi}{\lambda}d\left(1 + |\sin\theta_0|\right) = \frac{N-1}{N}\pi \qquad \text{or} \qquad \underline{\frac{\lambda}{d} = \frac{N}{N-1}\left(1 + |\sin\theta_0|\right)}$$

This last equality satisfies <u>both</u> cases for θ_0

(*a*) For end-fired, $\theta_0 = \pm\pi/2$ so that the above becomes $\underline{\dfrac{\lambda}{d} = \dfrac{2N}{N-1}}$

7.8.5. Endfired in water $N = 30$ $f = 300$ Hz $c = 1500$ m/s

(*a*) From (7.8.18) $\dfrac{\lambda}{d} = \dfrac{2N}{N-1}$ $\quad d = \dfrac{N-1}{2N}\dfrac{c}{f} = \dfrac{29}{2\cdot 30}\dfrac{1500}{30} = \underline{2.42 \text{ m}}$

(*b*) The major lobe points in the direction $\theta_0 = \pi/2$ rad and is <u>a straight line</u>

(*c*) From (7.8.16), the first null away from $\theta_0 = \pi/2$ occurs when θ

decreases from $\pi/2$ until $\dfrac{N}{2}kd\left(\sin\theta_1 - 1\right) = -\pi$ This gives

$$\sin\theta_1 - 1 = -\pi\frac{2}{N}\frac{\lambda}{2\pi}\frac{1}{d} = -\frac{1}{N}\frac{2N}{N-1} = -\frac{2}{29}$$

$$\sin\theta_1 = \frac{27}{29} \qquad \theta_1 = 69^\circ$$

so that Beam width $= 2\cdot\left(90^\circ - 69^\circ\right) = \underline{42^\circ}$

(*d*) The major lobe is piston-like with half angle $21^\circ = 0.367$ rad. Therefore from (7.6.25) *et seq.*

$$D \sim \frac{4}{(2\cdot 0.367/\pi)^2} = 73 \qquad \text{for} \qquad DI \sim 10\log 73 = \underline{19 \text{ dB}}$$

7.10.3. The elements are spaced along the x-axis so that in the spherical coordinates of Appendix A8(*c*) the angle θ of Section 7.8 is the complement of ϕ and thus

$$\hat{x}\cdot\hat{r} = \sin\theta = \frac{x}{r}$$

(*a*) $q(\vec{r}_0) = Q\left[\delta\left(\vec{r}_0 - \hat{x}d\right) + \delta(0) + \delta\left(\vec{r}_0 + \hat{x}d\right)\right]$

Express the integrals in (7.10.5) as $\mathbf{\Phi} = -\dfrac{1}{4\pi r}e^{j(\omega t - kr)}\left\{I_1 + I_2 + I_3 + \cdots\right\}$

$$I_1 = \int_{V_0} q(\vec{r}_0)\, dV_0 = 3Q$$

$$I_2 = jk\int_{V_0} q(\vec{r}_0)\,(\vec{r}_0 \cdot \hat{r})\, dV_0 = jkQ[+d\sin\theta + 0 - d\sin\theta] = 0$$

$$I_3 = -\frac{1}{2!}k^2Q\left[d^2\sin^2\theta + 0 + d^2\sin^2\theta\right] = -\frac{1}{2!}k^2Q\,2d^2\sin^2\theta$$

$$I_4 = 0$$

$$I_5 = \frac{1}{4!}\,k^4Q\,2d^4\sin^4\theta$$

$$\underline{\mathbf{\Phi} = \frac{Q}{4\pi r}e^{j(\omega t - kr)}\left\{3 + 2\left[-\frac{(kd\sin\theta)^2}{2!} + \frac{(kd\sin\theta)^4}{4!} - \dots\right]\right\}}$$

(*b*)

$$\mathbf{\Phi} = \frac{Q}{4\pi r}e^{j(\omega t - kr)}\{3 + 2[\cos(kd\sin\theta) - 1]\}$$

$$= \underline{\frac{Q}{4\pi r}e^{j(\omega t - kr)}\left\{3 - 4\sin^2\!\left(\tfrac{1}{2}kd\sin\theta\right)\right\}}$$

(*c*)

$$\sin 3\beta = \sin(2\beta + \beta) = \sin 2\beta\cos\beta + \cos 2\beta\sin\beta$$

$$= (\sin\beta\cos\beta + \cos\beta\sin\beta)\cos\beta + (\cos\beta\cos\beta - \sin\beta\sin\beta)\sin\beta$$

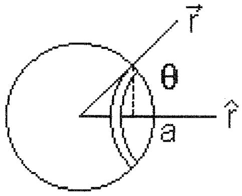
$$= 3\sin\beta - 4\sin^3\beta$$

Thus $\dfrac{\sin 3\beta}{\sin\beta} = 3 - 4\sin^2\beta$ and use this relation in the answer to (*b*)

$$\mathbf{\Phi} = \frac{3Q}{4\pi r}e^{j(\omega t - kr)}\left\{\frac{\sin\left(\tfrac{3}{2}kd\sin\theta\right)}{3\sin\left(\tfrac{1}{2}kd\sin\theta\right)}\right\}$$

Now use of $\mathbf{p} = -j\omega\rho_0\mathbf{\Phi}$ and $A = j\omega\rho_0 Q/4\pi$ shows that the above is identical with the formula for the three element array

7.10.7.



$$q(r_0) = A(a - r_0) \qquad 0 \le r_0 \le a$$
$$= 0 \qquad r_0 > a$$

(*a*)

$$\mathbf{\Phi}_1 = -\frac{A}{4\pi r}\, e^{j(\omega t - kr)} \int_0^a (a - r_0)\, 4\pi r_0^2\, dr_0$$

$$I_1 = \int_0^a (a - r_0)\, 4\pi r_0^2\, dr_0 = 4\pi\left(\frac{a}{3} r_0^3 - \frac{1}{4} r_0^4\right)\Bigg|_0^a = \frac{4\pi a^4}{12\, r}$$

$$\underline{\mathbf{\Phi}_1 = -\frac{A\, a^4}{12\, r}\, e^{j(\omega t - kr)}}$$

(*b*)

$$\mathbf{\Phi}_{12} = -jk\frac{A}{4\pi r}\, e^{j(\omega t - kr)} \int_{V_0} (a - r_0)(\vec{r}_0 \cdot \hat{r})\, dV_0$$

$$I_2 = \int_{V_0} (a - r_0)(\vec{r}_0 \cdot \hat{r})\, dV_0$$

$$= \int_0^\pi \int_0^a (a - r_0)\, r_0 \cos\theta\, 2\pi r_0 \sin\theta\, r_0\, dr_0 d\theta = 0$$

$$\underline{\mathbf{\Phi}_2 = 0}$$

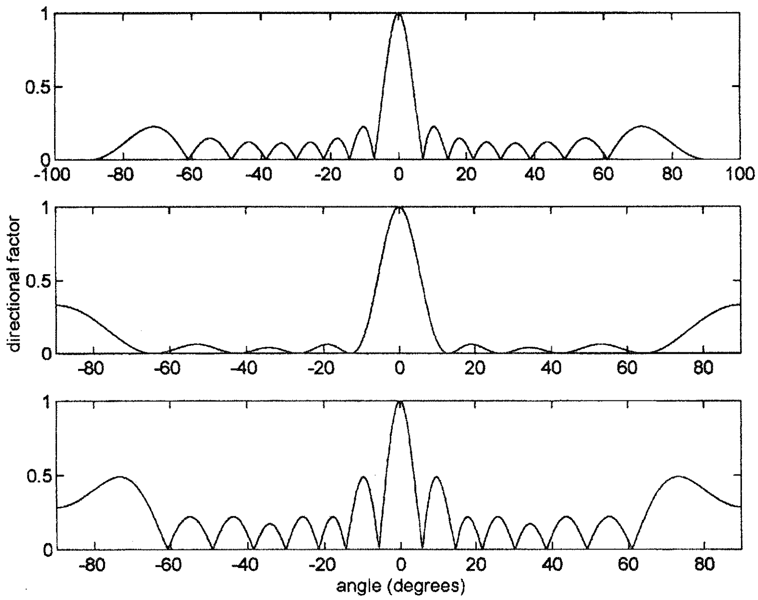
(*c*)

$$\mathbf{\Phi}_3 = \frac{1}{2!} k^2 \frac{A}{4\pi r}\, e^{j(\omega t - kr)} \int_0^\pi \int_0^a (a - r_0)\, 2\pi r_0^4 \cos^2\theta \sin\theta\, dr_0\, d\theta$$

$$= \frac{1}{2!} k^2 \frac{A}{4\pi r}\, e^{j(\omega t - kr)} \left(\frac{a}{5} a^5 - \frac{1}{6} a^6\right) 2\pi \left(-\frac{1}{3}\cos^3\theta\right)\Bigg|_0^\pi$$

$$\underline{= -\frac{A a^4}{180\, r}\, (ka)^2\, e^{j(\omega t - kr)}}$$

7.11.2C.



8.2.2. (*a*) From A10, $\eta = 1.85 \times 10^{-5}$ Pa·s and $\rho = 1.21$ kg/m^3 in air at 20°C and 1 atm. Since $p = \rho r T_K$ we have $p \propto \rho$ for constant T_K so that $\rho = 0.121$ kg/m^3 at 0.1 atm; $c = 343$ m/s is independent of p and η depends only on temperature

$$\tau_S = \frac{4}{3}\frac{1}{\rho_0 c^2}\eta = \frac{4}{3}\frac{1}{0.121 \cdot 343^2} 1.85 \times 10^{-5} = \underline{1.73 \times 10^{-9} \text{ s}}$$

(*b*) $$f = \frac{1}{2\pi\tau_S} = \underline{0.92 \times 10^7 \text{ Hz}}$$

(*c*)

$$\alpha_S = \frac{\omega}{c}\frac{1}{\sqrt{2}}\left[\frac{\sqrt{1+(\omega\tau)^2} - 1}{1+(\omega\tau)^2}\right]^{1/2} = \frac{1}{\sqrt{2}\cdot 1.73 \times 10^{-9}}\frac{1}{343}\left(\frac{\sqrt{2}-1}{2}\right)^{1/2}$$

$$= \underline{5.4 \times 10^5 \text{ Np/m}}$$

$$c_p = c\sqrt{2}\left[\frac{1+(\omega\tau)^2}{\sqrt{1+(\omega\tau)^2}+1}\right]^{1/2} = 343\sqrt{2}\left(\frac{2}{\sqrt{2}+1}\right)^{1/2} = \underline{442 \text{ m/s}}$$

(*d*) For 1 atm $\tau_S = 1.73 \times 10^{-10}$ s so that $\omega\tau_S = 0.10$

$\alpha_S = \underline{8.4 \times 10^4 \text{ Np/m}}$ $\quad c_p = \underline{344 \text{ m/s}}$

or, from Table 8.5.1, $$\alpha_S = \frac{0.99 \times 10^{-11}}{(2\pi \cdot 1.74 \times 10^{-9})^2} = \underline{8.3 \times 10^4 \text{ Np/m}}$$

8.2.5. (*a*) The linearized N-S squation is $\rho_0 \dfrac{\partial \vec{u}}{\partial t} = -\nabla p + \left(\frac{4}{3}\eta + \eta_B\right)\nabla(\nabla\cdot\vec{u}) - \eta\nabla\times\nabla\times\vec{u}$

For incompressible flow, $\rho = \rho_0$ and (5.3.3) yields $\nabla\cdot\vec{u} = 0$

Only shearing motion, so $\nabla p = 0$ N-S equation now simplifies to

$$\rho_0 \frac{\partial \vec{u}}{\partial t} = -\eta \nabla\times\nabla\times\vec{u}$$

Let $\hat{x}$ be $\parallel$ to the plate in the direction of the shear and $\hat{z}$ $\perp$ to the plate

$$\vec{u}(z,t) = u(z,t)\hat{x} \qquad \text{so that}$$

$$\nabla\times\vec{u} = -\hat{y}\frac{\partial u}{\partial z} \qquad \text{and} \qquad \nabla\times\nabla\times\vec{u} = \hat{x}\left[\frac{\partial}{\partial z}\left(-\frac{\partial u}{\partial z}\right)\right] = -\frac{\partial^2 u}{\partial z^2}\hat{x}$$

This gives

$$\rho_0 \frac{\partial u}{\partial t}\hat{x} = -\eta\left(-\frac{\partial^2 u}{\partial z^2}\right)\hat{x} \qquad \text{or} \qquad \underline{\frac{\partial^2 u}{\partial z^2} = \frac{\rho_0}{\eta}\frac{\partial u}{\partial t}}$$

(*b*) The boundary conditions are

$$u(0,t) = U_0 e^{j\omega t} \qquad \text{and} \qquad u \to 0 \qquad \text{as} \qquad z \to \infty$$

Try $\mathbf{u}(z,t) = U_0 e^{-\gamma z} e^{j\omega t}$ and substitute

$$\gamma^2 = \frac{\rho_0}{\eta} j\omega \qquad \gamma = \sqrt{\frac{\rho_0\omega}{\eta}}\, e^{j\pi/4} = \pm(1+j)\sqrt{\frac{\rho_0\omega}{\eta}}$$

Must choose the + sign to satisfy the boundary condition at ∞

$$\underline{u(z,t) = U_0 e^{-\sqrt{\rho_0\omega/2\eta}\,z}\cos\left(\omega t - \sqrt{\rho_0\omega/2\eta}\,z\right)}$$

(*c*)

Absorption coefficient $\alpha = \underline{\sqrt{\dfrac{\rho_0\omega}{2\eta}}}$

Phase speed $c_p = \dfrac{\omega}{\sqrt{\rho_0\omega/2\eta}} = \underline{\sqrt{\dfrac{2\eta\omega}{\rho_0}}}$

(*d*) Using values for air from the Appendix

$$\delta = \frac{1}{\alpha} = \sqrt{\frac{2\eta}{\rho_0\omega}} = \left(\frac{2\cdot 1.85\times10^{-5}}{1.21\cdot 2\pi}\right)^{1/2}\frac{1}{\sqrt{f}} = \frac{2.2\times10^{-3}}{\sqrt{f}}$$

At 10 Hz $\quad \delta = \underline{7.0\times10^{-4}\text{ m}}$

At 1 kHz $\quad \delta = \underline{7.0\times10^{-5}\text{ m}}$

8.6.4. air 500 Hz 60 dB *re* 20 μPa = 60 dB *re* 10^{-12} W/m² at 1000 ft

$I = 10^{-6}$ W/m² at 1000 ft = 306 m

(*a*) $W_0 = 2\pi r^2 I = 2\pi\cdot 306^2\cdot 1\times10^{-6} = \underline{0.588\text{ W/m}^2}$

(*b*) Table 8.5.1, $\alpha_c = 1.37\times10^{-11}\cdot 500^2 = \underline{3.43\times10^{-6}\text{ Np/m}}$

$W_0 = We^{2\alpha r} = 0.588\exp(2\cdot 3.43\times10^{-6}\cdot 306) = \underline{0.589\text{ W}}$

(c) Figure 8.6.2, 0% humidity

$$a = 1.4\times10^{-3}\ \text{dB/m} \quad \text{or} \quad \alpha = 1.4\times10^{-3}/8.7 = \underline{1.6\times10^{-4}\ \text{Np/m}}$$

$$W_0 = 0.588\exp(2\cdot1.6\times10^{-4}\cdot306) = \underline{0.648\ \text{W}}$$

(d) Figure 8.6.2, 100% humidity

$$a = 2.6\times10^{-3}\ \text{dB/m} \quad \text{or} \quad \alpha = 2.6\times10^{-3}/8.7 = \underline{3.0\times10^{-4}\ \text{Np/m}}$$

$$W_0 = 0.588\exp(2\cdot3.0\times10^{-4}\cdot306) = \underline{0.707\ \text{W}}$$

8.7.1. (a) From (8.2.4) η and η_B have dimensions $[\mathrm{s}]\left[\dfrac{\mathrm{kg}}{\mathrm{m}^3}\right]\left[\dfrac{\mathrm{m}}{\mathrm{s}}\right]^2$ and from (8.2.12)

α has dimensions $\left[\dfrac{1}{\mathrm{s}^2}\right]\left[\dfrac{\mathrm{m}^3}{\mathrm{kg}}\right]\left[\dfrac{\mathrm{s}}{\mathrm{m}}\right]^3[\mathrm{s}]\left[\dfrac{\mathrm{kg}}{\mathrm{m}^3}\right]\left[\dfrac{\mathrm{m}}{\mathrm{s}}\right]^2 = \underline{\mathrm{m}^{-1}}$

(b) pure water 40 kHz 4000 m

$$a(H_2O) = 4.9\times10^{-10}e^{(-T/26-Z/25)}f^2 = 4.9\times10^{-10}e^{-5/26}(4\times10^4)^2$$
$$= 0.647\ \text{dB/km}$$

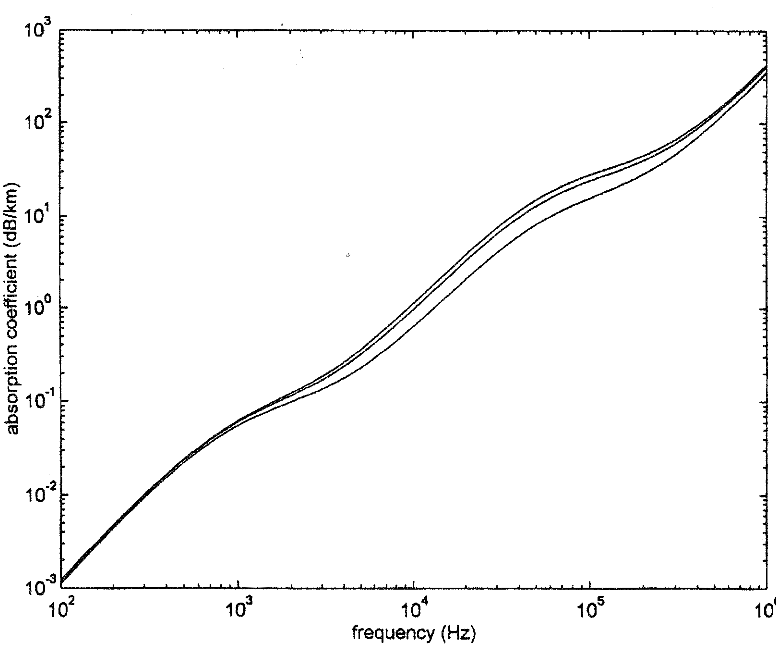
and thus $a(H_2O)\cdot r = \underline{2.6\ \text{dB over the 4 km path}}$

(c) a(Boric Acid) is negligible at 40 kHz

$$a(MgSO_4) = 22e^{5/14}\frac{(4\times10^4)^2}{(42000e^{5/18})^2+(4\times10^4)^2} = 10.8\ \text{dB/km}$$

$$ar = 2.6 + 10.8\cdot4 = \underline{46\ \text{dB over the 4 km path}}$$

8.7.3C. (b)



8.8.1. (8.8.1) is $\rho_0 \frac{\partial \vec{u}}{\partial t} + \nabla p = -\eta \nabla \times \nabla \times u$ Assume the geometry of the discussion

Neglecting losses intrinsic to propagation in the infinite medium, we can write

$$\vec{u} = \left[u_x(x,t) + u'(x,z,t)\right]\hat{x} \qquad \text{with} \qquad u_x = U_0 e^{j(\omega t - kx)}$$

$$p = p_x + p' \qquad \text{with} \qquad p_x = \rho_0 c U_0 e^{j(\omega t - kx)}$$

Since the primary field is assumed lossless, the only losses are those associated with the secondary field. Thus, $u_x\hat{x}$ satisfies Euler's equation so that all its terms vanish on the LHS of (8.8.1) and it makes no contribution on the RHS. Now, (8.8.1) reduces to

$$\rho_0 \frac{\partial(u'\hat{x})}{\partial t} + \nabla p' = -\eta \nabla \times \nabla \times (u'\hat{x})$$

On the RHS, $\nabla \times (u'\hat{x}) = -\frac{\partial u'}{\partial z}\hat{y}$ and $\nabla \times \nabla \times (u'\hat{x}) = -\frac{\partial^2 u'}{\partial z^2}\hat{x} - \frac{\partial^2 u'}{\partial x \partial z}\hat{z}$

so that, component by component, we have

$$\rho_0 \frac{\partial u'}{\partial t} + \frac{\partial p'}{\partial x} = \eta \frac{\partial^2 u'}{\partial z^2}$$

$$\frac{\partial p'}{\partial y} = 0$$

$$\frac{\partial p'}{\partial z} = \eta \frac{\partial^2 u'}{\partial x \partial z}$$

8.8.2. With $\mathbf{u}' \approx U_0 e^{-z/\delta} e^{j(\omega t - kz - z/\delta)}$ and from (8.8.3) the secondary pressure is

$$\mathbf{p}' \approx \eta \frac{\partial \mathbf{u}'}{\partial x} = \eta(-jk)\mathbf{u}' \qquad \text{so that} \qquad \frac{\partial \mathbf{p}'}{\partial x} \approx -k^2 \eta \mathbf{u}'$$

Then with the help of (8.8.5), $\left|\frac{\partial p'}{\partial x}\right| \bigg/ \left|\rho_0 \frac{\partial u'}{\partial t}\right| \sim \frac{k^2 \eta u'}{\rho_0 \omega u'} \sim k^2 \frac{\delta^2}{2} \sim \underline{\frac{1}{2}(k\delta)^2}$

8.10.7. $a_b = 0.01 \text{ cm} = 1 \times 10^{-4} \text{ m} \qquad z = 10 \text{ m} \qquad \mathscr{P}_0 = 2 \text{ atm}$

(*a*) (8.10.25) $\omega_0 = \frac{1}{1 \times 10^{-4}} \sqrt{\frac{3 \cdot 1.402 \cdot 2 \cdot 1.013 \times 10^5}{1026}} = 2.88 \times 10^5$

$\underline{f_0 = 46 \text{ kHz}}$

(*b*) (8.10.28) $Q = \frac{1}{\frac{2.88 \times 10^5}{1500} 1 \times 10^{-4} + \sqrt{2.88 \times 10^5}\, 1.6 \times 10^{-4}} = 9.52$

$$\sigma_S = 4\pi a^2 Q^2 = 4\pi(1\times10^{-4})^2 9.52^2 = \underline{1.14\times10^{-5}\ \mathrm{m}^2}$$

$$\sigma = \frac{4\pi a^2 Q}{k_0 a} = \frac{4\pi(1\times10^{-4})^2 9.52}{1.92\times10^{-2}} = \underline{6.23\times10^{-5}\ \mathrm{m}^2}$$

(c) $a = 0.01$ dB/m From (8.10.1) *et seq.*

$$a = 4.35 N\sigma = 0.01 \quad \text{or} \quad N = \underline{36.9\ \text{per m}^3}$$

(d) From Fig. 8.7.2, <u>the same</u>

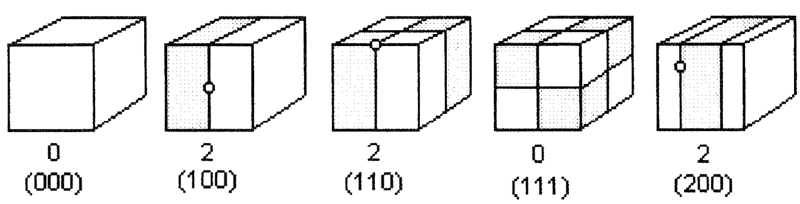
9.2.4. (a)

$$\omega_{lmn} = \sqrt{\left(\frac{l\pi}{L}\right)^2 + \left(\frac{m\pi}{L}\right)^2 + \left(\frac{n\pi}{L}\right)^2} \qquad l = 0,1,2,3,\cdots$$

$$f_{lmn} = \frac{c}{2L}\sqrt{l^2 + m^2 + n^2} \qquad m = 0,1,2,3,\cdots \quad n = 0,1,2,3,\cdots$$

$$\begin{aligned}
f_{000} &= 0 \\
f_{100} = f_{010} = f_{001} &= c/2L = 0.500\,c/L \\
f_{110} = f_{011} = f_{101} &= \sqrt{2}c/2L = 0.707\,c/L \\
f_{111} &= \sqrt{3}\,c/2L = 0.866 c/L \\
f_{200} = f_{020} = f_{002} &= 2c/2L = c/L
\end{aligned}$$

(b)



The values 0 and 2 give the number of 90 degree rotations of each box

(c) The three labelled (1 0 0), (0 1 0), (0 0 1) are degenerate
The three labelled (1 1 0), (1 0 1), (0 1 1) are degenerate
The three labelled (2 0 0), (0 2 0), (0 0 2) are degenerate

(d) Indicated by the open circles in the sketches of (b)

9.3.4. Water-filled cylindrical tank with rigid side and bottom, pressure release top. Height $H = 1$ m and radius $a = 1$ m. With z measured upward from the bottom and (r, θ) for cylindrical coordinates as in Appendix A7(c),

$$\mathbf{p}_{lmn} = J_m(k_{mn}r)\cos m\theta \cos k_{zl}z\, e^{j\omega_{lmn}t}$$

$$k_{zl}H = k_{zl} = \left(l - \tfrac{1}{2}\right)\pi \qquad l = 1,2,3,\ldots$$

$$k_{mn}a = k_{mn} = j'_{mn} \qquad m = 0,1,2,\ldots$$

$$n = 1,2,3,\ldots$$

(*a*) $f_{lmn} = \dfrac{c}{2\pi}\sqrt{\left(j'_{mn}\right)^2 + \left[\left(l-\tfrac{1}{2}\right)\pi\right]^2}$ and use Appendix A5(*d*) to obtain the j'_{mn}s

$$f_{101} = \frac{1500}{2\pi}\sqrt{0^2 + \left(\tfrac{1}{2}\pi\right)^2} = \underline{375\ \text{Hz}}$$

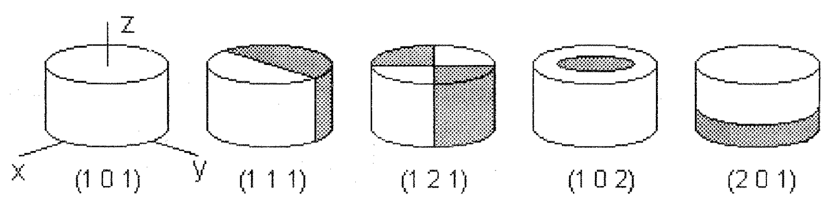
$$f_{111} = \frac{1500}{2\pi}\sqrt{1.84^2 + \left(\tfrac{1}{2}\pi\right)^2} = \underline{577\ \text{Hz}}$$

$$f_{121} = \frac{1500}{2\pi}\sqrt{3.05^2 + \left(\tfrac{1}{2}\pi\right)^2} = \underline{819\ \text{Hz}}$$

$$f_{102} = \frac{1500}{2\pi}\sqrt{3.83^2 + \left(\tfrac{1}{2}\pi\right)^2} = \underline{988\ \text{Hz}}$$

$$f_{201} = \frac{1500}{2\pi}\sqrt{0^2 + \left(\tfrac{3}{2}\pi\right)^2} = \underline{1124\ \text{Hz}}$$

(*b*)



(1, 0, 1)	$z = H$	
(1, 1, 1)	$z = H$	$\theta = 0°$
(1, 2, 1)	$z = H$	$\theta = 45°, 135°$
(1, 0, 2)	$z = H$	$r = (2.40/3.83)a = 0.63\,a$
(2, 0, 1)	$z = H/3, H$	

9.3.5C. $a = 10$ cm $\quad b = 50$ cm $\quad h = 10$ cm $\quad$ air-filled, rigid walls.
The general solution is

$$\mathbf{p}_{lmn} = \left[\mathbf{A}_{lmn}J_m(k_n r) + \mathbf{B}_{lmn}Y_m(k_n r)\right]\cos m\theta \cos k_{zl}z\, e^{j\omega_{lmn}t}$$

For the lowest radial mode $m = 0$, $l = 0$ so that $k_z = 0$, and

$$\frac{d}{dr}\left[\mathbf{A}J_0(kr) + \mathbf{B}Y_0(kr)\right] = 0 \qquad \text{at} \qquad r = a, b \qquad \text{so that}$$

$$\mathbf{A}J_1(ka) = \mathbf{B}Y_1(ka)$$

$$\mathbf{A}J_1(kb) = \mathbf{B}Y_1(kb)$$ If we define $ka = u$ then this gives

$$\frac{J_1(u)}{J_1(5u)} = \frac{Y_1(u)}{Y_1(5u)}$$ to be solved for the lowest value of u

While the problem is to be solved by computer, in this case we can obtain a solution fairly simply by hand. To get a starting point for a trial and error solution, first note that for this lowest radial mode there must be about half a wavelength standing between $r = a$ and $r = b$, so that $b - a = 0.40$ m $\sim \lambda/2$ or $f \sim c/\lambda = 343/0.80 \sim 430$ Hz for air and thus $u = ka \sim 0.79$. Then, from Appendix A5

u	$J_1(u)/J_1(5u)$		$Y_1(u)/Y_1(5u)$	
0.6	0.8455	$>$	-3.8817	
0.7	2.3945	$>$	-2.6894	
0.8	-5.5879	$<$	-2.4582	
0.9	-1.7564	$>$	-2.9001	-overshot, so back up and interpolate,
0.85	-2.49	$\approx$	-2.58	-gives a close estimate $f \approx 464$ Hz.

More exact interpolation yields the computer based result, $f = 462$ Hz.

9.5.4. Combining (9.5.3) and (9.5.4), $\omega/c = \sqrt{k_z^2 + (\omega_{lm}/c)^2}$ Now,

$$\frac{d\omega}{dk_z} = c_g = \frac{ck_z}{\sqrt{k_z^2 + (\omega_{lm}/c)^2}} = \frac{c\sqrt{(\omega/c)^2 - (\omega_{lm}/c)^2}}{\omega/c}$$ and thus

$$\frac{c_g}{c} = \sqrt{1 - \left(\frac{\omega_{lm}}{\omega}\right)^2}$$

9.6.6. Write $p_{lm} = P_{lm}(x,y)f(z,t)$ and drop the subscripts and arguments in what follows

(*a*)
$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)Pf = \frac{1}{c^2}\frac{\partial^2}{\partial t^2}Pf$$

$$f\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)P + P\frac{\partial^2 f}{\partial z^2} - P\frac{1}{c^2}\frac{\partial^2 f}{\partial t^2} = 0$$

$$\frac{1}{P}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)P + \frac{1}{f}\left(\frac{\partial^2}{\partial z^2} - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right)f = 0$$

But $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)P_{lm} = \left(\frac{\omega_{lm}}{c}\right)^2 P_{lm}$ and substitution gives

$$\frac{\partial^2 f_{lm}}{\partial z^2} - \frac{1}{c^2}\frac{\partial^2 f_{lm}}{\partial t^2} - \left(\frac{\omega_{lm}}{c}\right) f_{lm} = 0$$

(*b*) Write $g_{lm}(z,t) = f_{lm}(z,t)\cdot 1(t - z/c)$ then note that

$$\frac{\partial g}{\partial t} = \frac{\partial f}{\partial t}\cdot 1 - f\cdot 1' \qquad \frac{\partial^2 g}{\partial t^2} = \frac{\partial^2 f}{\partial t^2}\cdot 1 + 2\frac{\partial f}{\partial t}\cdot 1' + f\cdot 1''$$

$$\frac{\partial g}{\partial z} = \frac{\partial f}{\partial z}\cdot 1 - \frac{1}{c}f\cdot 1' \qquad \frac{\partial^2 g}{\partial z^2} = \frac{\partial^2 f}{\partial z^2}\cdot 1 - \frac{2}{c}\frac{\partial f}{\partial z}\cdot 1' + \frac{1}{c^2}f\cdot 1''$$

where $'$ indicates differentiation with respect to the argument of the function [$(t - z/c)$ in this case] and substitute into the answer to (*a*). After cancellations, with $\delta(t - z/c) = 1'(t - z/c)$, we have

$$-\frac{2}{c}\frac{\partial f}{\partial z}\delta(t - z/c) - \frac{2}{c^2}\frac{\partial f}{\partial t}\delta(t - z/c) = 0$$

which is true for all t including $t = z/c$ only if $\left.\frac{\partial f}{\partial z}\right|_{t=z/c} = -\frac{1}{c}\left.\frac{\partial f}{\partial t}\right|_{t=z/c}$

(*c*) For $f = f[t^2 - (z/c)^2]$ partial differentiations give

$$\frac{\partial f}{\partial z} = \frac{\partial f}{\partial[t^2-(z/c)^2]}\frac{\partial[t^2-(z/c)^2]}{\partial z} = \frac{2z}{c^2}f' \qquad \text{and} \qquad \frac{\partial f}{\partial t} = \frac{2t}{c}f'$$

which show that the required equality in (*b*) at $t = z/c$ is satisfied

(*d*) As an additional shorthand, define $T(z) = z/c$ and suppress its argument for notational simplicity. Our desired solution is

$$f(z,t)\cdot 1(t-T) = J_0\left(\omega_{lm}\sqrt{t^2-T^2}\right)\cdot 1(t-T)$$

From $\frac{\partial f}{\partial z} = \omega_{lm}\frac{-z/c^2}{\sqrt{t^2-(z/c)^2}}J_0'$ and $\frac{\partial f}{\partial t} = \omega_{lm}\frac{t}{\sqrt{t^2-(z/c)^2}}J_0'$ we see that f satisfies the equality of (*b*) at $t = z/c$ so it is worth proceeding. With $T = z/c$ the relevant partial derivatives are

$$\frac{\partial^2 f}{\partial z^2} = \left[\omega_{lm}\frac{-T/c}{\sqrt{t^2-T^2}}\right]^2 J_0'' + \left[\omega_{lm}\frac{-1/c^2}{\sqrt{t^2-T^2}} - \omega_{lm}\frac{(T/c)^2}{(t^2-T^2)^{3/2}}\right]J_0'$$

$$\frac{\partial^2 f}{\partial t^2} = \left[\omega_{lm}\frac{t}{\sqrt{t^2-T^2}}\right]^2 J_0'' + \left[\omega_{lm}\frac{1}{\sqrt{t^2-T^2}} - \omega_{lm}\frac{t^2}{(t^2-T^2)^{3/2}}\right]J_0'$$

Now substitute into the differential equation of (*a*) collect terms and simplify

$$J_0'' + \frac{J_0'}{\omega_{lm}\sqrt{t^2 - T^2}} + J_0 \overset{?}{=} 0$$

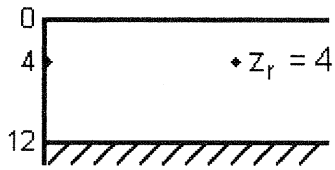
To check whether or not the questioned equality is satisfied, use the expressions in Appendix A4(*b*) for the derivatives of $J_0(u)$ with respect to its argument u

$\left(\text{where} \;\; u = \omega_{lm}\sqrt{t^2 - T^2} \;\; \text{in this case}\right)$ gives the relations

$$J_0{}' = -J_1 \qquad \text{and} \qquad J_0{}'' = -J_1{}' = -J_0 + \frac{J_1}{u}$$

The questioned equality is satisfied.

9.8.4.



$$\omega = 4\omega \qquad k = 4k_{z1}$$

$$k_{z1} = \frac{1}{2}\frac{\pi}{H} \qquad \sin k_{z1}z_r = \sin k_{z1}z_0 = 0.500$$

$$k_{z2} = \frac{3}{2}\frac{\pi}{H} \qquad \sin k_{z1}z_r = \sin k_{z1}z_0 = 1$$

$$k_{z3} = \frac{5}{2}\frac{\pi}{H} \qquad \text{evanescent}$$

$$\kappa_1 = \sqrt{(\omega/c)^2 - k_{z1}^2} = \sqrt{16-1}\,k_{z1} = \sqrt{15}(\pi/2H) = 0.507$$

$$\kappa_2 = \sqrt{(\omega/c)^2 - k_{z2}^2} = \sqrt{16-9}\,k_{z1} = \sqrt{7}(\pi/2H) = 0.346$$

(*a*)

$$\mathbf{p}(r,z_r,t) = -j\frac{2}{H}\sqrt{\frac{2\pi}{r}}\left\{\frac{1}{\sqrt{\kappa_1}}\sin^2\frac{\pi}{6}\,e^{-j\kappa_1 r} + \frac{1}{\sqrt{\kappa_2}}e^{-j\kappa_2 r}\right\}e^{j(\omega t + \pi/4)}$$

$$= -j\frac{2}{H}\sqrt{\frac{2\pi}{r}}\frac{1}{\sqrt{\kappa_2}}\left\{1 + \sqrt{\frac{\kappa_2}{\kappa_1}}\sin^2\frac{\pi}{6}\,e^{-j(\kappa_1-\kappa_2)r}\right\}e^{j(\omega t - \kappa_2 r + \pi/4)}$$

$$P(r,z_r) = \frac{2}{H}\sqrt{\frac{2\pi}{\kappa_2}}\frac{1}{\sqrt{r}}\left|1 + \left(\frac{\kappa_2}{\kappa_1}\right)^{1/2}\sin^2\frac{\pi}{6}\,e^{-j(\kappa_1-\kappa_2)r}\right|$$

Let $\quad A = \left(\dfrac{\kappa_2}{\kappa_1}\right)^{1/2}\sin^2\dfrac{\pi}{6} = \left(\dfrac{7}{15}\right)^{1/4}0.500^2 = 0.207$

$$\theta = (\kappa_1 - \kappa_2)r = 0.161r$$

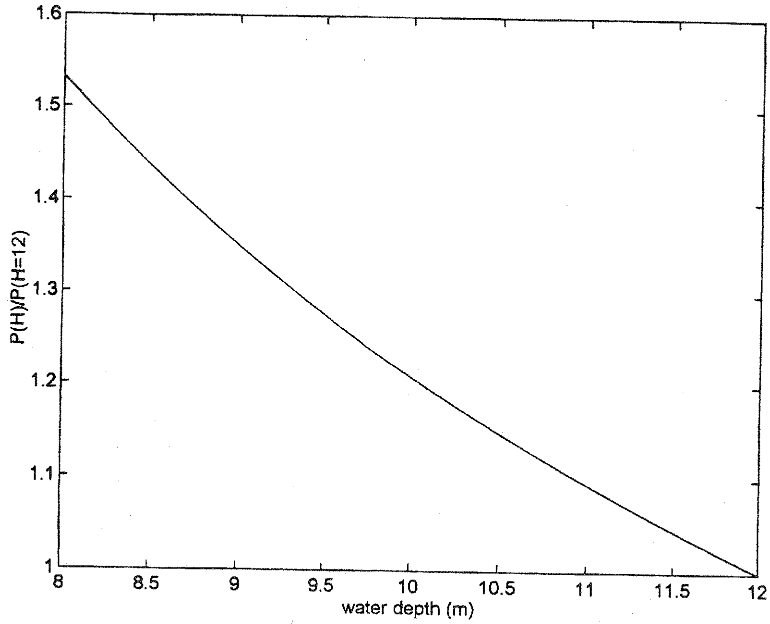
$$P(r,z_r) = \frac{2}{H}\sqrt{\frac{2\pi}{\kappa_2}}\frac{1}{\sqrt{r}}\left|1 + A\,e^{-j\theta}\right| = \frac{2}{H}\sqrt{\frac{2\pi}{\kappa_2}}\frac{1}{\sqrt{r}}\left[1 + A^2 + 2A\cos(0.161r)\right]^{1/2}$$

$$= \frac{0.710}{\sqrt{r}}\left[1 + 0.207^2 + 0.414\cos(0.161r)\right]^{1/2}$$

$$= \underline{\frac{0.725}{\sqrt{r}}\left[1 + 0.397\cos(0.161r)\right]^{1/2}}$$

(*b*) Phase shift of 180° when $\quad |\kappa_1 - \kappa_2|\,\Delta r = \pi \qquad \underline{\Delta r = \dfrac{\pi}{0.161} = 19.5 \text{ m}}$

9.8.7C.



9.8.8. The power radiated from the source can be calculated at a distance of 1 m,

$$\Pi(source) = 4\pi \cdot 1^2 \cdot I(1)$$

The power (averaged over depth H) trapped in the layer at large distances is

$$\Pi(r) = 2\pi r H\, I(r)$$

Conservation of emergy requires the two powers to be the same, so that

$$\frac{I(1)}{I(r)} = \frac{P^2(1)}{P^2(r)} = \frac{rH}{2} \qquad \text{or} \qquad \frac{P(1)}{P(r)} = \sqrt{\frac{rH}{2}}$$

9.9.2. (*a*) For β imaginary, $(\omega/c_2)^2 > \kappa_n^2$ and so define

$$\gamma_n = \sqrt{(\omega/c_2)^2 - \kappa_n^2} \qquad \text{and} \qquad k_{zn} = \sqrt{(\omega/c_1)^2 - \kappa_n^2}$$

Then the eigenfunctions can be written in the form

$$Z_{1n} = \sin k_{zn} z \qquad 0 \le z \le H$$

$$Z_{2n} = \frac{\sin k_{zn} H}{\sin \phi_n} \sin[\gamma_n(z - H) + \phi_n] \qquad z > H$$

(*b*) Application of the boundary conditions at $z = H$ as in (9.9.3) and executing the steps following (9.9.5) but with $\beta_n^2 = -\gamma_n^2$ results in

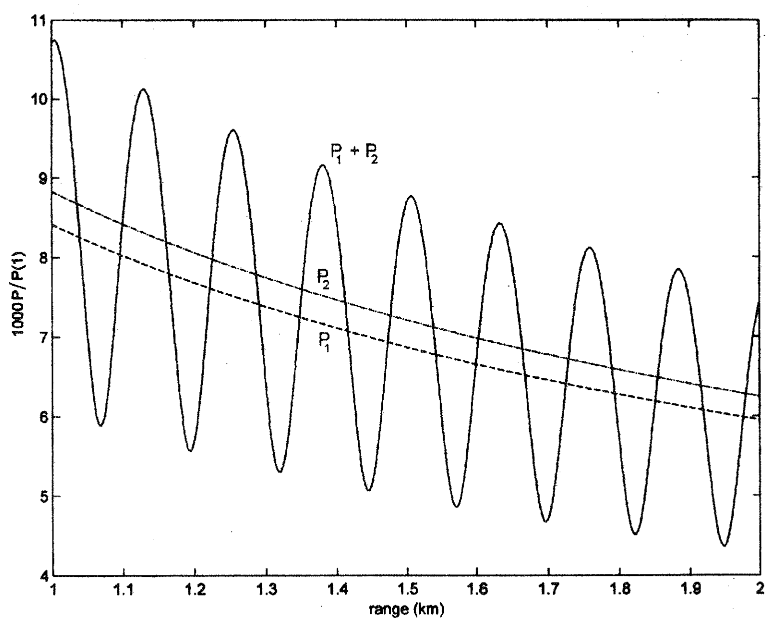
$$\tan y = b \frac{y \tan \phi_n}{\sqrt{-a^2 + y^2}}$$ with y, a and b as given in (9.9.7).

Solutions require $y \ge a$ so that $k_{zn} \ge (\omega/c_1)\sin\theta_c$ For any $y \ge a$ a value of ϕ_n can be found satisfying the above transcendental equation. Therefore, $\underline{k_{zn} \text{ is not quantized}}$.

(c) Substitute $k_{zn}(\kappa_n)$ from (a) into the answer for k_{zn} in (b) and solve for κ_n.

(d) The critical angle θ_n is given by $\cos\theta_n = \kappa_n / k_1$ and from the answer to (c) this gives $\cos\theta_n \le \cos\theta_c$ or $\underline{\theta_n \ge \theta_c}$

9.9.12C.



9.9.15 $SL = 200$ dB *re* 1 μPa $f = 3.5$ kHz $r = 20$ km $H = 100$ m
$\rho_1 = 1000$ kg/m^3 $\rho_2 = 1250$ kg/m^3 $c_1 = 1500$ m/s $c_2 = 1600$ m/s
source and receiver at depths of 50 m

(a) (9.9.9) $f_1 = \dfrac{c_1}{4H\sin\theta_c} = \dfrac{1500}{4\cdot 100\sqrt{1-(15/16)^2}} = 10.8 \text{ Hz} \ll 3.5 \text{ kHz}$

Substitution into (9.9.7) and (9.9.8) shows that

$\tan k_{z1}H \approx 0$ so that $k_{z1}H \approx \pi$ and from (9.9.14)

$A_1 = \sqrt{2/H}$

Also, from (9.9.4) $\kappa_1 = \sqrt{(\omega/c_1)^2 - k_{z1}^2} \approx \omega/c_1$

The lowest mode has a node at the bottom and a maximum at a depth of 50 m. For a source of $P(1) = 1$ μPa (at 1 m from the source) exciting just the lowest mode, (9.9.15) shows that the pressure amplitude at 20 km for source and receiver both at 50 m depth is

$$P(r) = \sqrt{\frac{2\pi}{\kappa_1 r}\frac{2}{H}(1\cdot 1)} = \sqrt{\frac{c}{fr}\frac{2}{H}}$$

At $r = 20$ km this gives $P(20\text{ km}) = 9.26\times 10^{-5}\ \mu$Pa. The loss in sound pressure level is $SPL(1) - SPL(20\text{ km}) = 20\log[1/(9.26\times 10^{-5})] = 80\text{dB}$

Thus

$$SPL(20\text{km}) = 200 - 80 = \underline{120\text{ dB } re\ 1\mu\text{Pa}}$$

(*b*) For just this mode present, the intensity level will be lower for other depths.

(*c*) For the lowest two normal modes, both are far above cutoff so that both k_zs can be approximated $k_{z1} = \pi/H$ and $k_{z2} = 2\pi/H$

$$\Delta\kappa = \kappa_1 - \kappa_2 = \sqrt{(\omega/c_1)^2 - k_{z1}^2} - \sqrt{(\omega/c_1)^2 - k_{z2}^2}$$

$$\approx k_1\left\{\left[1 - \tfrac{1}{2}(k_{z1}/k_1)^2\right] - \left[1 - \tfrac{1}{2}(k_{z2}/k_1)^2\right]\right\}$$

$$\approx \frac{1}{2}\frac{c_1}{\omega}\left(k_{z2}^2 - k_{z1}^2\right) = \frac{1}{2}\frac{c_1}{\omega}\left(\frac{\pi}{H}\right)^2(4-1) = 1.0\times 10^{-4}$$

There will be a 180° phase shift over a range interval Δr when $\Delta\kappa\Delta r = \pi$

$$\underline{\Delta r = \frac{\pi}{\Delta\kappa} = 31\text{ km}}$$

10.2.2. From (10.2.7) for $kL \ll 1$, $\mathbf{Z}_{mo} \approx -j\rho_0 cS\frac{1}{kL} = -j\left(\rho_0 c^2\frac{S}{L}\right)\frac{1}{\omega} = \underline{-j\frac{s_0}{\omega}}$

This is a spring of stiffness $s_0 = \rho_0 c^2 S/L$ A static compression of the gas column of cross-sectional area S and length L through a distance ΔL gives a stiffness $s = \Delta F/\Delta L = pS/\Delta L$ For the perfect gas within the tube, $p = \rho_0 c^2 s = \rho_0 c^2\Delta L/L$ so that $\Delta L = Lp/\rho_0 c^2$ and substitution gives $s = \rho_0 c^2 S/L$ in agreement with the value for s_0 obtained above.

10.3.1. $L = 1.0$ m $\quad a = 0.05$ m $\quad m = 0.015$ kg $\quad f = 150$ Hz $\quad \Xi_0 = 0.005$ m

(*a*) $kL = \frac{2\pi f}{c}L = \frac{2\pi\cdot 150}{343}1.0 = 2.75 \qquad ka = 0.137 \ll 1$

For $ka \ll 1$ use (10.2.10) and (10.2.4) to obtain mechanical impedances

$$\frac{\mathbf{Z}_{mL}}{\rho_0 cS} = \frac{1}{2}(ka)^2 + j\frac{8}{3\pi}ka = 0.0094 + j0.116 \qquad \text{(baffled)}$$

$$\frac{\mathbf{Z}_{m0}}{\rho_0 cS} = \frac{0.0094 + j0.116 + j\tan kl}{1 + j(0.0094 + j0.116)\tan kL} = \frac{0.0094 + j(0.116 - 0.413)}{1 + 0.048 - j0.0039}$$

$$= 0.009 - j0.28$$

and with $\rho_0 cS = 415\cdot\pi(0.05)^2 = 3.26$ the input mechanical impedance of the piston + tube is

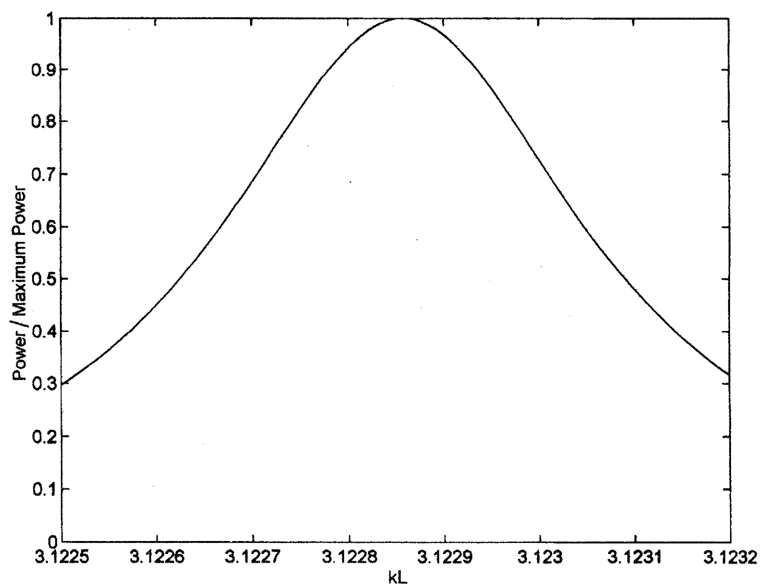
$$\mathbf{Z}_m = \mathbf{Z}_{m0} + j\omega m = 3.26\cdot(0.009 - j0.28) + j2\pi\cdot 150\cdot 0.015$$

$$= \underline{0.029 + j13.2 \ \ \text{N·s/m}}$$

(*b*) $F = U_0 Z_m = \omega \Xi_0 Z_m = 2\pi f\cdot 0.005\sqrt{0.029^2 + 13.2^2} = \underline{62\ \text{N}}$

(*c*) $\Pi = \frac{1}{2} R_m U_0^2 = \frac{1}{2} 0.029\cdot(2\pi f\cdot 0.005)^2 = \underline{0.32\ \text{W}}$

10.3.4C.



10.4.4. $r_1 = 1.48\times 10^6$ Pa·s/m $\qquad r_2 = 8.0\times 10^6$ Pa·s/m $\qquad f = 1$ kHz

(*a*) Because the concrete is assumed infinitely thick, there is only a transmitted wave, and since $r_2 > r_1$ the reflected pressure wave is in phase with the incident at the boundary. (Recall the discussion in Section 6.2.) Consequently, the phase angle $\theta = 0$ and (10.4.2) gives

$$\frac{B}{A} = \frac{8.0 - 1.48}{8.0 + 1.48} = 0.69 \qquad SWR = \frac{1 + B/A}{1 - B/A} = \underline{5.45}$$

(*b*) $20\log 5.45 = \underline{14.7\ \text{dB}}$

(*c*) $\lambda_1 = 1480/1000 = 1.48$ m From (10.4.6) the first node is at $k\Delta x = \pi/2$ where Δx is measured from the wall into the fluid. This is a quarter wavelength, so the first three nodes are located at distances from the wall of

$$\lambda/4 = \underline{0.37}\ \text{m} \qquad 3\lambda/4 = \underline{1.11}\ \text{m} \qquad 5\lambda/4 = \underline{1.85}\ \text{m}$$

10.6.2. Piston: mass m Tube: radius a and length L ($a \ll L$) with baffled end

(*a*) Assume no losses. For $ka \gg 1$ $\mathbf{Z}_{mL} \approx \rho_0 cS$ and (10.2.4) gives

$$\frac{\mathbf{Z}_{m0}}{\rho_0 cS} = \frac{\mathbf{Z}_{mL}/\rho_0 cS + j\tan kL}{1 + j(\mathbf{Z}_{mL}/\rho_0 cS)\tan kL} \approx 1 \qquad \text{and so}$$

$$\mathbf{Z}_m = \rho_0 cS + j\omega m$$

$$U_0 = \frac{F}{Z_m} = \frac{F}{\sqrt{(\rho_0 cS)^2 + (\omega m)^2}} \quad \text{and} \quad \Pi = \frac{1}{2}R_m U_0^2 = \underline{\frac{F^2}{2}\frac{\rho_0 cS}{(\rho_0 cS)^2 + (\omega m)^2}}$$

(b) For $ka \ll 1$ and $kL \ll 1$ use (10.2.10) with $\tan kL \approx kL$ to obtain $\mathbf{Z}_{mL}$ and then substitute into (10.2.4)

$$\frac{\mathbf{Z}_{m0}}{\rho_0 cS} \approx \frac{\frac{1}{2}(ka)^2 + j\left(\frac{8}{3\pi}ka + kL\right)}{\left(1 - \frac{8}{3\pi}ka\right) + j(ka)^2 kL} \approx \frac{1}{2}(ka)^2 + jkL$$

$$\mathbf{Z}_m \approx \tfrac{1}{2}\rho_0 cS(ka)^2 + j(\omega m + \rho_0 cSkL)$$

$$\Pi \approx \underline{\frac{1}{4}\rho_0 cS(ka)^2 \frac{F^2}{[\omega(m + \rho_0 SL)]^2}}$$ [Note $\rho_0 SL$ is the mass of gas in the tube]

10.8.2. $0.3 \times 0.4 \times 0.5$ m $\quad V = 0.06\ \text{m}^3 \quad L = 0.03$ m $\quad a = 0.1$ m
$m_c = 0.01$ kg $\quad s_c = 1000$ N/m $\quad$ air

(a) Assume both sides of the speaker opening are flanged

$$L' = L + 1.7a = 0.03 + 0.17 = 0.2\ \text{m}$$

$$S = \pi a^2 = 0.0314\ \text{m}^2$$

$$\omega_0 = c\sqrt{S/L'V} = 343\sqrt{0.0314/(0.2\cdot 0.06)}$$

$$f_0 = \omega_0/2\pi = \underline{88.3\ \text{Hz}}$$

(b) From (10.8.4) the stiffness of the resonator is

$$s = \rho_0 c^2 S^2/V = 415\cdot 343\cdot 0.0314^2/0.06 = 2340\ \text{N/m}$$

$$s_T = 2340 + 1000 = 3340\ \text{N/m}$$

From (10.8.1) $\quad m = \rho_0 SL' = 1.21\cdot 0.0314\cdot 0.2 = 0.0076$ kg

$$m_T = 0.01 + 0.0076 = 0.0176\ \text{kg}$$

So, $\quad \omega_c = \sqrt{s_T/m_T} = \sqrt{3340/0.0176} = 436$ rad/s

$$f_0 = \underline{69.3\ \text{Hz}}$$

(c)

$$f_{0c} = \frac{1}{2\pi}\sqrt{\frac{s_c}{m_c}} = \frac{1}{2\pi}\sqrt{\frac{1000}{0.01}} = \underline{50.3\ \text{Hz}}$$

(d) Amplitude $\Xi_0 = 0.002$ m so $\quad U_0 = \omega\Xi_0 = 436\cdot 0.002 = 0.872$ m/s

$$ka = \omega a/c = 436 \cdot 0.1/343 = 0.13 \qquad \text{and} \qquad k = 1.3 \text{ m}^{-1}$$

For $ka << 1$ we can approximate the piston radiation resistance,

$$R_r \approx \frac{\rho_0 c k^2}{2\pi} S^2 = \frac{415 \cdot 1.3^2}{2\pi} 0.0314^2 = 0.11 \text{ N·s/m}$$

and the radiated power is

$$\Pi = \frac{1}{2} R_r U_0^2 = \frac{1}{2} 0.11 \cdot 0.872^2 = \underline{0.042 \text{ W}}$$

(*e*) The pressure in the cavity comes from (10.8.3)

$$P_c = \rho_0 c^2 \frac{S \Xi_0}{V} = 415 \cdot 343 \frac{0.0314 \cdot 0.002}{0.06} = \underline{149 \text{ Pa}}$$

$$F = 0.4 \cdot 0.5 \cdot 149 = \underline{29.8\ N}$$

10.9.2. Room volume $V = 2.5 \times 4.0 \times 4.0 = 40 \text{ m}^3$

Wall thickness $L = 0.1$ m

Door area $S = 0.8 \times 2.0 = 1.6 \text{ m}^2$

Effective radius of door $a = (1.6/\pi)^{1/2} = 0.714$ m

(*a*) (10.8.2) et seq. and (10.8.8)

$$f_0 = \frac{343}{2\pi} \sqrt{\frac{1}{40} \frac{1.6}{0.1 + 1.6 \cdot 0.714}} = \underline{9.8 \text{ Hz}}$$

(*b*) (10.8.1), (10.8.4), and (10.9.5)

$$\frac{1}{s} = \frac{V}{\rho_0 c^2 S^2} = \frac{40}{415 \cdot 343 \cdot 1.6^2} = 1.10 \times 10^{-4}$$

$$C = \frac{S^2}{s} = \underline{2.81 \times 10^{-4} \text{ m}^3\text{/Pa}}$$

$$m = \rho_0 S L' = 1.21 \cdot 1.6 \cdot (0.1 + 1.6 \cdot 0.714) = 2.41$$

$$M = \frac{2.4}{1.6^2} = \underline{0.940 \text{ Pa·s}^2\text{/m}^3}$$

(*c*) The driver is within the room, and is therefore driving the equivalent simple harmonic oscillator at one end of the spring with the mass at the other end. See Fig. 1.12.2. The equivalent electrical circuit is the parallel combination of the reactances ωM and $-1/\omega C$. At 20 Hz

$$\mathbf{Z} = \frac{1}{\dfrac{1}{j\omega M} - \dfrac{1}{j/\omega C}} = j \frac{1}{8.47 \times 10^{-3} - 3.53 \times 10^{-2}}$$

$$= \underline{-j37.3 \text{ Pa·s/m}^3}$$

10.10.2. Use (10.10.7) with $\dfrac{\rho_0 c}{S} = \dfrac{\rho_1 c_1}{S_1}$ $\quad R_0 = \dfrac{\rho_2 c_2}{S_2}$ $\quad X_0 = 0$

(a)
$$T_{\mathrm{II}} = \frac{\rho_1 c_1}{S_1}\frac{\rho_2 c_2}{S_2}\frac{4}{\left(\dfrac{\rho_2 c_2}{S_2} + \dfrac{\rho_1 c_1}{S_1}\right)^2 + 0}$$
$$= \underline{\frac{4\rho_1 c_1 S_1 \rho_2 c_2 S_2}{(\rho_2 c_2 S_1 + \rho_1 c_1 S_2)^2}}$$

(b) $\underline{\rho_1 c_1 S_2 = \rho_2 c_2 S_1}$

10.10.5. $S = 0.3^2 = 0.09\ \mathrm{m}^2 \qquad a = 0.08\ \mathrm{m}$
$S_b = \pi a^2 = 0.020\ \mathrm{m}^2 \qquad L' = 0 + 1.7a = 0.136\ \mathrm{m}$

(a) The filter is most effective at resonance $\omega_0 = 2\pi f_0 = c\sqrt{\dfrac{S_b}{L'V}}$

$$V = \frac{c^2 S_b}{4\pi^2 f_0^2 L'} = \frac{343^2\cdot 0.020}{4\pi^2\cdot 30^2\cdot 0.136} = \underline{0.49\ \mathrm{m}^3}$$

(b) Use (10.10.16) where the side branch is the Helmholtz resonator. From (10.8.1), (10.8.4), (10.9.4), and (10.9.5), the acoustic reactance of the side branch, is

$$X_b = \rho_0\left(\frac{\omega L'}{S_b} - \frac{c^2}{\omega V}\right)$$

At 60 Hz, we have $\omega = 2\omega_0$ so that

$$X_b(60\ \mathrm{Hz}) = \rho_0 c\left(\frac{3}{2}\sqrt{\frac{L'}{VS_b}}\right) = \rho_0 c\cdot 1.5\sqrt{\frac{0.136}{0.49\cdot 0.020}} = 5.59\,\rho_0 c$$

Thermoviscous losses in the neck of the resonator can be considered negligible, so $R_b \approx 0$ and (10.10.16) becomes

$$T_{\mathrm{II}} = \frac{X_b^2}{(1/2S)^2 + X_b^2} = \frac{5.59^2}{5.56^2 + 5.59^2} = \underline{0.50\ \text{at 60 Hz}}$$

[From (10.8.10) and (10.10.16), the assumption that R_b is negligible is equivalent to $R_b = 2mc\alpha_w/S^2 << \rho_0 c/2S$ where α_w is given by (8.9.19). This can be verified with the help of the values in A10.]

10.10.9. a(duct) $= 0.6$ m $\quad S$(duct) $= \pi a^2 = 1.13\ \mathrm{m}^2 \quad a_b$(piston) $= 0.5$ m
For $f < 60$ Hz, $\lambda > 343/60 = 5.7$ m.
For these low frequencies, $\lambda >> a_b$ so that $R_b << X_b$ and

$$X_b = j\omega \frac{m}{S^2}$$

(*a*) Substituting into (10.10.16) and neglecting R_b

$$T_{\mathrm{II}} \approx \frac{\left(\frac{\omega m}{S^2}\right)^2}{\left(\frac{\rho_0 c}{2S}\right)^2 + \left(\frac{\omega m}{S^2}\right)^2} = \frac{\left(\frac{2S}{\rho_0 c}\frac{\omega m}{S^2}\right)^2}{1 + \left(\frac{2S}{\rho_0 c}\frac{\omega m}{S^2}\right)^2}$$

$$= \underline{\frac{x^2}{1+x^2} \quad \text{where} \quad x = \frac{2\omega m}{\rho_0 c S}}$$

(*b*) When $T_{\mathrm{II}} \le \frac{1}{2}$ then $x \le 1$ so that $2\omega m \le \rho_0 c S$

$$m \le \frac{\rho_0 c S}{2\omega} = \frac{415 \cdot 1.13}{2 \cdot 2\pi \cdot 60} = \underline{0.62 \text{ kg}}$$

11.3.3. ***SPL*(tone) = 140, 150, or 160 dB *re* 1 μPa**

***PSL*(noise) = 150 dB *re* 1 μPa/Hz$^{1/2}$**

Using Fig. 11.3.1 and $\oplus$ to represent combination of levels

(*a*) w = 1 Hz $SPL = 140 \oplus (150 + 10\log 1)$ = <u>150.4 dB *re* 1 μPa</u>
$\langle PSL\rangle_1$ = 150.4 - 10 log 1 = <u>150.4 dB *re* 1 μPa/Hz$^{1/2}$</u>
w = 10 Hz SPL = $140 \oplus (150 + 10\log 10)$ = <u>160 dB *re* 1 μPa</u>
$\langle PSL\rangle_{10}$ = 160 - 10 log 10 = <u>150 dB *re* 1 μPa/Hz$^{1/2}$</u>
w = 100 Hz $SPL = 140 \oplus (150 + 10\log 100)$ = <u>170 dB *re* 1 μPa</u>
$\langle PSL\rangle_{100}$ = 170 - 10 log 100 = <u>150 dB *re* 1 μPa/Hz$^{1/2}$</u>

(*b*) w = 1 Hz $SPL = 150 \oplus (150 + 10\log 1)$ = <u>153 dB *re* 1 μPa</u>
$\langle PSL\rangle_1$ = 153 - 10 log 1 = <u>153 dB *re* 1 μPa/Hz$^{1/2}$</u>
w = 10 Hz SPL = $150 \oplus (150 + 10\log 10)$ = <u>160.4 dB *re* 1 μPa</u>
$\langle PSL\rangle_{10}$ = 160.4 - 10 log 10 = <u>150.4 dB *re* 1 μPa/Hz$^{1/2}$</u>
w = 100 Hz $SPL = 150 \oplus (150 + 10\log 100)$ = <u>170 dB *re* 1 μPa</u>
$\langle PSL\rangle_{100}$ = 170 - 10 log 100 = <u>150 dB *re* 1 μPa/Hz$^{1/2}$</u>

(*c*) w = 1 Hz $SPL = 160 \oplus (150 + 10\log 1)$ = <u>160.4 dB *re* 1 μPa</u>
$\langle PSL\rangle_1$ = 160.4 - 10 log 1 = <u>160.4 dB *re* 1 μPa/Hz$^{1/2}$</u>
w = 10 Hz SPL = $160 \oplus (150 + 10\log 10)$ = <u>163 dB *re* 1 μPa</u>
$\langle PSL\rangle_{10}$ = 163 - 10 log 10 = <u>153 dB *re* 1 μPa/Hz$^{1/2}$</u>
w = 100 Hz $SPL = 160 \oplus (150 + 10\log 100)$ = <u>170.4 dB *re* 1 μPa</u>
$\langle PSL\rangle_{100}$ = 170.4 - 10 log 100 = <u>150.4 dB *re* 1 μPa/Hz$^{1/2}$</u>

(*d*) Until the *SPL* of the tone becomes similar to or greater than the *SPL* of the noise, the tone has little effect on the overall averaged *PSL* of the combination.

11.4.2. (*a*) • $P(D)$ and d' get smaller. The reduction of the signal amplitude reduces d' and for fixed $P(FA)$ the operating point shifts vertically down to a lower *ROC* curve.

(*b*) • Both $P(D)$ and $P(FA)$ get smaller because the instruction amounts to being more cautious in calling a detection. This means that $P(D)$ must be made smaller for the same d' so that the operating point slides down the *ROC* curve.

(*c*) • d' gets larger and $P(D)$ does also. $P(FA)$ may remain the same or get smaller, depending on the instructions to the subject..

(*d*) • The instruction is equivalent to that given in (*b*).

11.5.1. From (11.5.7) and (11.5.8) $$DT' = 5\log\frac{(d')^2}{w\tau} + \left|5\log\frac{\tau}{T_s}\right|$$

If $\tau > T_S$ then $$DT' = 5\log\frac{(d')^2}{w\tau} + 5\log\frac{\tau}{T_S} = 5\log\frac{(d')^2}{wT_S}$$

The DT' is a constant, independent of τ. The improvement in the processor from increasing τ is just balanced by the diminishing effectivemess of the filter with its shorter integration time. All of the signal of duration greater than T_S is ignored.

If $\tau < T_S$ then $$DT' = 5\log\frac{(d')^2}{w\tau} + 5\log\frac{T_S}{\tau} = 5\log\frac{(d')^2}{wT_S} + 10\log\frac{T_S}{\tau}$$

The DT' increases with decreasing τ because the filter, with its longer intergration time, is smoothing the output too much.

11.5.2. $\tau = 1$ s $\qquad P_{eN} = 2.83 \times 10^{-2}$ Pa

$P_{eS} = 0.02$ Pa $\qquad \sigma = 0.41 \times 10^{-2}$ Pa

$f = 200$ Hz $\qquad w = 100$ Hz

(*a*)
$$SPL_S = 20\log\frac{0.02}{20\times10^{-6}} = \underline{60\text{ dB } re\ 20\ \mu\text{Pa}}$$

$$SPL_N = 20\log\frac{2.83\times10^{-2}}{20\times10^{-6}} = \underline{63\text{ dB } re\ 20\ \mu\text{Pa}}$$

With Fig. 11.3.1 $\quad SPL_{S,N} = 60 \oplus 63 = \underline{64.7\text{ dB } re\ 20\ \mu\text{Pa}}$

(*b*) For energy detection use (11.5.6)

$$d' = (w\tau)^{1/2} S/N = \sqrt{100}\,(2.0/2.83)^2 = \underline{5.0} \qquad \text{(energy)}$$

For correlation detection use (11.5.5)

$$d' = (2w\tau S/N)^{1/2} = \sqrt{200}\,(2.0/2.83) = \underline{10.0} \qquad \text{(correlation)}$$

(*c*) From Fig. 11.4.2 with approximate extrapolation to the left of the curves

$P(FA) \sim \underline{5 \times 10^{-5}}$ (energy)

$P(FA) \sim \underline{5 \times 10^{-9}}$ (correlation)

(*d*) From (11.5.7) and (11.5.8) $\quad DT = 5\log[5.0^2/100] = \underline{-3\text{ dB}}$

(*e*) $DT = 5\log[5.0^2/100] + |5\log 0.5| = \underline{-1.5\ \text{dB}}$

$DT = 5\log[5.0^2/100] + |5\log 2| = \underline{-1.5\ \text{dB}}$

11.7.2. From Fig. 11.7.2 $w_{cb} = 100$ Hz

(*a*) From (11.5.7) and (11.5.8) $DT' = 5\log\dfrac{(d')^2}{w_{cb}\tau} + \left|5\log\dfrac{\tau}{T_S}\right|$

$$\tau > T_S \qquad DT' = 5\log\frac{(d')^2}{w_{cb}T_S}$$

$$\tau < T_S \qquad DT' = 5\log\frac{(d')^2}{w_{cb}T_S} + 10\log\frac{T_S}{\tau}$$

(*b*) $A(\text{signal only}) = A_S = \sqrt{A_{S,N}^2 - A_N^2}$

$DT' = 10\log(S/N) = 20\log(A_S/A_N)$

exp't	τ (s)	A_N (μPa)	A_S (μPa)	DT' (dB)
1	4	3.00×10^4	1.68×10^4	<u>-5.0</u>
2	1	3.00×10^4	1.68×10^4	<u>-5.0</u>
3	0.1	3.00×10^4	3.37×10^4	<u>+1.0</u>

(*c*) Since DT' is constant for expts 1 and 2, the formula for $\tau > T_S$ is valid. This means that we can use (11.5.7) with $DT' = DT$ and $t = T_S$ and obtain d' from the data

$d' = (A_{S,N} - A_N)/\sigma = 2$ Now,

$DT' = 5\log\dfrac{(d')^2}{w_{cb}T_S} = 5\log\dfrac{2^2}{100\cdot T_S} = -5$ so that $T_S = \underline{0.40\ \text{s}}$

[To check this result, substitute into the form for $\tau < T_S$ and note that $10\log(T_S/\tau) = +6.0$ dB.]

(*d*) $DT' = 5\log\dfrac{2^2}{w_{cb}\cdot 0.4}$

f =	100	200	500	1000	2000	5000	Hz
w_{cb} =	100	100	110	150	300	800	Hz
DT' =	<u>-5.0</u>	<u>-5.0</u>	<u>-5.2</u>	<u>-5.9</u>	<u>-7.4</u>	<u>-9.5</u>	<u>dB</u>

11.8.1. (*a*) 60 dB *re* 20 μPa at 100 Hz

From Fig. 11.7.1, $L_N = 50$ phon

From (11.8.1), $N = 0.046\times10^{L_N/30} = 0.046\times10^{5/3} = \underline{2.1\ \text{sone}}$

(b) For a loudness of 0.21 sone, the relation between loudness and loudness level is no longer given by (11.8.1) so use Fig. 11.8.1 to get L_N about 23 phon and then Fig. 11.7.1 to obtain L_I about 40 dB *re* 10^{-12} W/m^2

(c) For a loudness of 21 sone, use (11.8.1)

$$21 = 0.046 \times 10^{L_N/30} \quad \text{and} \quad L_N = 80 \text{ phon}$$

and from Fig. 11.7.1 L_I about 85 dB *re* 10^{-12} W/m^2

11.8.4. (a) 100 Hz at 60 dB gives 50 phon from Fig. 11.7.1 and 2.1 sone from (11.8.1)
200 Hz at 60 dB gives 60 phon and 4.6 sone
500 Hz at 55 dB gives 63 phon and 5.8 sone
500 Hz tone is the loudest

(b) Let $\oplus$ represent combination according to the nomogram of Fig. 11.3.1
$(60 \oplus 60) \oplus 53 = 63 \oplus 53 = 64$ dB *re* 20 μPa

(c) The tones are separated beyond the critical bands so the loudnesses add

$$\sum N_i = 2.1 + 4.6 + 5.8 = 12.5 \text{ sone} = 0.046 \times 10^{L_N/30}$$

$$L_N = 73 \text{ phon}$$

12.3.2. 200 × 50 × 30 ft $\qquad \bar{a} = 0.29$
$S = 2(200 \cdot 50 + 200 \cdot 30 + 50 \cdot 30) = 35 \times 10^3$ ft^2
$V = 200 \cdot 50 \cdot 30 = 300 \times 10^3$ ft^3

(a) $$A = \bar{a}S = 0.29 \cdot 35 \times 10^3 = 1.0 \times 10^4 \text{ ft}^2 \text{ (English sabines)}$$

$$T = \frac{0.049\,V}{A} = \frac{0.049 \cdot 3 \times 10^5}{1 \times 10^4} = 1.47 \text{ s}$$

(b) $$A = 1 \times 10^4/3.28^2 = 930 \text{ m}^2 \text{ (metric sabines)}$$

$$65 = 20 \log\left(\frac{P}{2 \times 10^{-4}}\right)$$

$$P = 0.365\ \mu\text{bar} = 0.0365 \text{ Pa}$$

$$\Pi = A\frac{P^2}{4\rho_0 c} = 930\,\frac{0.0365^2}{4 \cdot 415} = 7.1 \times 10^{-4} \text{ W}$$

(c) $$\Pi_1/\Pi_2 = A_1/A_2 = a_1/a_2$$

$$a_2 = a_1\frac{\Pi_2}{\Pi_1} = 0.29\frac{1 \times 10^{-4}}{7.1 \times 10^{-4}} = 0.0408$$

(d) $$T_2 = T_1 A_1/A_2 = T_1 a_1/a_2 = 1.47 \cdot 0.29/0.0408 = 10.4 \text{ s}$$

The long echo would drastically reduce intelligibility.

12.4.2. $a_c = 0.1 \qquad S_c = 6\cdot 10 = 60\ \text{m}^2$

$a_w = 0.05 \qquad S_w = 2(3\cdot 6 + 3\cdot 10) = 96\ \text{m}^2$

$a_f = 0.6 \qquad S_f = 6\cdot 10 = 60\ \text{m}^2$

$V = 3\cdot 6\cdot 10 = 180\ \text{m}^3$

(*a*) From (12.3.6) $\bar{a} = \dfrac{0.1\cdot 60 + 0.05\cdot 96 + 0.6\cdot 60}{60 + 96 + 60} = \underline{0.217}$

(*b*) $T = \dfrac{0.161V}{S\bar{a}} = \dfrac{0.161\cdot 180}{(60 + 96 + 60)\,0.217} = \underline{0.619\ \text{s}}$

(*c*) $T = \dfrac{0.161V}{-S\ln(1 - \bar{a}_E)} = \dfrac{0.161\cdot 180}{216\ln(1 - 0.217)} = \underline{0.548\ \text{s}}$ <u>(about 12% lower)</u>

12.5.1. 10 ft on a side

			125	500	2000 Hz
walls	= 400 ft^2	acoustic tile	0.10	0.55	0.65
ceiling	= 100 ft^2	acoustic plaster	0.07	0.40	0.65
floor	= 100 ft^2	carpeted floor	0.02	0.14	0.60

(*a*) $\sum Sa = 400\cdot 0.10 + 100\cdot(0.07 + 0.02) = 49\ \text{ft}^2$

$T = \dfrac{0.049V}{\sum Sa} = \dfrac{0.049\cdot 10^3}{49} = \underline{1.0\ \text{s}\quad \text{at 125 Hz}}$

(*b*) $\sum Sa = 400\cdot 0.55 + 100\cdot(0.40 + 0.14) = 274\ \text{ft}^2$

$T = \dfrac{0.049\cdot 10^3}{274} = \underline{0.18\ \text{s}\quad \text{at 500 Hz}}$

(*c*) $\sum Sa = 400(0.65) + 100(0.65 + 0.60) = 384\ \text{ft}^2$

$T = \underline{0.13\ \text{s}\quad \text{at 2000 Hz}}$

12.6.1. $T = 2$ s $\quad 10 \times 20 \times 50$ ft $\quad V = 10\cdot 20\cdot 50\ \text{ft}^3 = 1 \times 10^4\ \text{ft}^3 = 283\ \text{m}^3$

$SPL = 74$ dB *re* 20 μPa $\quad P_r = 0.1$ Pa

(*a*) $\Pi = 9.7\times 10^{-5}\dfrac{P_r^2 V}{T} = 9.7\times 10^{-5}\dfrac{0.1^2\cdot 283}{2} = \underline{1.4\times 10^{-4}\ \text{W}}$

(*b*) $T = 2 = \dfrac{0.049V}{A} = \dfrac{0.049\cdot(10\cdot 20\cdot 50)}{A} \qquad A = 245\ ft^2$

$\Delta SPL = 10 = 20\log(P_1/P_2)$

$\left(\dfrac{P_1}{P_2}\right)^2 = 10 = \dfrac{T_1}{T_2} = \dfrac{A_2}{A_1} \qquad A_2 = 10A_1 = 2450\ \text{ft}^2 \qquad \underline{\Delta A = 2205\ \text{ft}^2}$

(*c*) $T_2 = T_1/10 = \underline{0.2\ \text{s}}$

12.6.3. $S = 2(6{\cdot}7 + 7{\cdot}8 + 6{\cdot}8) = 292\ \text{ft}^2$ or $292/3.28^2 = 27.1\ \text{m}^2$

$V = 6{\cdot}7{\cdot}8 = 336\ \text{ft}^3$ or $336/3.28^3 = 9.52\ \text{m}^3$

$A = \bar{a}S = 0.02{\cdot}27.1 = 0.542\ \text{m}^2$ (metric sabines)

(a) From (12.2.4) and (12.2.7),

$$P^2 = \frac{4\Pi\rho_0 c}{A} = \frac{4\cdot 7.5\times 10^{-6}\cdot 415}{0.542} = 2.30\times 10^{-2}$$

$$P = 0.152\ \text{Pa} = 1.51\ \mu\text{bar}$$

$$SPL = 20\log(1.51/0.0002) = \underline{77.6\ \text{dB}\ re\ 0.0002\ \mu\text{bar}}$$

(b) If $SPL(t) = SPL(\infty) - 3\,\text{dB}$ then $P(t) = P/\sqrt{2}$ and $P^2(t) = P^2/2$

then from (12.2.6) $\dfrac{P^2(t)}{P^2} = 1 - e^{-t/\tau_E} = \dfrac{1}{2}$ so that $\dfrac{t}{\tau_E} = 0.693$

But $\tau_E = \dfrac{4V}{Ac} = \dfrac{4\cdot 9.52}{0.542\cdot 343} = 0.205\ \text{s}$ so that $\underline{t = 0.142\ \text{s}}$

(c) $A(\text{obs}) = A(\text{room}) = 0.542\ \text{m}^2 = 0.542\cdot 3.28^2\ \text{ft}^2$

$= \underline{5.83\ \text{ft}^2\ \text{(English sabines)}}$

12.8.2. $S_A = 20\cdot 50 + 2(40{\cdot}3) + 5(20 - 2{\cdot}3) = 1.31\times 10^3\ \text{m}^2$

$S = 20\cdot 50 + 2(15{\cdot}50) + 2(15{\cdot}20) + 2(40{\cdot}3) + 5\cdot 14 = 3.41\times 10^3\ \text{m}^2$

$S_W = 200\ \text{m}^3$

$V = 20\cdot 15\cdot 50 = 1.5\times 10^4\ \text{m}^3$

$A = S_A a_A + (S - S_W)a + S_W a_W \qquad T = \dfrac{0.161V}{A + 4mV}$

f(Hz)	a_A	a	a_W	A(m^2)	m	T(s)
125	0.40	0.14	0.18	1.01×10^3	-	2.39
250	0.55	0.10	0.06	1.05	0.0001	2.29
500	0.80	0.07	0.04	1.28	0.0002	1.87
1000	0.95	0.05	0.03	1.41	0.0008	1.66
2000	0.90	0.05	0.02	1.34	0.0025	1.62
4000	0.85	0.05	0.02	1.28	0.0081	1.37

(12.8.3) gives

$$1/T = 0.1 + 5.4\,S_T/V = 0.1 + 5.4\cdot 1.31\times 10^3/(1.5\times 10^4) = 0.572\ \text{s}^{-1}$$

or $T = 1.75$ s which is <u>quite close</u>

12.9.3. $4\times 6\times 10$ m

(a) $k = \sqrt{(\pi/L_x)^2 + (\pi/L_y)^2 + (\pi/L_z)^2} = \pi\sqrt{1/4^2 + 1/6^2 + 1/10^2} = 0.317\,\pi$

$$f = \frac{1}{2\pi}kc = \frac{0.317\cdot 343}{2} = \underline{54.3\ \text{Hz}}$$

$(b)\quad T = \dfrac{0.161\,V}{Sa} = \dfrac{0.161\cdot 4\cdot 6\cdot 10}{0.1\cdot 2\cdot(4\cdot 6 + 4\cdot 10 + 6\cdot 10)} = \underline{1.56\text{ s}}$

(c) From (12.9.18), $5 \sim \dfrac{4\pi V}{c^3}f^2 \qquad f \sim \sqrt{\dfrac{5\cdot 343^3}{4\pi\cdot 4\cdot 6\cdot 10}} \sim \underline{259\text{ Hz}}$

12.9.6. $5\times 5\times 5$ m $\quad v_x = 32$ (floor & ceiling) $\quad v_x = 160$ (walls)

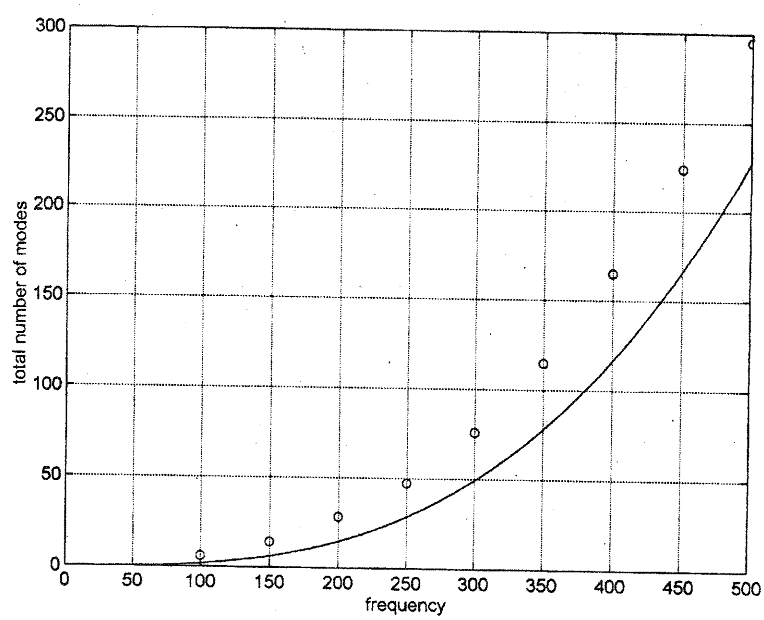
$$T = \frac{0.0201\,V}{\sum_{i=1}^{6}\epsilon_i S_i/v_i} = \frac{0.0201\cdot 5^3}{5^2\sum_{i=1}^{6}\epsilon_i S_i/v_i} = 0.1\Big/\sum_{i=1}^{6}(\epsilon_i/v_i)$$

(a) axial, striking floor and ceiling $\quad T = \dfrac{0.1}{0.5\dfrac{4}{160}+\dfrac{2}{32}} = \underline{1.33\text{ s}}$

(b) tangential, striking walls $\quad T = \dfrac{0.1}{\dfrac{4}{160}+0.5\dfrac{2}{32}} = \underline{1.78\text{ s}}$

(c) oblique $\quad T = \dfrac{0.1}{\dfrac{4}{160}+\dfrac{2}{32}} = \underline{1.14\text{ s}}$

12.9.9C.



13.2.1. (*a*)

f in Hz	BL(dB *re* 20 μPa)				
31.5	70				
		71.7			
63	67				
			72.7		
125	64				
		65.7			
250	61				
				73.0	
500	58				
		59.7			
1000	55				
			60.7		73.0 dB *re* 20 μPa
2000	52				
		53.7			
4000	49				
8000	46	46.0	46.0	46.0	

(*b*) Convert each band level to dBA with the help of Table13.2.1

f in Hz	dBA				
31.5	30.6				
		41.2			
63	40.8				
			53.9		
125	47.9				
		53.7			
250	52.4				
				60.7	
500	54.8				
		57.9			
1000	55				
			59.6		60.8 dBA
2000	53.2				
		54.9			
4000	50				
8000	44.9	44.9	44.9	44.9	

13.6.1. Directly from Fig.13.6.2. $\underline{L_{10} = 58 \text{ dBA}}$ $\underline{L_{50} = 48 \text{ dBA}}$ $\underline{L_{90} = 43 \text{ dBA}}$

13.8.1. For automobiles, use (13.8.1)

$$L_A = 71 + 23\log(v/88) = 71 - 9.6 = \underline{61 \text{ dBA at } 44 \text{ km/hr}}$$

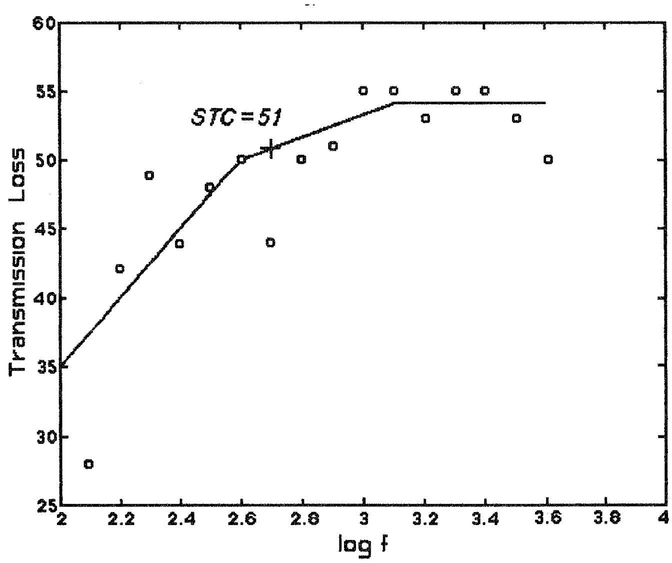
$$= 71 - 0 = \underline{71 \text{ dBA at } 88 \text{ km/hr}}$$

For motorcycles, use (13.8.2)

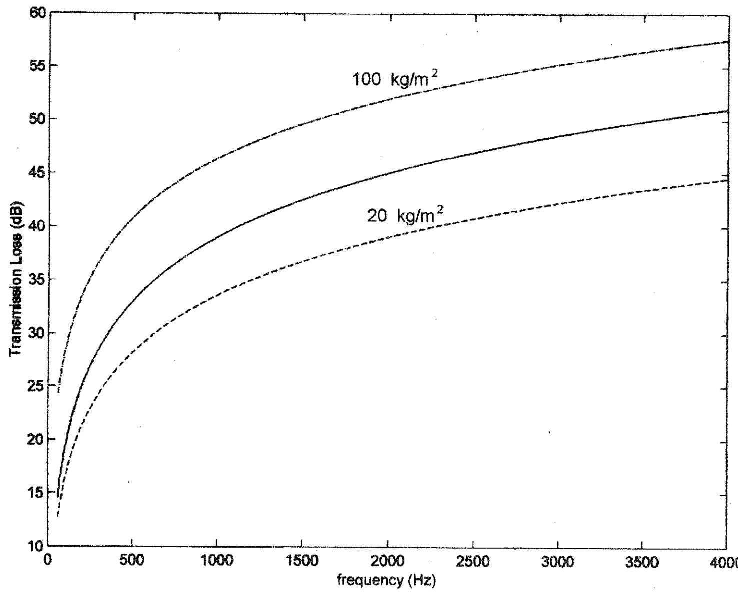
$$L_A = 78 + 25\log(v/88) = \underline{67 \text{ dBA at } 44 \text{ km/hr}}$$
$$= \underline{78 \text{ dBA at } 88 \text{ km/hr}}$$

13.13.1. $S = 30 \text{ m}^2$ $L_1 = 90$ dB and use (13.13.3)

f(Hz)	L_2	NR	A	10 log(S/A)	TL
125	65	25	15	3	28
160	51	39	15	3	42
200	44	46	16	2.7	49
250	48	42	17	2.5	44
315	44	46	19	2.0	48
400	42	48	20	1.8	50
500	47	43	25	0.8	44
630	40	50	28	0.3	50
800	39	51	30	0	51
1000	35	55	32	-0.3	55
1250	34	56	36	-0.8	55
1600	35	55	45	-1.8	53
2000	33	57	53	-2.5	55
2500	32	58	60	-3	55
3150	34	56	60	-3	53
4000	37	53	60	-3	50



13.15.2C.



14.3.2. (14.2.3) $\mathbf{Z}_{ms} = \left(1 - \mathbf{k}_c^2\right)\mathbf{Z}_{mo}$ and (14.2.12) $\mathbf{Z}_{mo} = \left(1 - \mathbf{k}_m^2\right)\mathbf{Z}_{ms}$

Combination gives $1 - \mathbf{k}_m^2 = \dfrac{1}{1 - \mathbf{k}_c^2}$ and for weak coupling $\mathbf{k}_m^2 \approx -\mathbf{k}_c^2$

so that $\underline{k_m \approx k_c}$

14.4.3. $R_{EB} = 5\ \Omega$

$R_W = 10\ \Omega$ at resonance

$R_{EF} = 12\ \Omega$ at resonance

from (14.4.23) $\eta_0 = \dfrac{R}{R_0 + R}\dfrac{R}{R_R}$ where

$$R_0 = R_{EB} \qquad R_R = \frac{\phi_M^2}{R_r} \qquad \text{and} \qquad R = R_{MOT} = \frac{\phi_M^2}{R_m + R_r}$$

from (14.4.16) $R_E(\text{wall}) = R_W = R_{EB} + R_{MOT}$

$$10 = 5 + \frac{\phi_M^2}{R_m + R_r}$$

$$R_{EF} = 12 = 5 + \frac{\phi_M^2}{R_r}$$

thence $\dfrac{\phi_M^2}{R_m + R_r} = 5$ and $\dfrac{\phi_M^2}{R_m} = 7$ so that $\dfrac{\phi_M^2}{R_m} = \dfrac{35}{2}$

and $\eta_{res} = \dfrac{5}{5+5} \dfrac{5}{35/2} = \dfrac{1}{7} = \underline{14.3\,\%}$

14.5.2. $l = N \cdot 2\pi a_c = 80 \cdot 2\pi \cdot 0.03/2 = 7.54$ m $\qquad m = 0.015$ kg

$R_0 = 3.2\ \Omega \qquad R_m = 1$ N·s/m

$L_0 = 0.2 \times 10^{-3}$ H $\qquad R_r = 1$ N·s/m

$B = 1$ T $\qquad \phi_M = Bl = 7.54 \qquad s = 1500$ N/m

displacement $\Xi = 1 \times 10^{-3}$ m at $f = 200$ Hz

assume X_r negligible

(a) $\mathbf{Z}_{EB} = R_0 + j\omega L_0 = 3.2 + j2\pi \cdot 200 \cdot 2 \times 10^{-4}$

$= \underline{3.2 + j0.25\ \ \Omega}$

$$\mathbf{Z}_{MOT} = \frac{\phi_M^2}{\mathbf{Z}_{mo} + \mathbf{Z}_r} = \frac{(Bl)^2}{R_m + j(\omega m - s/\omega) + R_r}$$

$$= \frac{(7.54)^2}{2 + j(400\pi \cdot 0.015 - 1500/400\pi)}$$

$$= \frac{57}{2 + j17.6} = \underline{0.36 - j3.20\ \ \Omega}$$

$\mathbf{Z}_E = \mathbf{Z}_{EB} + \mathbf{Z}_{MOT} = \underline{3.56 - j2.95\ \ \Omega}$

(b) (14.4.15) and (14.4.16) with $U = \omega\Xi = 400\pi \cdot 10^{-3} = 1.26$ m/s

$$U = \frac{\phi_M I}{|\mathbf{Z}_{mo} + \mathbf{Z}_r|} = \frac{|\mathbf{Z}_{MOT}|}{\phi_M} I$$

$$V = Z_E I = Z_E \frac{U\phi_M}{Z_{MOT}} = \sqrt{3.56^2 + 2.95^2}\ \frac{1.26 \cdot 7.54}{\sqrt{0.36^2 + 3.20^2}} = \underline{13.6\ \text{V}}$$

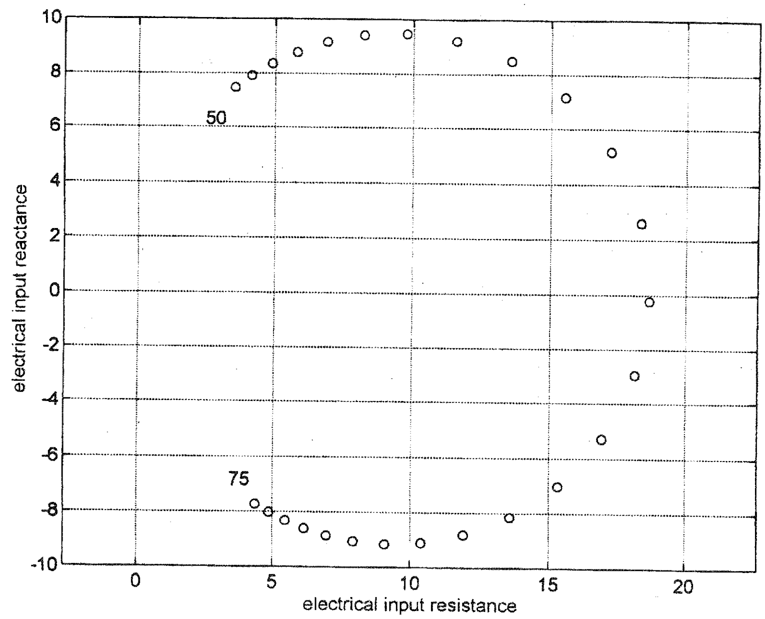
(c) $\Pi = \dfrac{U^2}{R_r} = 1.26^2 = \underline{1.58\ \text{W}}$

(d) $R = \phi_M^2 / (R_m + R_r) = 7.54^2/2 = \underline{28.5\ \text{Ohm}}$

$L = \phi_M^2 / s = 7.54^2/1500 = \underline{38\ \text{mH}}$

$C = m / \phi_M^2 = 0.015/7.54^2 = \underline{263\ \mu\text{F}}$

14.5.5C.



14.6.1. $a = 0.15$ m $\qquad f_0[\text{wall}] = 25$ Hz

$V = 0.1\ \text{m}^3 \qquad f_0[\text{cabinet}] = 50$ Hz

(*a*) (10.8.4) gives the stiffness of the cabinet volume

$$s_c = \frac{\rho_0 c^2 S^2}{V} = \frac{415\cdot 343\cdot(\pi 0.15^2)^2}{0.1} = 7.1\times 10^3\ \text{N/m}$$

Since $f_0 \propto s^{1/2}$ then if f doubles, s quadruples so we have $s_c + s = 4s$

$s = s_c/3 = \underline{2.37\times 10^3\ \text{N/m}}$

(*b*)

$$f_0[\text{wall}] = \frac{1}{2\pi}\left(\frac{s_c}{m+m_r}\right)^{1/2} = 25 \quad \text{so} \quad m + m_r = 0.096\ \text{kg}$$

(7.5.15) $\quad m_r = \dfrac{X_r}{\omega} = \dfrac{8}{3}\rho a^3 = 0.011$ kg $\quad$ and $\quad m = \underline{0.085\ \text{kg}}$

14.7.2. $a_c = 0.03$ m $\qquad S_0 = \pi a\cdot 0.03^2 = 2.83\times 10^{-3}\ \text{m}^2$

$m_c = 0.002$ kg $\qquad S_l = \pi\cdot 0.3^2 = 0.283\ \text{m}^2$

$f_0 = 300$ Hz $\qquad l = 1.0$ m

$s_c = m_c\omega_0^2 = 0.002\cdot(2\pi\cdot 300)^2 = 7100$ N/m $\qquad e^{2\beta\cdot 1} = 100$ so $\beta = 2.30\ \text{m}^{-1}$

(*a*) (14.7.8) $\quad \mathbf{Z}_m = S_0^2\,\mathbf{Z}(0) = R_m + jX_m = \rho_0 c S_0\left[\sqrt{1-(\beta/k)^2} + j\beta/k\right]$

$$X_m = 415 \cdot 2.83 \times 10^{-3} \frac{2.30}{\omega} 343 = \frac{926}{\omega}$$

Resonance when $\left(\omega m_c - s_c/\omega\right) + X_m = 0$

$$0.002\,\omega = \frac{7100 - 926}{\omega} \qquad \omega^2 = 3.09 \times 10^6$$

f = $\underline{280 \text{ Hz}}$

(b) $$\frac{\beta}{k} = \frac{2.30 \cdot 343}{2\pi \cdot 280} = 0.45$$

$$\mathbf{Z}_m = 415 \cdot 2.83 \times 10^{-3}\left[\sqrt{1 - 0.45^2} + j0.45\right] = 1.06 + j0.53 \quad \Omega$$

$$Z_m = 1.18 \ \Omega$$

$$\Pi = R_m U_0^2 = R_m\left(\frac{F}{Z_m}\right)^2 = 1.06 \cdot \left(\frac{5}{1.18}\right)^2 = \underline{19 \text{ W}}$$

14.8.1. See solution for Prob 5.12.11

14.9.2. a = 0.004 m $\qquad x_0$ = 0.001 cm = 1×10^{-5} m

ρ[steel] = 7700 kg/m³ $\quad$ so $\quad \rho_S$ = 7700·1 × 10^{-5} = 0.77 kg/m³

$\mathscr{T}$ = 10^4 N/m² $\qquad V_0$ = 150 V

(a) (4.4.12) and (4.2.3)

$$f = \frac{j_{01}}{2\pi a}\sqrt{\frac{(ScriptT)}{\rho_S}} = \frac{2.4}{2\pi \cdot 0.004}\sqrt{\frac{10^4}{7.7 \times 10^2}} = \underline{34.4 \text{ kHz}}$$

(b) (14.9.3) $\mathscr{M}_o \approx V_0 a^2/8x_0\mathscr{T}$ = 150·0.004²/(8·10^{-5}·10^4) = 3 × 10^{-3} V/Pa

$\mathscr{ML}$ = 20 log(3 × 10^{-3}/10) = $\underline{\text{-70 dB } re \text{ 1 V}/\mu\text{bar}}$

$\mathscr{ML}$ = 20 log(3 × 10^{-3}/1) = $\underline{\text{-50 dB } re \text{ 1 V}/\mu\text{Pa}}$

(14.9.1) C_0 = 27.8·0.004²/10^{-5} = 44.5 pF

$$\mathbf{Z}_{EB} = -j\frac{1}{\omega C_0} = -j\frac{1}{2\pi \cdot 10^4 \cdot 44.5 \times 10^{-12}} = \underline{-j3.6 \times 10^5 \ \Omega}$$

(c) $$ka = \frac{2\pi \cdot 10^4}{343} 0.004 = 0.73$$ and from Fig 14.8.1

$$P_0/P \sim 1.3 \qquad \text{so that} \qquad 20\log 1.3 = 2.3$$

Thus, $\mathscr{ML}$(0°, 10 kHz) = $\mathscr{ML}$ + 2.3 = $\underline{\text{-48 dB } re \text{ 1 V}/\mu\text{Pa}}$

14.11.1. ρ[Al] = 2700 kg/m³ $\qquad B$ = 0.25 T

l = 2.5 cm = 0.025 m $\qquad f$ = 250 Hz so that λ = 1.37 m

w = 0.4 cm = 0.004 m $\qquad P$[peak] = 2 Pa (or P_e = 1.41 Pa)

t = 0.001 cm = 1 × 10^{-5} m $\qquad L$ = 4 cm = 0.04 m [estimated]

S = lw = 0.4·2.5 = 1.0 cm² = 10^{-4} m²

m = ρtS = 2700·10^{-5}·10^{-4} = 2.7 × 10^{-6} kg

(a) From (14.11.3), for normal incidence and $kL = 2\pi fL/c = 0.18 \ll 1$

$$V = \mathcal{M}_o P = \frac{BlSL}{cm} P = \frac{0.25 \cdot 0.025 \cdot 0.04 \times 10^{-4}}{343 \cdot 2.7 \times 10^{-6}} 2 = \underline{5.4 \times 10^{-5}\ \text{V}}\ \text{[peak]}$$

(b) $\mathcal{M}_o = V_e/P_e = V[\text{peak}]/P[\text{peak}] = 5.4 \times 10^{-5}/2 = 2.7 \times 10^{-5}$ V/Pa so $\mathcal{ML}(1\ \text{V/Pa}) = 20 \log(2.7 \times 10^{-5}) = \underline{-91\ \text{dB}\ re\ 1\ \text{V/Pa}}$

(c) From (14.11.2) for $kL \ll 1$, $U = \dfrac{PSkL}{\omega m} = \dfrac{PSL}{cm}$ so that

$$\Xi = \frac{U}{\omega} = \frac{2 \cdot 10^{-4} \cdot 0.04}{2\pi \cdot 250 \cdot 343 \cdot 2.7 \times 10^{-6}} = \underline{5.5 \times 10^{-6}\ \text{m}}$$

14.12.2. $\mathcal{ML} = -40$ dB re 1 V/μbar $\quad S = 0.001\ \text{m}^2$

$V_0 = 12$ V $\quad s = 10^6$ N/m

$R_0 = 100\ \Omega$ $\quad P = 100\ \mu\text{bar} = 10$ Pa

(a) $\mathcal{M}_o = 10^{-40/20} = 0.01\ \text{V}/\mu\text{bar} = 0.1$ V/Pa and use (14.12.2)

$h = \mathcal{M}_o R_0 s/V_0 S = 0.1 \cdot 100 \cdot 10^6/(12 \cdot 0.001) = \underline{8.33 \times 10^8\ \Omega/\text{m}}$

(b) With (14.12.1) $\quad I = \dfrac{V_0}{R} = \dfrac{V_0}{R_0\left(1 + \dfrac{hS}{sR_0}p\right)} \approx \dfrac{V}{R_0}\left[1 - \dfrac{hS}{sR_0}p + \left(\dfrac{hS}{sR_0}p\right)^2\right]$

If $p = P\cos\omega t$ then $p^2 = \dfrac{P^2}{2} + \dfrac{P^2}{2}\cos 2\omega t$

$$\text{Ratio} = \left(\frac{hS}{sR_0}\right)^2 \frac{P^2}{2} \Big/ \left(\frac{hS}{sR_0}\right) P = \frac{1}{2}\frac{hSP}{sR_0}$$

$$= \frac{1}{2}\frac{P}{V_0}(ScriptM)_o = \frac{1}{2}\frac{10}{12}\frac{1}{10} = \underline{0.042}$$

14.13.1. $a = 0.02$ m $\quad \mathcal{T} = 5000$ N/m

$x_0 = 2 \times 10^{-5}$ m $\quad V_0 = 200$ V

Driven with 0.01 A at 1 kHz

(a) (14.9.3) $\mathcal{M}_o = V_0 a^2/8x_0\mathcal{T} = (200 \cdot 0.02^2)/(8 \cdot 2 \times 10^{-5} \cdot 5000) = 0.1$ V/Pa

$\mathcal{ML} = 20 \log 0.1 = \underline{-20\ \text{dB}\ re\ 1\ \text{V/Pa}}$

(b) (14.13.2) $\mathcal{M}_{oR}/\mathcal{S}_{IR} = 2\lambda/\rho_0 c$

$$S_{IR} = 0.1 \Big/ \frac{2 \cdot 343}{1000 \cdot 415} = 60.5\ \text{Pa/A at 1kHz}$$

(14.4.2) $P_{ax}(1) = \mathcal{S}_{IR} I = 60.5 \cdot 0.01 = 0.605$ Pa so that

$$SPL(1) = 20 \log 0.605 = \underline{4\ \text{dB}\ re\ 1\ \text{Pa}}$$

15.3.2. $f = 30$ kHz $\quad SPL = 140$ dB *re* 1 μPa at 1000 m

Figs 8.7.1 and 8.7.2 $\quad a = 7.4$ dB/km $= 7.4 \times 10^{-3}$ dB/m

(*a*) $$SPL = SL - 20\log r - ar$$

$$140 = SL - 20\log 1000 - 7.4\times10^{-3}\cdot 1000 = \underline{207.4 \text{ dB } re \text{ } 1\mu\text{Pa}}$$

(*b*) $$SPL = 207.4 - 20\log 2000 - 7.4\cdot 2 = \underline{127 \text{ dB } re \text{ } 1 \ \mu\text{Pa}}$$

(*c*) $$100 = 207.4 - 20\log r - 7.4\times10^{-3} r$$

$$20\log r + 7.4\times10^{-3} r = 107.4$$

r	20 log r	ar	SPL	
10000	80	74	154	
4000	72	27	99	
5000	74	37	111	
4500	73.1	33.3	106.4	
4600	73.3	33.9	107.2	
4700	73.4	34.6	108	$\underline{r = 4.6 \text{ km}}$

(*d*)

r	20 log r	ar	
100000	190	740	
10000	80	74	
12000	81.6	88.8	
11000	80.83	80.96	
10500	80.42	77.28	
10900	80.79	80.22	
10980	80.81	80.81	$\underline{r = 11.0 \text{ km}}$

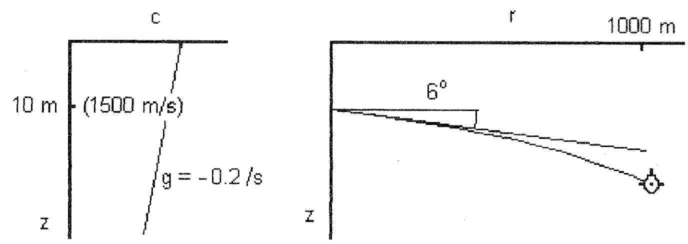
(*e*) $$TL = 20\log r + ar \qquad \log r = \ln r / \ln 10$$

$$\frac{d}{dr}(TL) = \frac{20}{2.3}\frac{1}{r} + a = \frac{8.7}{r} + a$$

Rates of increase are equal when $a = \dfrac{8.7}{r}$

$$r = \frac{8.7}{7.4\times10^{-3}} = \underline{1180 \text{ m}}$$

15.5.3.



(a) $x = 1000\tan 6^\circ = 105 \qquad d = 10 + 105 = \underline{115\ m}$

(b) $R = \dfrac{c}{g\cos\theta} = \dfrac{1500}{0.2\cos 6^\circ} = 7540\ \text{m}$

Use (15.4.3)

$$\Delta r = -7540\,(\sin 6^\circ - \sin\theta) = 1000$$

$$\sin\theta = \frac{1}{7540}(7540\sin 6^\circ + 1000) = 0.2372$$

$$\cos\theta = 0.97147$$

$$\Delta z = -7540(\cos\theta - \cos 6^\circ) = 174$$

$$d = 10 + 174 = \underline{184\ \text{m}}$$

15.5.6. $R = c_0/g = 1500/0.015 = 1\times 10^5$ m $\qquad f = 3$ kHz

$z_s = z_r = 7$ m $\qquad D = 100$ m

$a = 1.66\times 10^{-4}$ dB/m $\qquad b = 6$ dB/bounce

(a) $r_t = \dfrac{1}{8}2\sqrt{2RD}\sqrt{\dfrac{D}{D-z_s}} = \dfrac{D}{4}\sqrt{\dfrac{2D}{D-z_s}} = 25\sqrt{\dfrac{2\cdot 10^5}{93}} = \underline{1159\ \text{m}}$

(b) $r_s = 2\sqrt{2RD} = \underline{8944\ \text{m}}$

(c) Assume $r > r_t$

$$TL = 10\log r + 10\log r_t + (a + b/r_s)r$$

$$80 = 10\log r + 30.6 + (1.66\times 10^{-4} + 6/8944)r$$

$$10\log r + 8.33\times 10^{-4} r = 80 - 30.6 = 49.4$$

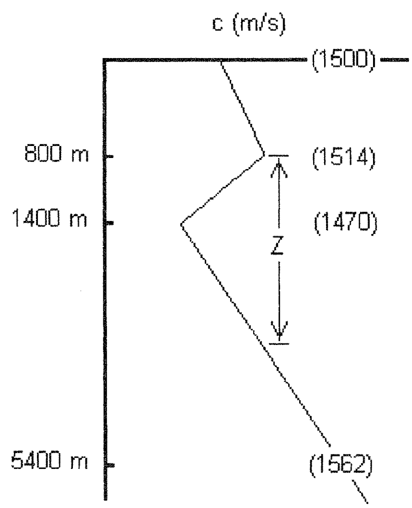
r	$10\log r$ +	8.33×10^{-4} r	= 49.4	
10000	40	8.33	48.3	
11000	40.4	9.2	49.6	$\underline{r = 11\ \text{km}}$

(d) From the discussion at the top of page 442,

$$r_t = D/2\theta_0 \qquad \theta_0 = \frac{100}{2 \cdot 1159} = 0.045 \text{ rad} = 2.47^\circ$$

So $\underline{2\theta_0 = 4.9^\circ}$

15.6.2.



$$g_1 = \frac{1470 - 1514}{1400 - 800} = -0.0733 \text{ s}^{-1}$$

$$g_2 = \frac{1562 - 1470}{5400 - 1400} = 0.0230 \text{ s}^{-1}$$

(*a*) By inspection, $\underline{1400 \text{ m}}$

(*b*)
$$\frac{\cos\theta_{max}}{1470} = \frac{1}{1514} \qquad \cos\theta_{max} = 0.97094$$

$$\theta_{max} = 0.24 \text{ rad} = \underline{13.8^\circ}$$

Cycle distance from (15.4.3)

Upper half-channel
$$\Delta r_1 = -\frac{1}{g_1}\frac{c(1400)}{\cos\theta_{1400}} = [\sin 13.8^\circ - \sin(-13.8^\circ)]$$

$$= \frac{1}{0.0733}\frac{1470}{0.97094}[2 \cdot 0.2385] = 9850 \text{ m}$$

Lower half channel
$$\Delta r_1 = -\frac{1}{g_2}\frac{c(1400)}{\cos\theta_{1400}} = [\sin(-13.8^\circ) - \sin 13.8^\circ]$$

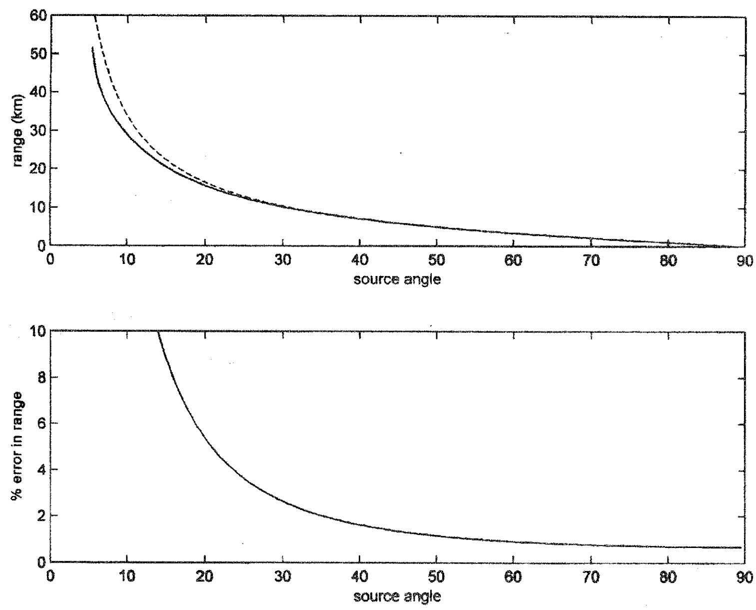
$$= -\frac{1}{0.0230}\frac{1470}{0.97094}[2 \cdot (-0.2385)] = 34.1 \text{ km}$$

$$\Delta r_1 + \Delta r_2 = 9.9 + 31.4 = \underline{41.3 \text{ km}}$$

(*c*) Use the equation in the line above (15.6.5) and note that $\theta_0 = \theta_{max}$

$$D_s = 1400 - 800 = 600 \text{ m}$$
$$D_r = (1514 - 1470)/0.0230 = 1910 \text{ m}$$
$$r_t = \frac{D_s + D_r}{2\theta_{\max}} = \frac{2510}{2\cdot 0.24} = \underline{5200 \text{ m}}$$

15.6.5C.



15.7.2. The [. . .]$^{1/2}$ term in (15.7.4) is the factor by which the pressure amplitude from surface interference reduces the free-field spherical spreading from the source. Expressing this factor as a *transmission anomaly TA*, the (positive) dB reduction in level, we have $TA = 20\log[1 + \mu^2 - 2\mu\cos(2khd/r)]^{1/2}$ Setting $\mu = 1$,

$$TA = -10\log\left[2\left(1 - \cos\frac{2khd}{r}\right)\right] \quad \text{But} \quad 1 - \cos\beta = 2\sin^2\frac{\beta}{2} \quad \text{so that}$$

$$TA = -10\log\left(4\sin^2\frac{khd}{r}\right) \quad \text{for} \quad \mu = 1$$

(*a*) $\log\left(4\sin^2\dfrac{khd}{r}\right) = -1$ or $\sin\dfrac{khd}{r} = \pm 0.158$ gives 10 dB reduction

Letting r decrease from infinitely large values, the smallest value of khd/r that satisfies the positive root gives the largest value of r beyond which the discrepancy always exceeds 10 dB. Using the small argument approximation,

$$\frac{khd}{r} = 0.158 \quad \text{or} \quad \underline{r = 6.33\,khd}$$

(*b*) $k = \frac{2\pi f}{c} = \frac{2\pi\cdot 500}{1500} = 2.09$ and $r = 6.33\cdot 2.09\cdot 10\cdot 20 = \underline{2640\ \mathrm{m}}$

15.8.1. $SL = 220$ dB *re* 1 μPa $\quad f = 1$ kHz
$EL = 110$ dB *re* 1 μPa $\quad r = 1000$ m

$$SL - 2TL + TS = EL$$

$$TL = 20\log r + ar = 20\log 1000 + 5.25\times 10^{-5}\cdot 1000 = 60\ \mathrm{dB}$$

$$TS = 110 - (220 - 2\cdot 60) = \underline{10\ \mathrm{dB}}$$

15.8.6. (*a*)

$$DT_e = 5\log\frac{d}{(w\tau)} \qquad DT_e' = 5\log\frac{d}{(w\tau)'} = DT_e' - 5$$

Thus $DT_e' - DT_e = 5\log\frac{(w\tau)'}{(w\tau)} = -5$

and $(w\tau)' = 10\,(w\tau)$ $\quad$ $\underline{10\ \text{times larger}}$

(*b*)

$$DT_c = 10\log\frac{d}{2(w\tau)} \qquad DT_c' = 10\log\frac{d}{2(w\tau)'} = DT_c' - 5$$

Thus $DT_c' - DT_c = 10\log\frac{(w\tau)'}{(w\tau)} = -5$

and $(w\tau)' = 3.16\,(w\tau)$ $\quad$ $\underline{3.16\ \text{times larger}}$

15.10.3. $f = 250$ Hz $\quad r = 10$ km
$SL = 150$ dB *re* 1 μPa $\quad P(FA) = 0.2\,\% = 2\times 10^{-3}$
$SS = 3$ $\quad$ – – – – –
$|\dot{R}| \le 14.7$ m/s $\quad$ (*e*) 5 processors (1 *Hz* each) same *overall* $P(FA)$

(*a*) (15.9.13) $\quad w_p = 1.33\dot{R}F = 1.33\cdot 14.7\cdot 0.250 = \underline{4.9\ \mathrm{Hz}}$

(*b*) $DNL = NSL + 10\log w - DI = 68 + 10\log 4.9 - 0 = \underline{75\ \mathrm{dB}\ re\ 1\ \mu\mathrm{Pa}}$

(*c*) $TL = 20\log 10^4 + 5.84\times 10^{-6}\cdot 10^4 = 80\ \mathrm{dB}$

$SL - TL = DNL + DT$

$DT = 150 - 80 - 75 = \underline{-5\ \mathrm{dB}}$

(*d*) Fig 11.4.2 $\quad d = (d')^2 \approx 10$ $\quad$ for Gaussian distributions, equal std dev

$$DT_e = 5\log\frac{d}{w\tau} = -5 \qquad \frac{w\tau}{d} = 10 \qquad \tau = \frac{10\cdot 10}{4.9} = \underline{20\ \mathrm{s}}$$

(*e*) $DNL = 68 + 10\log 1 - 0 = \underline{68\ \mathrm{dB}\ re\ 1\ \mu\mathrm{Pa}}$

$DT = 150 - 80 - 68 = \underline{2\ \mathrm{dB}}$

$P(FA)$ for each processor is now $2\times 10^{-3}/5 = 4\times 10^{-4}$ and $d \approx 13$

$$DT_e = 5\log\frac{13}{1\cdot\tau} = 2 \qquad \frac{13}{\tau} = 2.5 \qquad \tau = \underline{5.2\ \mathrm{s}}$$

15.10.7. 90 beams 20 parallel processors $w = 0.1$ Hz $\tau = 10$ s

(a) (90 beams)·(10 s per beam) = 900 s = <u>15 min</u>

(b) (90 beams)·(20 freq intervals/beam) = <u>1800 frequency-bearing cells</u>

(c) 1 FA/hr $\rightarrow$ 1 FA per 4 searches so $P(FA) = 1/(4\cdot 1800) = \underline{1.39 \times 10^{-4}}$

(d) From Fig 11.4.2 $d = (d')^2 = 14.4$

$$DT_e = 5\log\frac{d}{w\tau} = 5\log\frac{14.4}{0.1\cdot 10} = \underline{5.8\ \text{dB}}$$

15.11.4. $DNL(noise) = 80$ dB *re* 1 μPa

$RL = 113 - 10\log r - 10^{-4}r$

(a)

r	$DNL(n)$	RL	$DNL(n) \oplus RL$
500	80	<u>86.0</u>	<u>87.0 dB *re* 1 μPa</u>
1000	80	<u>82.9</u>	<u>84.7</u>
2000	80	<u>79.8</u>	<u>82.9</u>
5000	80	<u>75.5</u>	<u>81.3</u>
10000	80	<u>72.0</u>	<u>80.6</u>
20000	80	<u>68.0</u>	<u>80.2</u>

(b) $r \leq \underline{2000\ \text{m}}$

15.12.1. See Prob. 9.8.8 for the proof. The discrepancy between this TL and that given by (15.12.7) is $10\log(H/2) - 10\log(H/\pi) = 10\log(\pi/2) = \underline{2\ \text{dB}}$

15.12.4. $\rho = 1.79 \times 10^3$ kg/m^3 $\cos\theta_c = 1500/1540$

$c = 1.54 \times 10^3$ m/s $\theta_c = 0.23$ rad

$H = 90$ m

Evaluating (15.12.6) before making the approximation leading to (15.12.11) gives

$S' = \dfrac{r}{2H}\displaystyle\int_{2H/r}^{\theta_c} d\theta = \dfrac{r}{2H}\theta_c - 1$ rather than $S = \dfrac{r}{2H}\theta_c$ so that $S = S' + 1$.

For TL to be accurate within 1 dB at range r, (15.12.1) shows that we must have $(1+2S) \leq 1.1\cdot(1+2S')$ or $S \geq 5$ and thus $\underline{r \geq 4\ \text{km}}$.

16.2.1. Assume $\mathbf{p} = P e^{j(\omega t - \mathbf{k}x)}$ with $\mathbf{k} = k - j\alpha$

For the linearized solution to (16.2.12), we use

$$c^2\Box_L^2\mathbf{p} = c^2\left(1 + \tau\frac{\partial}{\partial t}\right)\frac{\partial^2\mathbf{p}}{\partial x^2} - \frac{\partial^2\mathbf{p}}{\partial t^2} = 0$$

Substitute the assumed solution, expand, and collect real and imaginary results

(1) $k^2 - \alpha^2 + 2\omega\tau\alpha k = (\omega/c)^2$

(2) $k^2 - \alpha^2 - 2\alpha k/\omega\tau = 0$

Elimination of $(k^2 - \alpha^2)$ between (1) and (2) gives

$$2\omega\tau\alpha k + 2\alpha k/\omega\tau = (\omega/c)^2$$

$$\alpha k = \left(\frac{\omega}{c}\right)^2 \frac{\omega\tau}{2[1+(\omega\tau)^2]} = \left(\frac{\omega}{c}\right)^2 \frac{\omega\tau}{2}[1-(\omega\tau)^2+\cdots]$$

and through $O(\omega\tau)$ we have $\quad \underline{\alpha k = \frac{1}{2}\left(\frac{\omega}{c}\right)^2 \omega\tau}$

Then, substituting this result into (1) and retaining terms correct through $O(\omega\tau)^2$

$$k^2 \approx \left(\frac{\omega}{c}\right)^2 + \alpha^2 - \left(\frac{\omega}{c}\right)^2(\omega\tau)^2 \approx \left(\frac{\omega}{c}\right)^2 + \left[\frac{1}{4}\left(\frac{\omega/c}{k}\right)^2\left(\frac{\omega}{c}\right)^2 - \left(\frac{\omega}{c}\right)^2\right](\omega\tau)^2$$

$$\approx \left(\frac{\omega}{c}\right)^2\left[1-\frac{3}{4}(\omega\tau)^2\right]$$

and

$$k = \frac{\omega}{c}\left[1-\frac{3}{8}(\omega\tau)^2+\cdots\right]$$

Thus, through $O(\omega\tau)$ $\quad \underline{\alpha = \omega^2\frac{\tau}{2c}} \quad$ and $\quad \underline{k = \frac{\omega}{c}}$

16.2.2. (*a*) First, let $\tau = 0$ so that (16.2.12) becomes

$$\left(c^2\frac{\partial^2}{\partial x^2} - \frac{\partial^2}{\partial t^2}\right)\mathbf{q} = A\sin mkx\, e^{jn\omega t} \qquad \text{and assume} \qquad \mathbf{q} = \mathbf{B}'\sin mkx\, e^{jn\omega t}$$

substitution gives

$$\left[-c^2(mk)^2 + (n\omega)^2\right]\mathbf{B}' = A$$

$$\mathbf{B}' = \frac{A/\omega^2}{n^2-m^2} \rightarrow \infty \qquad \text{when} \qquad n = m$$

Now with losses assume $\quad \mathbf{q} = \mathbf{B}\sin m\mathbf{k}x\, e^{jn\omega t} \quad$ where $\quad \mathbf{k} = k - j\alpha$

$$c^2\left(1+\tau\frac{\partial}{\partial t}\right)\frac{\partial^2\mathbf{q}}{\partial x^2} - \frac{\partial^2\mathbf{q}}{\partial t^2} = A\sin m\mathbf{k}x\, e^{jn\omega t}$$

$$\left[-c^2(m\mathbf{k})^2 - jn\omega c^2(m\mathbf{k})^2 + (n\omega)^2\right]\mathbf{B} = A$$

Expanding $\mathbf{k}$ and collecting terms

$$\left[\!\left[c^2\left\{-\left[(mk)^2+(m\alpha)^2\right] - 2n\omega\tau(m\alpha)(mk) + \left(\frac{n\omega}{c}\right)^2\right\} + jc^2\left\{2m\alpha\, mk - n\omega\tau\left[(mk)^2-(m\alpha)^2\right]\right\}\right]\!\right]\mathbf{B} = A$$

Discard terms of order higher than $O(\omega\tau)$ and use $\quad \omega/k = c$

$$(ck)^2\left[(n^2-m^2) + j(2m^2\alpha/k - n\omega\tau m^2)\right]\mathbf{B} = A$$

$$\mathbf{B} = \frac{A}{(mkc)^2\left[(n/m)^2 - 1 + j(2\alpha/k - n\omega\tau)\right]}$$

B sharply peaks and has maximum amplitude when $n = m$

(*b*) Thus, q is significant only when $n \approx m$ and this means that the above term of amplitude A on the RHS of (16.2.12) generates a significant particular solution only when $n \approx m$. Thus, all the nonlinearly generated terms important on the RHS of (16.2.10) must have $n \approx m$ for all admissable values of n and m. Given q, each v is related through the velocity potential, so each relevant term in v has the same $n \approx m$ as the associated q. So

$$c^2\nabla^2 f \approx \frac{\partial^2 f}{\partial t^2} \approx -(n\omega)^2 f$$

for all important terms in the RHS of (16.2.10) and this gives

$$c^2\Box_L^2 q = -\frac{1}{2}\frac{\partial^2}{\partial t^2}(\gamma q^2 + v^2 - q^2 + v^2)$$

$$= -\frac{1}{2}\frac{\partial^2}{\partial t^2}\left[(\gamma - 1)q^2 + 2v^2\right]$$

(1) If $q = M\cos(n\omega t \pm nkx)$ then $v = \pm M\cos(n\omega t \pm nkx)$ and

$$\frac{\partial^2 q^2}{\partial t^2} = \frac{\partial^2 v^2}{\partial t^2}$$

(2) If $q = M\cos n\omega t \sin nkx$ then $v = M\sin n\omega t \cos nkx$ and again we have the second partials equal.

By extension (or example) all choices of cartesian standing and traveling waves with $n \approx m$ lead to the same result so that

$$\underline{c^2\Box_L^2 q = -\frac{\gamma + 1}{2}\frac{\partial^2 q^2}{\partial t^2}}$$

16.3.4. (*a*) From (5.11.13) and (5.11.19), for $ka > 5$ we can assume that the particle velocity amplitude falls off as $1/r$. If $\vec{\mathbf{u}} = u(r,t)\,\hat{r}$ describes the particle velocity field initiated at the surface of the source of radius $r = a$ then

$$u = \frac{Ua}{r}\sin\left[\omega\left(t - \frac{r - a}{c + \beta u}\right)\right] \qquad \text{for } kr > 5$$

Differentiating with respect to r gives

$$\frac{\partial u}{\partial r} = -\frac{Ua}{r^2}\sin\left[\cdots\right] + \frac{Ua}{r}\cos\left[\cdots\right]\frac{\partial}{\partial r}\left[\cdots\right]$$

and evaluation at $u = 0$ results in

$$\left(\frac{\partial u}{\partial r}\right)_{u=0} = -\frac{Ua}{r}\,\omega\left[\frac{\partial}{\partial r}\left(\frac{r-a}{c+\beta u}\right)\right]_{u=0} = -\frac{Ua}{r}\,\omega\left[\frac{1}{c+\beta u} - \frac{r-a}{(c+\beta u)^2}\,\beta\,\frac{\partial u}{\partial r}\right]_{u=0}$$

$$= -\frac{Ua}{r}\,\omega\left[\frac{1}{c} - \beta\frac{r-a}{c^2}\left(\frac{\partial u}{\partial r}\right)_{u=0}\right]$$

Solving for the partial derivative,

$$\left(\frac{\partial u}{\partial r}\right)_{u=0} = -\frac{\omega/c}{\dfrac{r}{Ua} - \dfrac{\omega}{c^2}\beta(r-a)}$$

The discontinuity distance ℓ is reached when the denominator vanishes. The value of r at that distance is $r = \ell + a$ where ℓ is measured from the source surface at $r = a$ Defining the mach number $M = U/c$ of the source surface and solving for ℓ results in

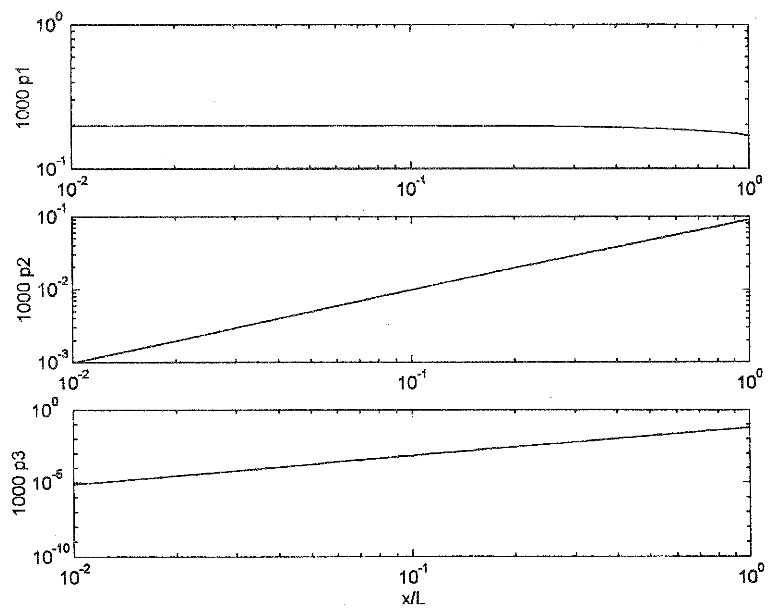
$$\ell \to \frac{a}{\beta Mka - 1}$$

(*b*) $\beta Mka \gg 1$ is equivalent to $ka \gg \dfrac{1}{\beta M}$ and in this limit, $\ell \to \dfrac{1}{\beta Mk}$

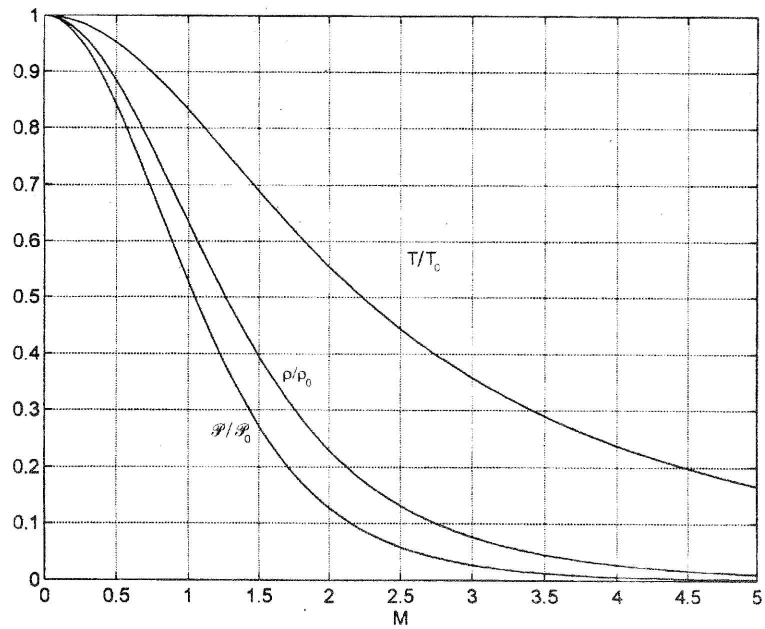
(*c*) Using (16.3.7) gives $\Gamma = \dfrac{M\beta}{\alpha/k} = \dfrac{1}{\alpha}\left(\dfrac{1}{\ell} + \dfrac{1}{a}\right)$

(*d*) From $ka > 5$ for (*a*), at 1 kHz we have $a \geq 5\dfrac{c}{\omega} = \dfrac{5\cdot 343}{2\pi\cdot 1000} = 0.27$ m

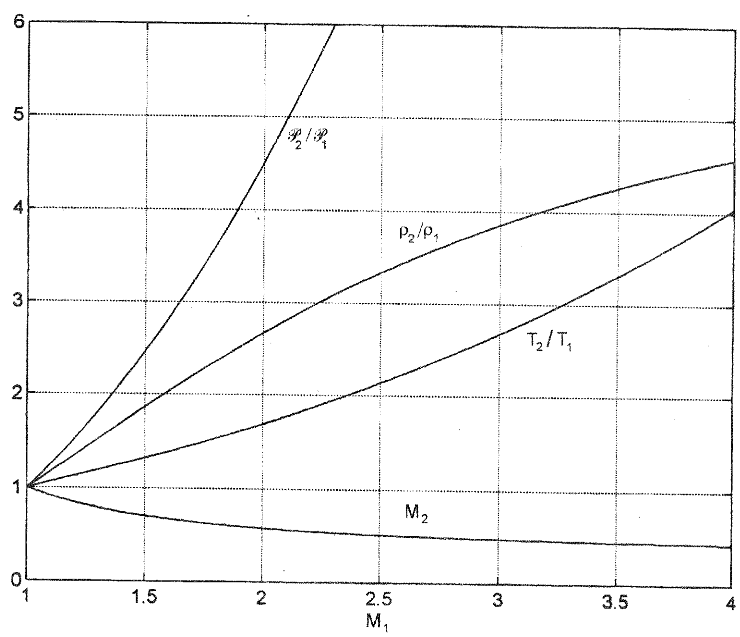
16.5.2C.



17.1.1C.



17.1.2C.



17.1.3. $p \;=\; 800$ mbar $\qquad \mathscr{P}_1 \;=\; 985$ mbar

$T_1 \;=\; 273 + 22 \;=\; 295$ K $\qquad c_1 \;=\; (1.4 \cdot 287 \cdot 295)^{1/2} \;=\; 344$ m/s (*a*) (17.1.14) $$\frac{(ScriptP)_2}{(ScriptP)_1} = \frac{985 + 800}{985} = \frac{7M_1^2 - 1}{6} \qquad \underline{M_1 = 1.302}$$

(*b*) (17.1.12) $$\frac{T_2}{T_1} = \frac{(M_1^2 + 5)(7M_1^2 - 1)}{36M_1^2} = 1.192 \qquad T_2 = \underline{351.8K}$$

(*c*) (17.1.16) $$\frac{u_b}{c_1} = \frac{5(M_1^2 - 1)}{6M_1} = 0.445 \qquad u_b = 153.2 \text{ m/s}$$

(5.6.5) or (17.1.4) $$c_2 = 344\sqrt{\frac{351.8}{295}} = 375.7 \text{ m/s}$$

(17.1.6) $$\mathscr{P}_0 = \mathscr{P}_2(1 + M_b^2/5)^{7/2} = (\mathscr{P}_1 + p)[1 + (u_b/c_2)^{7/2}/5]$$

$$= (985 + 800)\cdot\left[1 + \frac{1}{5}\left(\frac{153.2}{375.7}\right)^2\right]^{7/2} = \underline{2002 \text{ mbar}}$$

stagnation overpressure $p_{stag} = \mathscr{P} - \mathscr{P}_1 = 2002 - 985 = \underline{1017 \text{ mbar}}$

(*d*) (17.1.15) $$\frac{\rho_2}{\rho_1} = \frac{6M_1^2}{M_1^2 + 5} = \underline{1.52}$$

(*e*) From Appendix A10(*c*), $c_{\mathscr{P}} = 1.01 \times 10^3$ J/(kg·K) For (A9.24) we need $C_V = c_{\mathscr{P}}M/\gamma$ (here M is the molecular weight). Then, from A1 and A10,

$$C_V = \frac{1.01 \times 10^3 \cdot 28.964}{1.4} = 2.09 \times 10^4 \text{ J/(kmol·K)} = 20.9 \text{ J/(mol·K)}$$

$$S_2 - S_1 = 20.9\left[\ln\left(\frac{985 + 800}{985}\right) + 1.4\ln\left(\frac{1}{1.52}\right)\right] = \underline{0.1742 \text{ J/(mol·K)}}$$

or $\Delta S = 0.1742 \cdot 1000/28.96 = \underline{6.02 \text{ J/(kg·K)}}$

17.3.6. $T = 273 + 25 = 298$ K $\qquad \tau = 2.0$ ms

$\mathscr{P}_1 = 0.950$ bar $\qquad \mathscr{J}/S = 0.45$ bar·ms

$p(0) = 0.80$ bar $\qquad c_1 = 343(298/293)^{1/2} = 346$ m/s

(*a*) Solve (17.2.3) by trial and error for b with $(\mathscr{J}/S)/[p(0)\tau\} = 0.281$

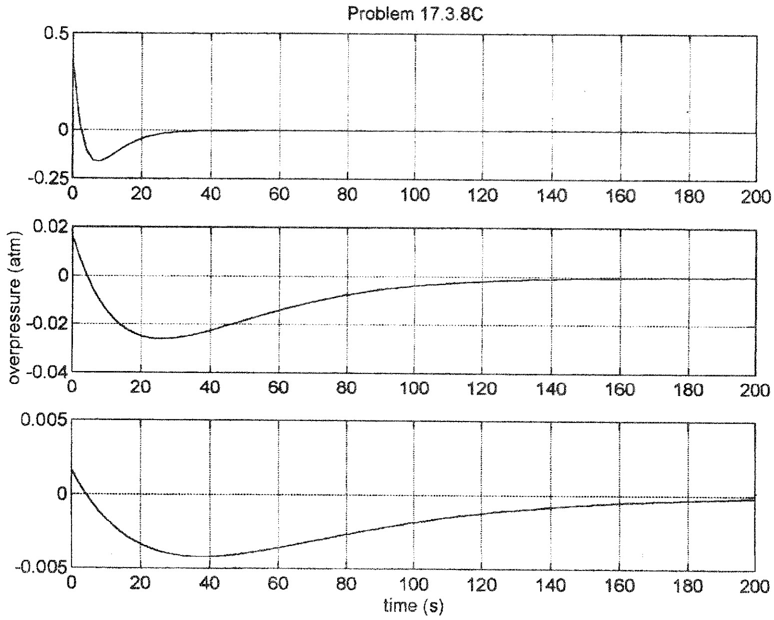
$\underline{b = 2.0}$

(*b*) (17.3.2) $$M_1 = \sqrt{\frac{6}{7}\frac{0.80}{0.950} + 1} = \underline{1.31}$$

(*c*) (17.1.16) $$u_b = 346\frac{5(1.31^2 - 1)}{6\cdot 1.31} = \underline{158 \text{ m/s}}$$

(*d*) (17.1.12) $$T_2 = T_1\frac{(M_1^2 + 5)(7M_1^2 - 1)}{36M_1^2} = 298\cdot 1.197 = \underline{357 \text{ K}} = 84°\text{C}$$

17.3.8C.



17.3.11. (*a*) Taking the limit for large R in (17.3.1)

$$\frac{P(0)}{(ScriptP)_1} \to \frac{9.7}{R^{9/4}}\left[\left(\frac{R}{7.2}\right)^3\right]^{5/12} \propto \frac{R^{5/4}}{R^{9/4}} = \underline{\frac{1}{R}}$$

(*b*) Taking the limit for large R in (17.3.5)

$$\langle u_1\rangle_R \to \frac{1960}{R^{9/10}}\left[\left(\frac{R}{5.2}\right)^{14/5}\right]^{2/7}\left[\left(\frac{R}{70}\right)^2\right]^{1/20} = 1960\frac{1}{5.2^{4/5}}\frac{1}{70^{1/10}} = \underline{343\ \text{m/s}}$$

(*c*) Yes. Both are consistent with the linear limit.

17.5.3. t_a = 11.01 s for small explosions and 10.55 s for large explosion

(*a*) Assume M_1 = 1 for small explosions

$$r = c_1 t_a = 343 \cdot 11.01 = \underline{3776\ \text{m}}$$

(*b*) $\langle u_1\rangle_R = \dfrac{3776}{10.55} = 358.0$ m/s Solve (17.3.5) and get R = 60 m

$$r = W^{1/3}R \qquad W = \left(\frac{3776}{60}\right)^3 = 2.5\times 10^5 \text{ kg TNT} = 250 \text{ tonne TNT}$$

With cratering, $W(\text{true}) \sim 1.5\,W$ = $\underline{374 \text{ tonne TNT}}$

(*c*) (17.3.1) $p(0) = \underline{1.37\times10^{-2}\ \text{bar}} = 1.39\times10^9\ \mu\text{Pa}$

(*d*) (17.3.2) and (17.1.16)

$$M_1 = \sqrt{\frac{6}{7}\frac{0.0137}{1.0325} + 1} = 1.00578$$

$$u_b = 343\,\frac{5(1.00578^2 - 1)}{6 \cdot 1.00578} = \underline{3.29\ \text{m/s}}$$

or, for M_1 very close to 1 and approximating $\mathscr{P}_1 \approx 1.0$ bar

from (17.3.2) $\quad M_1^2 - 1 = \dfrac{6}{7}\dfrac{p(0)}{(ScriptP)_1} \approx \dfrac{6}{7}\dfrac{0.0137}{1} = 0.0117$

and then from (17.1.16) $\quad u_b \approx 343\,\dfrac{5 \cdot 0.0117}{6 \cdot 1} = \underline{3.34\ \text{m/s}}$

ANSWERS TO EVEN-NUMBERED PROBLEMS

1.3.2. (*a*) -75 cm/s^2. (*b*) 3.0 cm. (*c*) 15 cm/s.
1.4.2. (*a*) $\frac{1}{2}(m + m_s/3)U^2$. (*b*) $(m + m_s/3)d^2x/dt^2 + sx = 0$
1.5.2. (*a*) $(x^2 + y^2)^{1/4}\cos[\frac{1}{2}\tan^{-1}(y/x) + n\pi]$, $(x^2 + y^2)^{1/4}$, $\frac{1}{2}\tan^{-1}(y/x) + n\pi$.
(*b*) $A\cos(\omega t + \phi)$, A, $\omega t + \phi$. (*c*) $2\cos\theta$, $2\cos\theta$, 0.
1.5.4. (*a*) $\sqrt{(x^2 + y^2)}$. (*b*) $\sqrt{(X^2 + Y^2)}$. (*c*) $\sqrt{[(x^2 + y^2)(X^2 + Y^2)]}$. (*d*) $xX - yY$.
(*e*) $\tan^{-1}[(xY + Xy)/(xX - yY)]$. (*f*) $(xX + yY)/(X^2 + Y^2)$
1.6.4. $+u_0/\omega_d$ if ϕ chosen as $-\pi/2$ ($-u_0/\omega_d$ if ϕ chosen as $+\pi/2$).
1.7.2. $\omega_0 = \sqrt{(s/m)}$, $\omega \to \infty$, $\omega \to 0$, $\omega_\pm = [\omega_0^2 + (R_m/2m)^2]^{1/2} \pm R_m/2m$.
1.7.4. (*a*) $d^2x/dt^2 + \omega_0^2 x = \frac{1}{2}\omega_0^2 at^2$. (*c*) $\frac{1}{2}X\omega_0^2 = F/m$.
1.7.6C. (*b*) 2.04, 2.48 rad/s (*c*) 0.707.
1.8.2. (*a*) $[F/(\omega m - s/\omega)](\cos\omega_0 t - \cos\omega t)$.
1.10.2. (*a*) 2.3 cm, 11.5 cm/s, 9.1 mW. (*b*) $-85.4°$.
(*c*) 2.25 Hz, 10.1 cm, 1.43 m/s, 1.43 W. (*d*) 5.05, between 2.04 and 2.49 Hz.
1.10.4. $\sqrt{[\omega_0^2 + (R_m/2m)^2]} \pm R_m/2m$ simplifies to $\omega_0 \pm R_m/2m$ for $R_m/2m << \omega_0$.
1.12.2. (*a*) $j(\omega m - s/\omega)$, $\sqrt{(s/m)}$. (*b*) $R + j\omega m$, 0. (*c*) $R - js/\omega$, ∞.
1.15.2. (*a*) $\mathscr{F}\delta(t)/R_m$, $\mathscr{F}\cdot 1(t)/R_m$.
1.15.4. $(\mathscr{F}/\omega_0 m)(\sin\omega_0 t)\cdot 1(t)$, $(\mathscr{F}/m)(\cos\omega_0 t)\cdot 1(t)$.
1.15.6. (*a*) $(F/\omega_0^2 m)(1 - \cos\omega_0 t)\cdot 1(t)$, $(F/\omega_0 m)(\sin\omega_0 t)\cdot 1(t)$.
1.15.8. $(1/2\pi)\omega/[\omega^2 - (w - jb)^2]$
1.15.12. (*b*) $\Delta t/2\pi$.

2.8.2. $2A/(1 + \sqrt{2})$
2.9.2. $R_m + \rho_L c + j(\omega m - s/\omega)$
2.9.4. (*a*) 112 Hz. (*b*) 0.97 N.
2.10.2. (*a*) c, $f/2$. (*b*) $c/2$, $f/2$. (*c*) $c/\sqrt{2}$, $f/\sqrt{2}$. (*d*) $c/\sqrt{2}$, $f/\sqrt{2}$. (*e*) $c/2$, $f/2$.
2.11.2. (*a*) 6.26 m/s. (*b*) 0.48, 3.21 Hz. (*c*) 12.9.
2.11.4. (*a*)-(*c*) 54.2, 158, 183 Hz.
2.12.2. (*a*) 15 m/s, 0.48 Hz, 0.2 cm^{-1}. (*b*) 0, 0; 2 cm, 6 cm/s.
(*c*) $-1.8\cos^2 3t$, $1.8\sin^2 3t$. (*d*) 2.83×10^{-6} J.
2.13.2. (*a*) No. (*b*) Yes.

3.4.2. All form orthogonal sets.
3.5.2. (*a*), (*b*) 157.2, 156.7 Hz where $Y = 1.95 \times 10^{11}$ has been used.
3.6.2. (*a*) 5.37 kHz. (*b*) 12.4 cm from 27 g mass. (*c*) 1.53.
3.6.4. $\tan kL = sL/\omega mc$
3.8.2. (*a*) $t/\sqrt{12}$. (*b*) $w/\sqrt{12}$. (*c*) $wt/[(w^2 + t^2)^{1/2}\sqrt{6}]$ im.
3.9.2. 328 kHz
3.11.2. $f_n/f_1 = 1, 2.76. 5.40$; $v_n/v_1 = 1, 1.66, 2.32$; $\lambda_n = 132.8, 80.0, 57.1$ cm;
nodes at 0, 100; 0, 50, 100; 0, 35.6, 64.4, 100 cm.
3.11.4C. (*a*) See answers to Prob. 3.11.2.
3.12.2. (*a*) -1.362, -0.982, -1.000, ... (*b*) -1.018, -1.000, -1.000, ...

3.13.2. 1.63 (or 1.72), depending on which values from A10*a*.

4.3.2. (1,2) 1.265, (1,3) 1.612, (2,1) 1.844, (1,4) 2.000, (2,2) 2.000
4.3.4. (*a*) $A_{nm}\sin(n\pi x/L)\cos(m\pi z/L)$, $n = 1, 2, 3, \ldots, m = 0, 1, 2, \ldots$.
(*b*) $f/(c/L)$ = (1,0) 0.5, (1,1) 0.707, (2,0) 1.00, (2,1) 1.12, (1,2) 1.12.
4.4.2. (*a*) $A_m J_m(j'_{mn}r/a)\cos(m\theta)\exp(j\omega_{mn}t)$, $\omega_{mn} = j'_{mn}c/a$, $c = \sqrt{(\mathcal{T}/\rho_S)}$, $k_{mn}a = j'_{mn}$.
(*c*) $f/\sqrt{(\mathcal{T}/\rho_S)}$ = 0.293 (1,1), 0.486 (2,1), 0.610 (0,2).
4.4.4. (*a*) 243 (0,1) , 386 (1,1), 517 (2,1), 556 (0,2) Hz. (*b*) (0,2) 10.87 cm.
4.5.2. 0.435.
4.7.2.

B	ka_1	ka_2	ka_3
0	2.40	5.52	8.65
1	2.54	5.54	8.66
2	2.67	5.55	8.67
5	3.02	5.59	8.68
10	3.49	5.68	8.69

Lower frequencies shifted upward more strongly than higher frequencies.
4.9.2. (*a*) 4000 N/m. (*b*) 6.95 kHz. (*c*) 2.84×10^{-8} m. (*d*) 1.41×10^{-8} m.
4.9.4. (*c*) Identical with $\omega_1 = \omega_0$ and the same Q's.
4.10.8. With $k_n a = j_{0n}$ and $\omega_n = k_n c$,

$$y(r,\theta,t) = \frac{1}{\pi a^2}\sum_n \frac{J_0(k_n r)}{\omega_n\left[J_1(k_n a)\right]^2}\sin\omega_n t$$

4.11.2. 6360 N/m.
4.11.4. 1.23 kHz.

5.2.2. $r = \mathcal{R}/(\rho V/n)$, 8314.3 J/(kmol·K).
5.2.4. (*a*) 28.965 kg/kmol. (*b*) 287.0 J/(kg·K)
5.4.2. (*b*) $-\nabla\mathcal{P} = \rho(\partial\vec{u}/\partial t) + (\vec{u}\cdot\nabla)\vec{u}$. (*c*) Infinite.
5.5.2. (*c*) For $\vec{u} = u(r,t)\hat{r}$

$$\frac{\partial^2 u}{\partial r^2} + \frac{2}{r}\frac{\partial u}{\partial r} - \frac{2}{r^2}u = \frac{1}{c^2}\frac{\partial^2 u}{\partial t^2}$$

$$\frac{\partial^2 p}{\partial r^2} + \frac{2}{r}\frac{\partial p}{\partial r} = \frac{1}{c^2}\frac{\partial^2 p}{\partial t^2}$$

5.6.2. (*a*) 1509.4 m/s. (*b*) 2.35 m/s·C°.

5.7.2. Fractional changes in ρ_0 must be small over distances of a wavelength.
5.7.6. $|(1/\rho_0)(\partial\rho_0/\partial t)| \ll \omega |s|$
5.9.2. (*a*) $\Delta T = [(\gamma - 1)/\gamma](T_0/\mathscr{P}_0)p$. (*b*) 0.076 C°.
5.11.2. 89.0, 79.6, 28.6, 3.12°; 7.60, 74.8, 364, 414 Pa·s/m.
5.11.6C. 2.1
5.12.2. (*a*) 2.83×10^{-3} Pa. (*b*) 9.64×10^{-9} W/m^2. (*c*) 6.82×10^{-6} m/s.
(*d*) 2.40×10^{-8} kg/m^3.
(*e*) 6.35×10^{-9} m. (*f*) 1.98×10^{-8}.
5.12.4. (*a*), (*b*) 180 dB *re* 1 μPa, 154 dB *re* 20 μPa.
5.12.6. (*a*) 2.92×10^{-8} J/m^3, 0.0632 Pa. (*b*) 4.55×10^{-5} J/m^3, 316 Pa.
5.12.8. (*a*) 13.0 kW/m^2. (*b*) 123 dB *re*1 μbar. (*c*) 8.66×10^{-5}. (*d*) $z = 10$ m.
5.12.10. 160 dB *re* 1 μPa/V, 134 dB *re* 20 μPa/V.
5.12.12. - 106 dB *re* 1 V/μbar, - 180 dB *re* 1 V/20 μPa.
5.13.2C. 1.29
5.13.4. (*a*) - 0.500. (*b*) - 0.293. (*c*) 0
5.14.2. (*a*) $[(\sin\theta_0)/\varepsilon]\sin(\varepsilon x/\cos\theta_0)$. (*b*) $(c_0\cos\theta_0)/(1 - \frac{1}{2}\sin^2\theta_0)$., $1 - \frac{1}{8}\theta_0^4$.
(*c*) 1%. (*d*) 1475, 1475, 1475, 1475, 1474.8 m/s.
(*e*) 0, 116, 233, 581, 1158 m; 0, 22.4, 22.4, 22.4, 22.1 km. (*f*) The higher sound speeds encountered by the steeper rays almost exactly counteract the longer ray paths so the times of flight remain nearly the same.
5.14.6. 5.66×10^{-8}, 5.66×10^{-8}
5.14.12. (*a*) 13.7°. (*b*) 1.67 km.
5.15.2. (*a*) $u_i\partial f/\partial x_i$. (*b*) $\partial(u_i\partial u_j/\partial x_i)/\partial x_j$. (*c*) $\partial(u_j\partial u_i/\partial x_i)/\partial x_j$.

6.2.2. 1.88×10^6, 1.26×10^6 Pa·s/m.
6.2.4. (*a*) 4.82, 1.20, 3.62 mW/m^2. (*b*) 3.62 mW/m^2.
(*c*) $\cos(\omega t - k_1x) + 2\cos k_1x\cos\omega t$.
(*d*) 0, 1.20 mW/m^2. (*e*) Yes.
6.3.2. (*a*) 6.94 cm, 5550 m/s. (*b*) 0.49
6.3.4. 8.35×10^6 Pa·s/m. (*b*) 1.04×10^4 kg/m^3, 800 m/s,
6.3.6. (*a*) (6.3.7) and same equation with subscripts 1 and 3 exchanged.
(*b*) Except for special cases, not the same. (*c*) R_{II} and T_{II} remain unchanged.
(*d*) Independent of direction of approach normal to layer.
6.3.8. (*a*) 68.5 ms. (*b*) 0.120, 0.437. (*c*) 335°.
6.4.2. 167 kg/m^3, 3000 m/s.
6.4.4. (*a*) 28.5°. (*b*) 104 Pa. (*c*) 4.2 Pa. (*d*) 0.00176
6.4.6. (*a*) 69.6°. (*b*) 65.4°. (*c*) 0.077
6.4.8. (*a*) If $c_2 < c_1$, $-\{1 - 2\beta(\rho_2/\rho_1)/\sqrt{[(c_1/c_2)^2 - 1]}\}$;
if $c_2 > c_1$, $-\{1 - j2\beta(\rho_2/\rho_1)/\cos\theta_c\}$.
(*b*) $1 - (\rho_1/\rho_2)\sqrt{\{8(\theta_c - \theta_i)/\sqrt{[(c_2/c_1)^2 - 1]}\}}$.
(*c*) $1 - j(\rho_1/\rho_2)\sqrt{\{8(\theta_i - \theta_c)/\sqrt{[(c_2/c_1)^2 - 1]}\}}$. (*d*) $\mathbf{R} \to \pm e^{-\delta}$ where $\mathbf{R} = \pm(1 - \delta)$.
Note that these are lowest-order results. For complex **R**, they give $|\mathbf{R}| > 1$, but this is only because second- and higher-order terms have been neglected.
6.6.2. $\{(\cos\theta_i - 1)^2 + [\omega\rho_S/(\rho_1c_1)]^2\cos^2\theta_i\}/\{(\cos\theta_i + 1)^2 + [\omega\rho_S/(\rho_1c_1)]^2\cos^2\theta_i\}$.

6.8.2. (*b*) Increased by 6 dB.

7.1.2. $\rho_0 caU_0/r$, aU_0/r, $\rho_0 ca^2U_0^2/2r^2$, $2\pi\rho_0 ca^2U_0^2$.
7.1.4. (*a*) 3.18 mW/m^2. (*b*) 1.63 Pa. (*c*) 0.0041 m/s. (*d*) 1.63×10^{-6} m. (*e*) 1.15×10^{-5}.
7 .2.2. (*b*) Yes.
7.3.2. (*a*) 1. (*b*) 14. (*c*) 14.4°. (*d*) - 13.6 dB.
7.4.2. (*b*) For $r > 2a$ the agreement is within 6%.
7.4.4. 2.5% high.
7.4.6C. (*c*) 37.3 cm.
7.6.2. (*a*) 28.6, 61.0°. (*b*) 1.67 m/s. (*c*) 6, 6 dB.
7.6.4. (*a*) 12.7 W. (*b*) 0.98 cW/m^2. (*c*) 5.9 dB. (*d*) 1.0×10^{-3} kg.
7.6.6. (*a*) 31 cm/s. (*b*) 15.3 N.
7.6.8. (*a*) 199 m. (*b*) 12 dB. (*c*) 11 dB. (*d*) about 25%, 1 dB, insignificant.
7.8.2. (*a*) 4.83 m. (*b*) 3.96°. (*c*) Plane of $\theta = 0$. (*d*) 15 dB.
7.8.6. ≥ 6, 0.8 dB.
7.8.8. (*a*) π. (*b*) $\cos^2[(\pi/2)\sin\theta]$.
7.10.4. Agrees.
7.10.6. $(Q/4\pi r)(kd)^2 \exp[j(\omega t - kr)]$.
7.10.12. $[3Q/4\pi r][1/(1\cdot3) + (ka)^2/(3\cdot5\cdot2!) - (ka)^4/(5\cdot7\cdot4!) + \ldots] \exp[j(\omega t - kr)]$.

8.2.2. (*a*) 1.73×10^{-9} s. (*b*) 92 MHz. (*c*) 5.4×10^5 Np/m, 442 m/s. (*d*) 8.4×10^4 Np/m, 344 m/s.
8.2.4. (*a*) 2.5×10^{25} molecules/m^3. (*b*) 462 m/s. (*c*) 9.7×10^{-8} m. (*d*) 2.1×10^{-10} s.
8.3.2. $\tau = \tau_S$.
8.4.2. $\frac{1}{2}(\mathscr{P}_0/\gamma)[\gamma/(\gamma - 1)]^2(\Delta T/T_K)^2$.
8.5.2. 0.710, 0.743.
8.6.2. (*a*) 0.362. (*b*) 267 m/s, consistent. (*c*) 276 m/s, consistent.
8.6.4. (*a*) 0.588 W. (*b*) 3.43×10^{-6} Np/m, 0.589W. (*c*) 1.6×10^{-4} Np/m, 0.648 W. (*d*) 3.0×10^{-4} Np/m, 0.707 W.
8.6.6. 1710 K, 2.36×10^{-20} J.
8.7.2. (*a*) 3.7×10^7 m. (*b*) 9.1×10^4 m. (*c*) 1.8×10^5, 3.5×10^3 m. (*d*) 7.2×10^3, 160 m. (*e*) 520, 50 m.
8.9.2. (*a*)-(*c*) 0.149, 1.10×10^{-4}, 2.88×10^{-3} dB/m.
8.9.6. (*a*) 1.56×10^{-4} m. (*b*) 339 m/s. (*c*) 0.042 Np/m. (*d*) 0.73 dB.
8.10.2. (*a*) Combine (8.10.1, 7, 16, 17), $\sigma = 2\alpha/N$. (*b*) 828, 187 Hz. (*c*) 3.3×10^{-13}, 3.2×10^{-12}, 4.6×10^{-12} m^2.
8.10.6. (*a*) $\omega_0 m_r/R_r$. (*b*) $\omega_0 m_r/R_m$.
8.10.8C. 24

9.2.2. (001), 0.250 c/L; (101) (011), 0.559 c/L; (002) (111), 0.750 c/L.
9.2.4. (*a*) [(000), 0 c/L]; (100) (010) (001), 0.500 c/L; (110) (101) (011), 0.707 c/L; (111), 0.866 c/L; (200) (020) (002), 1.000 c/L. (*c*) See (*a*). (*d*) Center of a wall for (100) etc.; center of an edge for (110) etc.; any position like (0, $L/4$, $L/4$) for (200) etc.

9.3.4. (*a*) (101) 375, (111) 577, (121) 819, (102) 988, (201) 1124 Hz.
(*b*) For all five modes, $z = 1$ m (top surface); if $\gamma_{mn} = 0$, then also $\theta = \pi/2$ for (111) and $\theta = \pi/4$, $\pi/2$ for (121); r = 0.63 m for (102); $z = 0.33$ m for (201)

9.4.2. (*a*), (*b*) 3.55 kHz, node at 20 cm, (antinode at 0 cm); 5.08, 0, 20, (9.3); 6.52, 0, 20, (11.6). (*c*) Only the first one.

9.5.2. (*a*) 0, $c/2L$, $(c/2L)\sqrt{2}$, c/L, $(c/2L)\sqrt{5}$.

9.6.2. 0 for (010) (110) (011); 0.707 for (001) (101).

9.6.4. (*a*) (11) (12) (21) (13) (31) (14) (41) (22) (23) (32) (33).
(*b*) (11) (13) (31) (33). (*c*) All have unit relative amplitude.

9.8.2. (*a*) 1. (*b*) $(0.508/\sqrt{r})\sin(0.131z)$.

9.8.4. (*a*) $(0.725/\sqrt{r})\sqrt{[1 + 0.397\cos(0.161r)]}$. (*b*) 19.5 m.

9.9.2. (*a*) $Z_{1n} = \sin k_{zn}z$, $Z_{2n} = [(\sin k_{zn}H)/(\sin \phi_n)]\sin[\gamma_n(z - H) + \phi_n]$ with $\gamma_n = [(\omega/c_2)^2 - \kappa_n^2]^{1/2}$. (*c*) $\kappa_n \le (\omega/c_1)\cos\theta_c$.

9.9.10. (*a*) 20.4°. (*b*) 89.7, 269.0, 448.3 Hz. (*c*)

f/f_1	k_{z1}	κ_1	A_1	k_{z2}	κ_2	A_2
1	0.131	0.353	0	--	--	--
2	0.187	0.729	0.340	--	--	--
3	0.207	1.109	0.362	0.393	1.059	0
4	0.218	1.489	0.373	0.434	1.441	0.364

10.2.2. (*a*) $-j(\rho_0 c^2 S/L)/\omega$. (*b*) Static adiabatic compression.

10.2.4. (*b*) - 0.8%, - 0.8%, - 0.8%.

10.3.2. (*a*) 164.5 Hz. (*b*), (*c*) 140, 6.34 W.

10.3.4C. (*b*) From (10.3.8), 6400; from (10.2.4), 6800 ± 400.

10.4.2. (*a*), (*b*) 50.5, 62.5 dB *re* 20 μPa. (*c*) 9.5 cm.

10.4.4. (*a*) 5.45. (*b*) 14.7 dB. (*c*) 0.37, 1.11, 1.85 m.

10.5.2C. (*c*) At 30 kHz, R_{m0} in error by 7.5% and X_{m0} by 20%; $\alpha/k = 1 \times 10^{-3}$, $\alpha L = 0.5$

10.6.2. $\frac{1}{2}\rho_0 cSF^2/[(\rho_0 cS)^2 + (\omega m)^2]$. (*b*) $\frac{1}{4}\rho_0 cS(ka)^2F^2/[\omega(m + \rho_0 SL)]^2$.

10.6.4C. (*a*) 486, 533 Hz.

10.8.2. (*a*)-(*c*) 88.3, 69.3, 50.3 Hz. (*d*) 0.042 W. (*e*) 149 Pa, 29.8 N.

10.9.2. (*a*) 9.80 Hz. (*b*) 2.8×10^{-4} m³/Pa, 0.94 Pa·s²/m³. (*c*) $-j37.3$ Pa·s/m³.

10.10.2. (*a*) $4\rho_1 c_1 S_1 \rho_2 c_2 S_2/(\rho_1 c_1 S_2 + \rho_2 c_2 S_1)^2$. (*b*) $\rho_2 c_2 S_1 = \rho_1 c_1 S_2$.

10.10.4. (*a*) $4S^2/(2S + S_b)^2$. (*b*) $4SS_b/(2S + S_b)^2$. (*c*) 0.64, 0.32.
(*d*) No, reflected back toward the source.

10.10.6. (*a*) 0.58. (*b*) 0.26.

10.10.8. 34, 66, 0.8 mW.

10.10.10. 0.41 m.

10.10.12C. $0.175 < S_2/S_1 < 5.82$.

10.10.14C. 113, 224, 450, 563, 788, 901 Hz

10.11.2. Identical with S the analog of r_1 and $(S_1 - S)$ of r_2.
10.11.4. (*b*) $LV/S = 3.31$ m^2.
10.11.6. (*a*), (*b*) (6.3.9) with $(S_1 - S)/S = r_2/r_1$. (*c*) $[2/(a + 1/a)]^2$ with $a = (S_1 - S)/S$.
10.11.8C. (*b*) Vary S and/or L, V, and S_b, but keep $S_b/L'V$ constant.

11.2.2. (*a*) 13. (*b*) $f_n = f_1 \times 2^{(n-1)/12}$. (*c*) - 0.11%. (*d*) +0.11%.
11.3.2. (*a*) 0.142, 0.126, 0.224, 0.178, 0.224, 0.158 Pa.
(*b*) (4.86, 3.83, 12.1, 7.63, 12.1, 6.0) $\times 10^{-5}$ W/m^2.
(*c*) 77.0, 76.0, 81.0, 79.0, 81.0, 78.0 dB.
(*d*) 57.0, 53.0, 55.0, 50.0, 49.0, 43.0 dB (*e*) 87 dB *re* 20 μPa. (*f*) Same.
11.3.4. (*a*) 40, 33, 30 dB *re* 10^{-12} W/m^2. (*b*) 63.6 dB *re* 10^{-12} W/m^2.
11.3.6. (*a*) 68.6, 66.4, 68.6 dB *re* 10^{-12} W/m^2. (*b*) 64.5, 66.8, 74.5 dB *re* 10^{-12} W/m^2.
11.4.2. (*a*) $P(D)$ and d' get smaller, $P(FA)$ same.
(*b*) $P(D)$ and $P(FA)$ get smaller, d' same.
(*c*) $P(D)$ and d' get larger, change in $P(FA)$ undetermined. (*d*) Same as (*b*).
11.4.4. (*a*) 0.01, 0.34, 0.12. (*b*) 2, 3, 1.
11.5.2. (*a*) 60, 63, 64.7 dB *re* 20 μPa. (*b*) 5.0 (energy), 10.0 (correlation).
(*c*) ~ 5×10^{-5} (energy), ~ 5×10^{-9} (correlation). (*d*) - 3 dB. (*e*) - 1.5, - 1.5 dB.
11.5.4. Compare well.
11.6.2. 3.10, 2.78, 2.38, 1.76, 1.01, 0 cm.
11.7.2. (*a*) $DT' = 5 \log[(d')^2/(\omega_{cr} T_s)]$ for $\tau > T_s$,
$DT' = 5 \log[(d')^2/(\omega_{cr} T_s)] + 10 \log(T_s/\tau)$ for $\tau < T_s$. (*b*) - 5.0, - 5.0, 1.0 dB.
(*c*) 0.40 s. (*d*) - 5.0, - 5.0, - 5.2, - 5.9, - 7.4, - 9.5 dB.
11.7.4. (*a*) $\frac{1}{2}a_2 P + a_1 P \cos \omega t + \frac{1}{2}a_2 P^2 \cos 2\omega t$. (*b*) $\frac{1}{2}a_2(P_1^2 + P_2^2)$ at 0 Hz, $a_1 P_1$ at ω_1,
$a_2 P_2$ at ω_2, $\frac{1}{2}a_2 P_1^2$ at $2\omega_1$, $\frac{1}{2}a_2 P_2^2$ at $2\omega_2$, $a_2 P_1 P_2$ at $\omega_1 + \omega_2$ and at $\omega_1 - \omega_2$.
11.7.6. 177 Hz if 6, 7, 8 harmonics; 247 Hz if 4, 5, 6 harmonics.
11.8.2. (*a*) 67, 73, 75, 84, 75, 65 phon. (*b*) 25, 38, 44, 54, 45, 35 phon.
11.8.4. (*a*) 500 Hz. (*b*) 64 dB *re* 20 μPa. (*c*) 73 phon.

12.3.2. (*a*) 1.47 s. (*b*) 7.1×10^{-4} W. (*c*) 0.0408. (*d*) 10.4 s, greatly reduced.
12.3.4. (*a*) 6.1×10^{-5} W. (*b*) 0.166. (*c*) 1.12 s. (*d*) 57.5 dB *re* 20 μPa.
12.3.6. (*b*) 4.7×10^{-3} m^{-1}. (*c*) about 1×10^{-2} m^{-1}. (*d*) 1.05 s [using m from (*b*)]
12.4.2. (*a*) 0.217. (*b*) 0.619 s. (*c*) 0.548 s, about 12% lower.
12.4.4. (*a*) 2.04 m. (*b*) 169, (*c*) 0.24 s, 0.29.
12.4.6. (*c*) Totally implausible; open a window and $T = 0$ (??).
12.5.2. (*a*) 70 dB *re* 20 μPa. (*b*) 66 phon. (*c*) 62 dB *re* 20 μPa, 66 phon.
12.6.2. - 6.4 dB.
12.7.2. 0.79 m.
12.8.2. 2.39 s (125 Hz), 2.29 (250), 1.87 (500), 1.66 (1 k), 1.62 (2 k), 1.37 (4 k); good.
12.9.2. (*a*)-(*c*) 34.3, 48.5, 59.4 Hz.
12.9.4. (*a*) 34.3, 48.5, 59.4, 68.6 Hz.
(*b*) All six sides and bisecting the cube in 1, 2, and 3 dimensions.
12.9.6. (*a*)-(*c*) 1.33, 1.78, 1.14 s.

12.9.8. (*a*) 192 Hz, 100. (*b*) 22 Hz, 21.
12.9.10. Within the triangle with vertices at $(X, Y) = (1, 1)$, $(3/2, 5/2)$, $(9/5, 11/5)$.

13.2.2. (*a*) 39, 50, 62, 73, 78, 80, 81, 74, 64 dB *re* 20 μPa.
(*b*) 75, 75, 62, 73, 78, 80, 81, 73, 63 dB *re* 20 μPa.
13.3.2. (*a*)-(*c*) 64, 47, 46 dBA. (*d*) 96%. (*e*) Somewhat high (about 1 out of 40 unintelligible).
13.6.2. 94 dBA.
13.8.2. 1.44.
13.15.2C. The approximation consistently overestimates the *TL* values predicted by (13.15.6) except at the lowest value, 10 kg/m^3 at 60 Hz.

14.3.2. (*a*) $1 - \mathbf{k}_m^2 = 1/(1 - \mathbf{k}_c^2)$. (*b*) $|\,\mathbf{k}_m\,| \approx |\,\mathbf{k}_c\,|$.
14.4.2. 160 dB *re* 1 μPa/V, 134 dB *re* 20 μPa/V.
14.4.4. (*a*)-(*d*) 110, 82, 95, 106 dB *re* 10^{-12} W/m^2. (*e*) 2.2 W.
14.5.2. (*a*) $(3.2 + j0.25)$, $(0.36 - j3.20)$, $(3.56 - j2.95)$ Ω. (*b*) 13.6 V. (*c*) 1.58 W. (*d*) 28.5 Ω, 38 mH, 263 μF.
14.5.4. (*a*) 1.1 W. (*b*) 93 dB *re* 20 μPa.
14.6.2. (*a*) 99 Hz. (*b*) 1.58, 0.81, 0.43 W.
14.7.2. (*a*) 280 Hz. (*b*) 19 W
14.7.4. (*a*) 18.3. (*b*) 5.9×10^{-4} m^3/s. (*c*) 3.3×10^{-5} m. (*d*) 31%. (*e*) 1.22 V.
14.7.6. (*b*) x to p, t to x, R_m/m to 2β, ω_0 to $\pm k$, and ω_d to $\pm\kappa$.
14.8.2. - 180 dB *re* 1 V/μbar.
14.9.2. (*a*) 34.4 kHz. (*b*) - 70 dB *re* 1 V/μbar, - 50 dB *re* 1 V/Pa, $-j3.6 \times 10^5$ Ω. (*c*) - 48 dB *re* 1 V/Pa.
14.10.2. (*a*), (*b*) - 86, - 88 dB *re* 1 V/Pa.
14.11.2. $2PS[\cos kx - \cos k(L - x)]\cos \omega t$.
14.11.4. (*b*) $[\sin(\tfrac{1}{2}kL \cos \theta)]/(\tfrac{1}{2}kL)$. (*c*) 2.78. (*d*) 5.1 kHz.
14.12.2. (*a*) 8.33×10^8 Ω/m. (*b*) 0.042 (ratio of peak amplitudes).
14.13.2. (*a*) - 134 dB *re* 1 V/μbar. (*b*) 60dB *re* 1 μbar.
14.13.4. - 48 dB *re*1 V/Pa.

15.3.2. (*a*), (*b*) 207, 127 dB *re* 1 μPa. (*c*)-(*e*) 4.6, 11, 1.2 km.
15.4.2. (*b*) - 0.4 s^{-1}. (*c*) - 0.13 C$^\circ$/m. (*d*) 610 m. (*e*) 9.4°.
15.5.4. (*a*) 2.96. (*b*) 774 m. (*c*) 6.969 km. (*d*) $11.3^\circ/(10n \pm 1)$, $n = 1, 2, 3, \ldots$.
15.5.6. (*a*)-(*c*) 1.159, 8.944, 11 km. (*d*) 4.9°.
15.6.2. (*a*) 1400 m. (*b*) 13.8°, 41.3 km. (*c*) 5.2 km.
15.7.2. (*a*) 6.33 *khd*. (*b*) 2.64 km.
15.8.2. 7 dB.
15.8.4. 0.91 km.
15.8.6. (*a*), (*b*) 10, 3.16 times larger.
15.9.2. 76 dB *re* 1 μPa.
15.10.2. (*a*) $4.66/\sqrt{R}$. (*b*) 1.47 kHz.. (*c*) 21.7 km.
15.10.4. (*a*), (*b*) 10.24, 1.024 Hz. (*c*) Between 15.4 and 12.3 m/s.

15.10.6. (*a*)-(*d*) 140, 143, 150, 161 dB *re* 1 μPa.
(*e*) Broadband, broadband and 300 Hz tone, broadband, 600 Hz tone.
15.11.2. (*a*) 10 log R_Π + 20 log z - 6 dB. (*b*) 11.6 dB *re* 1 μPa.
15.11.4. (*a*) 86, 83, 80, 76, 72, 68 and 87, 85, 83, 81, 81, 80 dB *re* 1 μPa. (*b*) $\leq$ 2 km.
15.11.6. (*a*), (*b*) 4.65, 2.25 km.
15.12.4. 4 km.

16.3.4. (*a*) $a/(\beta Mka - 1)$. (*c*) $(1/\ell - 1/a)/\alpha$. (*d*) $a > 0.27$ m.

17.1.4. (*a*) 3.69 bar. (*b*) 2.03.
17.1.8. (*a*) Multiply u_1 and M_1 by $\sin(\alpha)$ and multiply u_2 and M_2 by $\sin(\alpha - \delta)$.
[This holds also for (17.1.11) through (17.1.15)]
(*b*) 1.76 atm, 67°C, 11° deflection.
17.2.2. (*a*) $p(0)\tau(1/b - 1/b^2)$. (*c*) $b < 1$.
17.2.4. (*a*) $417M_1$ Pa·s/m. (*b*) Yes.
17.3.2. (*a*) 6.53×10^5 J. (*b*) 0.142 kg TNT equivalent.
17.3.6. (*a*) 2.0. (*b*) 1.31. (*c*)158 m/s. (*d*) 357 K.
17.3.12. (*a*) $\propto 1/R$. (*b*) 345 m/s. (*c*) Both consistent.
17.5.2. (*a*) 56 mbar. (*b*) 15.2 s. (*c*) 1.56 s. (*d*) 12.9 m/s. (*e*) 31.9 bar·ms

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